

श्री
Balaji

ADVANCED PROBLEMS IN

Algebra

for
JEE
Main & Advanced

by:

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KOTA



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Solutions

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SECTION-1

SINGLE CHOICE QUESTIONS

- If the equation $\sin x + \cos x = y^2 - y + a$ has no solution in x and y , then
 - $a < -2$
 - $a \in (-1, 1)$
 - $a < \frac{1}{4} - \sqrt{2}$
 - $a > \sqrt{2} + \frac{1}{4}$
- The set of values of 'a' for which the equation $\cos^4 x - \sin^4 x + \cos 2x + a^2 + a = 0$ will have atleast one real solution is
 - $[-2, 1]$
 - $[-1, 2]$
 - $[-1, 1]$
 - $[1, 2]$
- The solution set of the inequality $\sqrt{x} - 3 \leq \frac{2}{\sqrt{x} - 2}$ is
 - $[0, 1] \cup [4, 16]$
 - $[0, 1] \cup (4, 16]$
 - $(0, 1) \cup (4, 16)$
 - $(0, 1] \cup (4, 16]$
- Let $a, b, c \in \mathbb{R}$, $a \neq 0$ such that a and $4a + 3b + 2c$ have the same sign. Then the equation $ax^2 + bx + c = 0$ must have
 - both roots in $(1, 2)$
 - no roots in $(1, 2)$
 - not both roots in $(1, 2)$
 - exactly one root in $(1, 2)$
- Let $x = \frac{\sqrt{7} + \sqrt{3}}{\sqrt{7} - \sqrt{3}}$, $y = \frac{\sqrt{7} - \sqrt{3}}{\sqrt{7} + \sqrt{3}}$, then the value of $x^4 + y^4 + (x + y)^4$ is equal to
 - 527
 - 1254

6. Let 'p' is a root of the equation $x^2 - x - 3 = 0$. Then the value of $\frac{p^3 + 1}{p^5 - p^4 - p^3 + p^2}$ is equal to
- (A) $\frac{4}{3}$ (B) $\frac{4}{9}$
 (C) $\frac{2}{9}$ (D) $\frac{2}{3}$
7. Given that the equation $(x - 19)(x - 97) = p$ has real roots α and β . Then the minimum real root of the equation $(x - \alpha)(x - \beta) = -p$ is
- (A) 13 (B) 16
 (C) 18 (D) 19
8. Let a, b, c are real numbers with $a^2 + b^2 + c^2 > 0$. Then the equation $x^2 + (a + b + c)x + (a^2 + b^2 + c^2) = 0$ has
- (A) 2 positive real roots (B) 2 negative real roots
 (C) 2 real roots with opposite sign (D) no real roots
9. Let a, b are two different positive integers and the two quadratic equations $(a - 1)x^2 - (a^2 + 2)x + (a^2 + 2a) = 0$ and $(b - 1)x^2 - (b^2 + 2)x + (b^2 + 2b) = 0$ have one common root. Then the value of $\frac{a^b + b^a}{a^{-b} + b^{-a}}$ is equal to
- (A) 256 (B) 64
 (C) 16 (D) 72
10. If p, q_1 and q_2 are real numbers with $p = q_1 + q_2 + 1$, then which of the following must be correct about the equations $E_1 : x^2 + x + q_1 = 0$ and $E_2 : x^2 + px + q_2 = 0$,
- (A) Nothing can be said about roots of the two equations.

(B) atleast one of the equation has distinct real roots.

(C) atleast one must have imaginary roots.

(D) atleast one must have real roots of oposite sign.

Quadratic Equations

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11. Given that the solution set of the quadratic inequality $ax^2 + bx + c > 0$ is $(2, 3)$.

Then the solution set of the inequality $cx^2 + bx + a < 0$ will be

(A) $\left(\frac{1}{3}, \frac{1}{2}\right)$

(B) $(-\infty, 2) \cup (3, \infty)$

(C) $\left(-\infty, \frac{1}{3}\right) \cup \left(\frac{1}{2}, \infty\right)$

(D) Nothing can be said

12. Complete set of real values of k for which the inequality $kx^2 - kx - 1 < 0$ holds for any real x , satisfy

(A) $k \in (-4, 0)$

(B) $k \in (-4, 0]$

(C) $k \in [-4, 0)$

(D) $k \in [-4, 0]$

13. The poduct of all values of x satisfying the equation

$$\frac{1}{x^2 + 2x} + \frac{1}{x^2 + 6x + 8} + \frac{1}{x^2 + 10x + 24} = \frac{1}{5} - \frac{1}{x^2 + 14x + 48} \text{ is}$$

(A) -80

(B) 40

(C) -10

(D) -20

14. Let x^1, x^2, x^3 be the roots of the equation $x^3 + 3x + 5 = 0$. Then the value of

expression $\left(x_1 + \frac{1}{x_1}\right)\left(x_2 + \frac{1}{x_2}\right)\left(x_3 + \frac{1}{x_3}\right)$ is equal to

(A) $-\frac{29}{5}$

(B) -5

(C) $-\frac{24}{5}$

(D) -1

15. The value of maximum real root minus the minimum real root of the equation

$$(x^2 - 5)^4 + (x^2 - 7)^4 = 16 \text{ is}$$

(A) $\sqrt{5} + \sqrt{7}$

(B) $2\sqrt{5}$

(C) $\sqrt{28}$

(D) $4\sqrt{2}$

16. Three real numbers x, y, z are such that $x^2 + 6y = -17$, $y^2 + 4z = 1$ and $z^2 + 2x = 2$.

Then the value of $x^3 + y^3 + z^3$ is equal to

(A) 30

(B) -24

(C) -36

(D) -28

17. The degree of the polynomial $p(x)$ which satisfies the condition

$$(x)p(x-1) = (x-15)p(x) \text{ is :}$$

(A) 4

(B) 15

(C) 7

(D) can not be determined

18. The equations $x^2 - 4x + k = 0$ and $x^2 + kx - 4 = 0$, where k is a real number, have exactly one common root. Then the number of values of k is

(A) 0

(B) 1

(C) 2

(D) 3

19. Suppose 1, 2, 3 are the roots of the equation $x^4 + ax^2 + bx = c$. Then the value of c is

(A) 25

(B) 8

(C) 24

(D) 36

20. If a, b are the roots of the quadratic equation $x^2 + \lambda x - \frac{1}{2\lambda^2} = 0$, where λ is a real parameter. Then the minimum value of $a^4 + b^4$ is
- (A) $2\sqrt{2}$ (B) $2(\sqrt{2} - 1)$
 (C) $\sqrt{2}$ (D) $2 + \sqrt{2}$
21. If a, b, c are complex numbers and $a + b + c = ab + bc + ca = abc = 1$, then find $a + b^4 + c^4$
- (A) 1 (B) 0
 (C) 3 (D) 8

Quadratic Equations

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22. Let $p(x) = x^6 + ax^5 + bx^4 + x^3 + bx^2 + ax + 1$. Given that 1 is a root of $p(x) = 0$ and -1 is not. What is the maximum number of distinct real roots that p could have
- (A) 6 (B) 5
 (C) 4 (D) 3
23. If the roots of quadratic equation $ax^2 + bx + c = 0$ are $\frac{m}{m-1}$ and $\frac{m+1}{m}$, then $(a+b+c)^2$ is equal to
- (A) $b^2 - ac$ (B) $4(b^2 - 4ac)$
 (C) $b^2 - 4ac$ (D) $\frac{b^2 + c^2}{a^2}$
24. The set of all real values of 'a' for which both the roots of the equation $x^2 - 1 = 0$ lie between the roots of the equation $x^2 + (3a - a^2)x - 3a^3 = 0$ is equal to
- (A) $(-\infty, -1)$ (B) $\left(-1, -\frac{1}{3}\right) \cup (1, \infty)$
 (C) $(-1, 1)$ (D) $\left(-\frac{1}{3}, 1\right)$

(C) $(1, \infty)$

(D) $\left(-\frac{1}{3}, 1\right]$

25. The complete set of real values of 'a' for which the smaller root of the equation $x^2 + 2ax - 3 = 0$ lies in the interval $(-1, 1)$ is

(A) $(1, \infty)$

(B) $(0, 1)$

(C) $\left(-\frac{1}{4}, \infty\right)$

(D) $a \in (2, \infty)$

26. If the ratio of the roots of equation $ax^2 + 2bx + c = 0$ is same as the ratio of the roots of equation $px^2 + 2qx + r = 0$, where a, b, c, p, q, r are non zero real numbers. Then

the value of $\left(\frac{b^2}{q^2}\right)\left(\frac{p}{a}\right)\left(\frac{r}{c}\right)$ is equal to :

(A) 4

(B) 2

(C) 1

(D) -1

27. The complete set of real values of 'a' for which the inequality $ax^2 - (3 + 2a)x + 6 > 0$, $a \neq 0$ holds for exactly three negative integral values of x is

(A) $\left[-\frac{5}{4}, -\frac{3}{4}\right]$

(B) $\left[-2, -\frac{5}{4}\right]$

(C) $(-1, 0)$

(D) $\left[-1, -\frac{3}{4}\right]$

28. If the equation $ax^2 + bx + c = x$, $a, b, c \in \mathbb{R}$ and $a \neq 0$, has no real roots, then the equation $a(ax^2 + bx + c)^2 + b(ax^2 + bx + c) + c = x$ will have

(A) 2 distinct real roots

(B) no real roots

(C) 2 equal real roots

(D) nothing can be said

29. The equation $x^2 - 4ax + 1 = 0$ has real roots given by α and β , where a is real. Then the complete set of values of a for which $\alpha \geq a$ and $\beta \geq 0$ is

(A) $[1, \infty)$

(B) $[2, \infty)$

(C) $(-\infty, -\frac{1}{2}] \cup [\frac{1}{2}, \infty)$

(D) $[\frac{1}{2}, \infty)$

30. If the quadratic equation $ax^2 - bx + 7 = 0$ does not have two distinct real roots, then the minimum value of $a + b$ is equal to :

(A) -8

(B) -7

(C) -6

(D) -5

31. If α and β are roots of the equation $ax^2 + bx + c = 0$, then the roots of the equation $ax^2 - bx(x + 1) + c(x + 1)^2 = 0$ are

(A) $\frac{\alpha}{\alpha+1}, \frac{\beta}{\beta+1}$

(B) $\frac{-\alpha}{\alpha+1}, \frac{-\beta}{\beta+1}$

(C) $\frac{\alpha}{1-\alpha}, \frac{\beta}{1-\beta}$

(D) $\frac{\alpha}{\alpha-1}, \frac{\beta}{\beta-1}$

32. If the equation $(6a + 3b + 4c)x^2 + (11a + 8b + 7c)x + (3c + 5a + 5b) = 0$ has equal real roots, where a, b, c are positive real numbers. Then a, b, c are in

(A) A.P.

(B) G.P.

(C) H.P.

(D) nothing can be said

33. Let a, b , are real numbers such that $a + b = 5$. Then the equation $x^2 - ax - b = 0$ must have for all real values of a

(A) equal real roots

(B) imaginary roots

(C) distinct real roots

(D) nothing can be said

34. If both roots of the equation $x^2 - 2(a - 1)x + 2a + 1 = 0$ are positive, then

(A) $a < 2$

(B) $a \geq 4$

(C) $1 \leq a \leq 4$

(D) $1 < a < 2$

35. The complete set of real values of p for which both roots of the equation $x^2 + 2(p - 3)x + 9 = 0$ lie in $(-6, 1)$ is

(A) $\left[6, \frac{27}{4}\right)$

(B) $\left(\frac{27}{4}, 9\right]$

(C) $[6, 9)$

(D) $(-2, 0] \cup (2, 9)$

36. the complete set of real values of 'a' for which the equation $9^x - (4a)3^x + 4 - a^2 = 0$ has an unique root in the interval $(0, 1)$ is

(A) $(-12, 0)$

(B) $(0, 11)$

(C) $(-11, -1) \cup \left\{\frac{2}{\sqrt{5}}\right\}$

(D) $(-13, -5) \cup \left\{\frac{2}{\sqrt{5}}\right\}$

37. Let a, b, c be real numbers, $a \neq 0$, if α is a root $a^2x^2 + bx + c = 0$, β is a root of $a^2x^2 - bx - c = 0$ and $0 < \alpha < \beta$, then the equation $a^2x^2 + 2bx + 2c = 0$ has a root γ that always satisfies

(A) $\gamma < \alpha$

(B) $\gamma > \beta$

(C) $\alpha < \gamma < \beta$

(D) $\gamma = \frac{\alpha + \beta}{2}$

38. If the maximum and minimum values of $y = \frac{x^2 - 3x + c}{x^2 + 3x + c}$ are 7 and $\frac{1}{7}$ respectively, $c \in \mathbb{R}$. Then the value of c is equal to

(A) 2

(B) 4

(C) 3

(D) 5

39. The complete set of real values of 'a' for which there are distinct real numbers x, y satisfying the equations $x = a - y^2$ and $y = a - x^2$ is

(A) $\left[\frac{3}{2}, \infty\right)$

(B) $\left[\frac{5}{2}, \infty\right)$

(C) $\left[\frac{7}{4}, \infty\right)$

(D) $(0, \infty)$

40. If a, b, c, p, q, r are non-zero real numbers, such that $a < b < c$ and

$f(x) = (x-a)(x-b)(x-c) - p^2(x-a)^2 - q^2(x-b)^2 - r^2(x-c)^2$, then $f(x) = 0$ must have
(A) exactly 1 real root (B) exactly 3 distinct real roots

(C) 2 equal and 1 distinct real roots (D) nothing can be said

41. If $ax^2 + bx + 8 = 0$, $a, b \in \mathbb{R}$, $a \neq 0$ has no distinct real roots, then the least value of $4a + b$ is

(A) -4

(B) -3

(C) -2

(D) -1

SECTION-2

ONE OR MORE THAN ONE CORRECT QUESTIONS

1. Let the inequalities $y(-1) > -4$, $y(1) < 0$ and $y(3) > 5$ are known to hold for $y = ax^2 + bx + c$, then

(A) $b > 3$

(B) $b < 2$

(C) $a < 0$

(D) $a > \frac{1}{8}$

2. Let $ax^2 + bx + c$ is integer for all integral values of x , then which of the following must be true ?

(A) c is integer

(B) $b = \frac{n}{2}, n \in \mathbb{I}$

(C) $a = \frac{n}{2}, n \in \mathbb{I}$

(D) a, b both are integers

3. If the value of quadratic trinomial $ax^2 - bx + c$ is an integer for $x = 0, x = 1$ and $x = 2$, then the value of the given trinomial is an integer for

(A) $x = 2017$

(B) $x = 2018$

(C) $x = -4$

(D) $x = -2017$

4. How many roots does the equation $x^4 = 5x + 2a$ possess depending on 'a' ?

(A) if $a < -\frac{15}{16}\sqrt[3]{10}$

(B) 2 if $a > -\frac{15}{16}\sqrt[3]{10}$

(C) 3 if $a = -\frac{15}{16}\sqrt[3]{10}$

(D) 1 if $a = -\frac{15}{16}\sqrt[3]{10}$

5. How many roots does the equation $x^3 + ax + 2 = 0$ possess depending on 'a'?

(A) 1 if $a \in (-\infty, -3)$

(B) 3 if $a \in (-\infty, -3)$

(C) 2 if $a = -3$

(D) 1 if $a \in (-3, \infty)$

6. How many solutions does the system of equations $|x| + |y| = 1, x^2 + y^2 = a^2$ possess depending on 'a'?

(A) if $|a| < \frac{1}{\sqrt{2}}$

(B) 0 if $|a| > 1$

(A) $0 < |a| < \sqrt{2}$

(B) $0 < |a| < 1$

(C) 4 if $|a| = \frac{1}{\sqrt{2}}, 1$

(D) 8 if $\frac{1}{\sqrt{2}} < |a| < 1$

7. The values of 'a' for which the curves $y = 1 + \frac{x^2}{a^3}$ and $y = 4\sqrt{x}$ possess only one point in common is/are

(A) $a \in \left(0, \frac{1}{3}\right)$

(B) $a \in (-\infty, 0)$

(C) $a = \frac{1}{3}$

(D) $a \in \left(\frac{1}{3}, \infty\right)$

8. The greatest value of the function $f(x) = \frac{1}{2bx^2 - x^4 - 3b^2}$ on the interval $[-2, 1]$ depending on the parameter 'b' is/are

(A) $-\frac{1}{3b^2}$ if $b \in [0, 2]$

(B) $\frac{1}{4b - 4 - 3b^2}$ if $b \in [0, 4]$

(C) $\frac{1}{8b - 16 - 3b^2}$ if $b \leq 2$

(D) $-\frac{1}{3b^2}$ if $b \geq 2$

9. The greatest value of the function $f(x) = x^4 - 6bx^2 + b^2$ on the interval $[-2, 1]$ depending on the parameter b is

(A) b^2 if $b \geq \frac{2}{3}$

(B) $16 - 24b + b^2$ if $b \leq \frac{2}{3}$

(C) $4 - 12b + b^2$ if $0 \leq b \leq \frac{4}{3}$

(D) $16 - 24b + b^2$ if $b \geq \frac{2}{3}$

10. Let $f(x) = ax^2 + bx + c$, $a \neq 0$, a, b, c are integers. Let $f(1) = 0$, $50 < f(7) < 60$ and $70 < f(8) < 80$, then

(A) $f(10) = 135$

(B) $f(3) = 4$

(C) $f(5) = 20$

(D) $f(-2) = 21$

11. Let x satisfy the equation $\frac{\sqrt{a} + \sqrt{x-b}}{\sqrt{b} + \sqrt{x-a}} = \sqrt{\frac{a}{b}}$, $a, b > 0$, then

(A) $x \in [a, \infty)$ if $a = b$

(B) Number of values of x is exactly one if $a > b$

(C) Number of values of x is zero if zero if $a \neq b$

(D) Number of values of x is exactly one if $a < b$

12. Given that a, b, c are positive distinct real numbers such that quadratic expressions $ax^2 + bx + c$, $bx^2 + cx + a$ and $cx^2 + ax + b$ are always non-negative. Then the

expression $\frac{a^2 + b^2 + c^2}{ab + bc + ca}$ can never lie in

(A) $(-\infty, 2]$

(B) $(-\infty, 1]$

(C) $(2, 4)$

(D) $[4, \infty)$

13. Given that the equation $mx^2 - 2(m+2)x + m+5 = 0$, $m \in \mathbb{R}$ has no real root. Then the equation $(m-6)x^2 - 2(m+2)x + (m+5) = 0$ can have

(A) Two equal real roots

(B) Two distinct real roots

(C) No real roots

(D) Exactly one real root

14. The possible values of 'b', $b \in \mathbb{R}$ for which the equations $2017x^2 + bx + 7102 = 0$ and $7102x^2 + bx + 2017 = 0$ have a common root is/are

(A) -9119

(B) -10879

(C) 9119

(D) 10879

15. The equation $(x-a)(x-a-b) = 1$, where a, b are positive constants has

(A) One root less than a and the other is greater than a .

(B) One root less than $a+b$ and the other is greater than $a+b$.

(C) One root less than a and the other is greater than $a+b$.

(D) Roots lying between a and $a+b$.

16. Let p, q are integers and the two roots of the equation $x^2 - \left(\frac{p^2 + 11}{9}\right)x + \frac{15}{4}(p + q) + 16 = 0$ are p and q . Then
- (A) $p = 4$ (B) $p = 13$
(C) $q = 13$ (D) $q = 7$
17. Let $M = 3x^2 - 8xy + 9y^2 - 4x + 6y + 13$, where $x, y \in \mathbb{R}$, then
- (A) M can not be equal to zero (B) M must be negative
(C) $M > 2$ (D) M must be positive
18. The equation $8x^4 - 16x^3 + 16x^2 - 8x + a = 0$, $a \in \mathbb{R}$ has
- (A) Atleast two real roots $\forall a \in \mathbb{R}$.
(B) At least two imaginary roots $\forall a \in \mathbb{R}$.
(C) The sum of all non-real roots equal to 2, if $a > \frac{3}{2}$.
(D) The sum of all non-real roots equal to 1, if $a \leq \frac{3}{2}$.
19. If a, b, c are the roots of the cubic equation $x^3 - 3ax^2 + 3bx - c^3 = 0$, then which of the following may be possible?
- (A) $a = 3$ (B) $b = 6$
(C) $c = 0$ (D) $a = b = c$
20. Consider the equation $\frac{4x^2 + x + 4}{x^2 + 1} + \frac{x^2 + 1}{x^2 + x + 1} = \frac{31}{6}$, has
- (A) The number of distinct real roots equal to 4.
(B) The number of distinct real roots equal to 3.
(C) The sum of all distinct real roots of the equation is -2 .
(D) The sum of all distinct real roots of the equation is -1 .

21. Given that the equation $\frac{x}{x+1} + \frac{x+1}{x} = \frac{4x+a}{x(x+1)}$, $a \in \mathbb{R}$, has only one real root, then
- (A) Number of values of a is 3. (B) Number of values of a is 1.
- (C) Sum of all values of ' a ' is $\frac{1}{2}$. (D) Sum of all values of ' a ' is $\frac{13}{2}$.
22. Let $f(x) = x^3 - 3x + b$ and $g(x) = x^2 + bx - 3$, where b is a real number. If the equations $f(x) = 0$ and $g(x) = 0$ have a common root, then
- (A) Number of possible values of b is 3.
- (B) Number of possible value of b is 2.
- (C) Sum of all possible values of b is 0.
- (D) Sum of all possible values of b is 2.
23. Consider the equation $3x^7 - x^4 - 30x^5 + 10x^2 + 3x^3 - 1 = 0$ then
- (A) The minimum real root of equation is $(-\sqrt{3} - \sqrt{2})$.
- (B) The minimum real root of equation is $(-\sqrt{3} + \sqrt{2})$.
- (C) The maximum real root of equation is $(\sqrt{3} + \sqrt{2})$.
- (D) The number of positive roots of equation is 3.
24. The equation $x^4 - 9x^3 + 2(10 - a)x^2 + 9ax + a^2 = 0$ for x where ' a ' is real parameter has
- (A) 4 distinct real roots if $a > -4$, $a \neq 0$.
- (B) 2 distinct real roots if $-\frac{25}{4} < a < -4$.
- (C) No real roots if $a < -\frac{25}{4}$.
- (D) 3 distinct real roots for 2 values of a .

25. The number of real roots of the equation $\sqrt{|1-x|} = kx$, where k is a parameter is

- (A) 1, if $k > \frac{1}{2}$ (B) 2, if $k = \frac{1}{2}$
 (C) 3, if $0 < k < \frac{1}{2}$ (D) 1, if $k \leq 0$

26. The complete set of values of real parameter 'm' for which the equation $m \sin^2 x + (m-1)\sin x + (m-2) = 0$ has

- (A) No real roots is $(-\infty, 1) \cup \left(1 + \frac{2}{\sqrt{3}}, \infty\right)$
 (B) 5 roots in $[0, 2\pi]$ is $\{2\}$
 (C) 4 roots in $[0, 2\pi]$ is $(1, 2) \cup \left(2, \frac{\sqrt{3}+2}{\sqrt{3}}\right)$
 (D) 2 roots in $[0, 2\pi]$ is $\left\{-1, 1, \frac{\sqrt{3}+2}{\sqrt{3}}\right\}$

27. The set of values of real number 'a' for which the equation $a^3 + a^2|a+x| + |a^2x+1| = 1$ has not less than 4 different solutions which are integers can be

- (A) $\left(\frac{1}{2}, 1\right)$ (B) $(-3, -1)$
 (C) $(-\infty, -3]$ (D) $\left[-\frac{1}{\sqrt{3}}, \frac{1}{2}\right]$

28. The possible value(s) of 'p' for which the equations $ax^2 - px + ab = 0$ and $x^2 - ax - bx + ab = 0$ may have a common root, given that a, b are non zero real numbers is/are :

- (A) $b^2 + a$ (B) $a + ab$
 (C) $a^2 + b$ (D) $b + ab$

29. If $\sin 10^\circ$ is a root of the equation $4ax^3 - 3ax + b = 0$, where a, b are real parameters, $a \neq 0$, then the remaining two roots are

- (A) $\sin 130^\circ$ (B) $\sin 40^\circ$

(C) $\sin 250^\circ$

(D) $\sin 200^\circ$

30. Suppose a and b are two positive real numbers such that the roots of the cubic equation $x^3 - ax + b = 0$ are all real. If α is a root of this cubic with minimum absolute value, then

(A) $\alpha < \frac{b}{a}$

(B) $\alpha > \frac{b}{a}$

(C) $\alpha \leq \frac{3b}{2a}$

(D) $\alpha > \frac{3b}{2a}$

Quadratic Equations

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31. If the equations $ax^2 + bx + c = 0$ and $cx^2 + bx + a = 0$, $a, b, c \in \mathbb{R}$ and $ac \neq 0$ have a common non real root, then

(A) $a = -c$
(C) $|b| > 2|a|$

(B) $a = c$
(D) $|b| < 2|a|$

32. If the equation $ax^2 + bx + c = 0$, where $a, b, c \in \mathbb{R}$ and $a > 0$ has two real roots α and β such that $\alpha < -2$ and $\beta > 2$, then

(A) $c < 0$

(B) $a - |b| + c < 0$

(C) $4a + 2|b| + c < 0$

(D) $6a + |b| + c < 0$

33. Let the expression $\frac{(ax - b)(dx - c)}{(bx - a)(cx - d)}$ takes all real values when x is real, a, b, c, d are all distinct real parameters. Then which of the following is/are possible

(A) $a^2 > b^2$ and $c^2 < d^2$

(B) $a^2 > b^2$ and $c^2 > d^2$

(C) $a^2 < b^2$ and $c^2 > d^2$

(D) $a^2 < b^2$ and $c^2 < d^2$

34. The possible set of real values of ' a ' for which, for all x not exceeding unity in absolute value, the inequality $\frac{ax - a(1-a)}{a^2 - ax - 1} > 0$ is valid

(A) $(1, \infty)$

(B) $\left(-\infty, \frac{-1 + \sqrt{5}}{2}\right)$

(C) $\left(-\infty, \frac{-1 - \sqrt{5}}{2}\right)$

(D) $(2, \infty)$

35. Given that system of inequalities $x^2 + 4x + 3 \leq \alpha$ and $x^2 - 2x \leq 3 - 6\alpha$ have a unique solution, then
- (A) Sum of all possible values of α is equal to 0.
 (B) Number of all possible values of α is 2.
 (C) Sum of all possible values of α is equal to -1 .
 (D) Number of all possible values of α is 3.
36. If $ax^2 + bx + c = 0$, $a \neq 0$ be an equation with integral coefficients and $D > 0$ be its discriminant. Then the equation $b^2x^2 - Dx - 4ac = 0$ must have
- (A) Two integral roots
 (B) Two irrational roots
 (C) Two rational roots
 (D) At least one integral root

37. If all the roots of the equation $x^4 - 12x^3 + ax^2 + bx + 81 = 0$ where $a, b \in \mathbb{R}$ are positive, then
- (A) $b + 2a = 0$
 (B) $b + a = -54$
 (C) $a = 54$
 (D) $a = 27$
38. If each pair of equations $x^2 + ax + 2 = 0$, $x^2 + bx + 6 = 0$ and $x^2 + cx + 3 = 0$ has a common root, then $a + b + c$ can be equal to
- (A) -3
 (B) 3
 (C) -12
 (D) 12
39. Let p and q be real numbers such that the parabola $y = x^2 - 2px + q$ has no common point with the x -axis. Let there exist points A and B on the parabola such that AB is parallel to the x -axis and $\angle AOB = 90^\circ$ ('O' is origin), then possible values of q is/are
- (A) -1
 (B) $\frac{3}{17}$
 (C) $\frac{1}{4}$
 (D) $\frac{1}{2}$
40. Let the equation $x^n + px^2 + qx + r = 0$, where $n \geq 6$, $r \neq 0$ has roots $\alpha_1, \alpha_2, \alpha_3, \dots, \alpha_n$

and $S_k = \sum_{i=1}^n a_i^k$, where k is an natural number, then

(A) $S_n + pS_2 + qS_1 + nr = 0$

(B) Roots of the equation can not all be real

(C) $S_n = -nr$

(D) $S_n = r$

41. The three roots of equation $x^4 - px^3 + qx^2 - rx + s = 0$, where $p, q, r, s \in \mathbb{R}$ and $s < 0$, are $\tan A$, $\tan B$ and $\tan C$ where A, B, C are angles of a triangle. Then the fourth root of the equation can be equal to :

(A) $\frac{p + \sqrt{p^2 - 4s}}{2}$

(B) $\frac{p - \sqrt{p^2 - 4s}}{2}$

(C) $\frac{p + r}{1 - q + s}$

(D) $\frac{p - r}{1 - q + s}$

SECTION-3

COMPREHENSION BASED QUESTIONS

COMPREHENSION: (1 TO 3)

Given that $a > 0$, $|ax^2 + bx + c| \leq 1$, if $-1 \leq x \leq 1$, $a, b, c \in \mathbb{R}$ and $ax + b$ has its maximum value 2, when $-1 \leq x \leq 1$. Then :

1. $a =$

(A) 3

(B) 1

(C) 2

(D) 4

2. $b =$

(A) -1

(B) 2

(C) 1

(D) 0

3. $c =$

(A) -1

(B) 0

(C) 1

(D) 2

COMPREHENSION: (4 TO 6)

Consider a function $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = x^3 + ax^2 + bx + c$, where a, b, c are real such that $\max |f(x)| \leq 2$ for all $x \in [-2, 2]$.

4. $f(x)$ must be

(A) odd function

(B) even function

(C) neither odd nor even function

(D) nothing can be said

5. $b =$

(A) 1

(B) 0

(C) -2

(D) -3

6. $f(1)$ is equal to

(A) -2

(B) -1

(C) 1

(D) 2

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Advance Problems in Algebra for JEE

COMPREHENSION: (7 TO 8)

Let $f(x) = x^3 + ax^2 + bx + c$ and $g(x) = x^3 + bx^2 + cx + a$, where a, b, c are integers with $c \neq 0$. Let $f(1) = 0$ and the roots of $g(x)$ are squares of the roots of $f(x)$. Then

7. $a^3 + b^2 + c =$

(A) -1

(B) 1

(C) -3

(D) 3

8. $ab + bc + ca =$

(A) -1

(B) 3

(C) 1

(D) -3

COMPREHENSION: (9 TO 11)

Consider the equation $x^4 - (k - 1)x^2 + (2 - k) = 0$. The complete set of possible values of real k for which the equation has

9. Four distinct real roots is

(A) $(-\infty, 2)$

(B) $(2\sqrt{2} - 1, 2)$

(C) $(\sqrt{2} - 1, 2\sqrt{2} - 1)$

(D) $(2, \infty)$

10. 3 distinct real roots is

(A) $\{2\}$

(B) $\{\sqrt{2} - 1, 2\}$

(C) $\{\sqrt{5} - 1\}$

(D) $\{2\sqrt{2}, \sqrt{3} - \sqrt{2}\}$

11. 2 distinct real roots is

(A) $(0, 2)$

(B) $(-\infty, 2\sqrt{2} - 1)$

(C) $(2, \infty)$

(D) $\{2\sqrt{2} - 1\} \cup (2, \infty)$

COMPREHENSION: (12 TO 13)

Consider the inequality $9 - x^2 > |x - a|$, where a is a real number. Then the complete set of values of 'a' for which the given inequality has

12. Atleast one negative solution is

(A) $(-8, 8)$

(B) $(-9, 9)$

(C) $(-3, 8)$

(D) $\left(-\frac{37}{4}, 9\right)$

13. Only positive solutions is

(A) $\left(-9, \frac{41}{4}\right)$

(B) $\left(9, \frac{41}{4}\right)$

(C) $\left(9, \frac{37}{4}\right)$

(D) (8, 9)

COMPREHENSION: (14 TO 15)

Consider a function, $f(\theta) = \tan^2\theta + (a + 1)\tan\theta - (a - 3)$, $\theta \in \mathbb{R}$ where a is a real parameter. The complete set of values of 'a' for which

14. $f(\theta) > 0 \forall \theta \in \left(0, \frac{\pi}{2}\right)$ is

(A) $[-3, 3]$

(B) $(-3 - \sqrt{5}, 0)$

(C) $[3 + 2\sqrt{5}, \infty)$

(D) $(-3 - 2\sqrt{5}, 3]$

15. $f(\theta) < 0 \forall \theta \in \left(0, \frac{\pi}{2}\right)$ is

(A) $(0, \infty)$

(B) $(-\infty, -3)$

(C) $(3 + 2\sqrt{5}, \infty)$

(D) Null set

COMPREHENSION: (16 TO 18)

If rational number $\frac{p}{q}$, $q \neq 0$ and p, q are relatively prime integers is a root of polynomial function, then p is a divisor of constant term and q that of leading coefficient. Therefore the possible rational zeroes are formed by listing the factors of constant term over the factors of leading coefficient.

16. The sum of all possible rational x-intercepts of the curve $y = 6x^4 - 13x^3 - 35x^2 - x + 3$ is
- (A) $-\frac{1}{3}$ (B) -1 (C) $-\frac{11}{6}$ (D) $-\frac{13}{6}$
17. The number of possible rational roots of the equation $x^4 + (2p_1 + 1)x^3 + (2p_2 + 1)x^2 + (2p_3 + 1)x + (2p_4 + 1) = 0$, p_1, p_2, p_3, p_4 are integers is
- (A) 0 (B) 2
(C) 4 (D) nothing can be said
18. The number of possible integral roots of the equations $ax^2 + bx + c = 0$, where a, b, c are prime numbers greater than 2 is
- (A) 0 (B) 1
(C) 2 (D) nothing can be said

COMPREHENSION: (19 TO 21)

Consider the equation $x^4 + (\alpha - 1)x^3 + x^2 + (\alpha - 1)x + 1 = 0$, where ' α ' is a real parameter. Then

19. The given equation has 2 positive and 2 negative roots if
- (A) $\alpha \leq -\frac{1}{2}$ (B) $\alpha \geq \frac{5}{2}$ (C) $-\frac{1}{2} \leq \alpha \leq \frac{5}{2}$ (D) None of these
20. The given equation has 2 distinct negative roots if
- (A) $\alpha < -\frac{1}{2}$ (B) $\alpha > \frac{5}{2}$ (C) $-\frac{1}{2} < \alpha < \frac{5}{2}$ (D) $-\frac{5}{2} < \alpha < \frac{1}{2}$
21. The given equation has no real roots if
- (A) $\alpha < -\frac{1}{2}$ (B) $\alpha > \frac{5}{2}$ (C) $-\frac{1}{2} < \alpha < \frac{5}{2}$ (D) $-\frac{5}{2} < \alpha < \frac{1}{2}$

COMPREHENSION: (22 TO 23)

Given that the two quadratic equations $E_1 : x^2 - ax + b = 0$ and $E_2 : x^2 - px + q = 0$, where a, b, p, q are real parameters have a common root.

22. If the other roots of equations are reciprocal to each other, then $(q - b)^2$ is equal to

(A) $(p - a)^2$

(B) $b(p - a)^2$

(C) $q(p - a)^2$

(D) $bq(p - a)^2$

23. If the equation E_2 has equal roots, then $b + q$ is equal to

(A) ap

(B) $\frac{ap}{2}$

(C) $2ap$

(D) $4ap$

COMPREHENSION: (24 TO 26)

One root of the equation $x^4 - 5x^3 + ax^2 + bx + c = 0$ is $3 + \sqrt{\quad}$. If all the roots of the equation are real given that a, b, c are rational parameters, then

24. The greatest value of 'a' is equal to

(A) $\frac{7}{4}$

(B) $\frac{5}{4}$

(C) 2

(D) $\frac{9}{4}$

25. The least value of 'b' is equal to

(A) $\frac{13}{2}$

(B) 6

(C) $\frac{9}{2}$

(D) $\frac{11}{2}$

26. The greatest value of 'c' is equal to

(A) $\frac{3}{4}$

(B) $\frac{5}{4}$

(C) $\frac{7}{4}$

(D) $\frac{9}{4}$

COMPREHENSION (27 TO 29) :

Let x_1, x_2, x_3, x_4 be the roots of the equation $x^4 + ax^3 + bx^2 + cx + d = 0$. If $x_1 + x_2 = x_3 + x_4$ and $a, b, c, d \in \mathbb{R}$, then

27. If $a = 2$, then $b - c$ is equal to

- (A) 0 (B) 1 (C) 2 (D) 4

28. If $b < 0$, then how many different real values of 'a', we may have?

- (A) 0 (B) 1 (C) 2 (D) 3

29. If $b + c = 1$ and $a \neq -2$, then for real values of 'a', the complete set of real values of 'c' is

- (A) $(-\infty, 1]$ (B) $[1, \infty)$ (C) $\left(-\infty, \frac{1}{4}\right]$ (D) $\left(-\infty, \frac{5}{4}\right)$

COMPREHENSION (30 TO 31)

Let $f(x) = x^4 + ax^3 + bx^2 + ax + 1$ be a polynomial where a, b are real numbers, then

30. If $f(x) = 0$ has two different pairs of equal roots, then the least value of $a + b$ is

- (A) 0 (B) 1 (C) 2 (D) 4

31. If all roots of $f(x) = 0$ are imaginary and $b = -1$, then the complete set of values of 'a' is

- (A) $\left(-\frac{1}{2}, \frac{1}{2}\right)$ (B) $(-1, 1)$ (C) $\left(-\frac{5}{2}, \frac{5}{2}\right)$ (D) $(-3, 3)$

COMPREHENSION (32 TO 33) :

Let m be a real number, such that the roots x_1 and x_2 of the equation $x^2 + (m - 4)x + m^2 - 3m + 3 = 0$ are real numbers, then

32. The value of m for which $x_1^2 + x_2^2 = 6$ is equal to

(A) $-1 - \sqrt{5}$

(B) $\sqrt{5} + 1$

(C) $\sqrt{5} - 1$

(D) $1 - \sqrt{5}$

33. The maximum value of $\frac{mx_1^2}{1-x_1} + \frac{mx_2^2}{1-x_2} + 8$ is equal to

(A) $\frac{11}{3}$

(B) 1

(C) $\frac{49}{9}$

(D) $\frac{121}{9}$

COMPREHENSION (34 TO 35) :

Consider functions $f(x) = \frac{x^2 + 4x + 3}{x^2 + 7x + 14}$ and $g(x) = \frac{x^2 - 5x + 10}{x^2 + 5x + 20}$, for all real values of x , then

34. The greatest value of $f(x)$ is equal to

(A) 3

(B) 4

(C) 1

(D) 2

35. The greatest value of function $(g(x))^{f(x)}$ is equal to

(A) 9

(B) 4

(C) 8

(D) 16

SECTION-4

MATCH THE COLUMN

1.	Column-I		Column-II
(A)	If a, b, c are length of sides of a triangle, then the roots of the equation $a^2x^2 + (b^2 + a^2 - c^2)x + b^2 = 0$ are	(P)	of opposite signs
(B)	If a, b, c are unequal positive numbers and b is A.M. of a and c , then the roots of the equation $ax^2 + 2bx + c = 0$ are	(Q)	both positive
(C)	If $a \in \mathbb{R}$, then the roots of the equation $x^2 - (a + 1)x - a^2 - 4 = 0$ are	(R)	both negative
(D)	If a, b, c are unequal positive numbers and b is H.M. of a and c , then the roots of the equation $ax^2 + 2bx + c = 0$ are	(S)	real and distinct
		(T)	imaginary

2. Let α, β, γ be three numbers such that $\alpha + \beta + \gamma = 2$, $\alpha^2 + \beta^2 + \gamma^2 = 6$ and $\alpha^3 + \beta^3 + \gamma^3 = 11$, then

	Column-I		Column-II
(A)	$\alpha^4 + \beta^4 + \gamma^4$ is equal to	(P)	13
(B)	$\alpha^5 + \beta^5 + \gamma^5$ is equal to	(Q)	26
(C)	$(\alpha^2 - 4)(\beta^2 - 4)(\gamma^2 - 4)$ is equal to	(R)	57
(D)	$\alpha^6 + \beta^6 + \gamma^6$ is equal to	(S)	110

(D)	$\alpha + \beta + \gamma$ is equal to	(S)	119
		(T)	129

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3.

	Column-I		Column-II
(A)	The possible integral values of λ for which $(\lambda - 2)x^2 + 8x + (\lambda + 4) > 0 \forall x \in \mathbb{R}$ is/are	(P)	2
(B)	The equation $x^2 + 2(a^2 + 1)x + (a^2 - 14a + 48) = 0$ possesses roots of opposite signs, then the value of 'a' can be	(Q)	3
(C)	If the equation $ax^2 + 2bx + 4c = 16$ has no real roots and $a + c > b + 4$, then the integral value of 'c' can be equal to	(R)	4
(D)	the possible values of x satisfying the equation $ x^2 - x - 6 = x + 2$ is/are	(S)	5
		(T)	7

4.

	Column-I		Column-II
(A)	The possible value(s) of 'a' for which the largest value of $\sin^2 x - 2a \sin x + a + 3$ is 7 is/are	(P)	-3
(B)	The possible value(s) of 'a' for which the largest value of $x^4 - ax^2 + 2a - 1$ for $x \in [-1, 2]$ is 9 is/are	(Q)	1
(C)	The possible value(s) of 'a' for which the largest value of $x^4 - ax^2 + 2a - 1$ for $x \in [-1, 2]$ is 9 is/are	(R)	2
(D)	The possible value(s) of 'a' for which the largest value of $x^4 - ax^2 + 2a - 1$ for $x \in [-1, 2]$ is 9 is/are	(S)	3
		(T)	4

(C)	The possible value(s) of 'a' for which the equation $\tan^4 x - 3\tan^2 x + (a - 1) = 0$ has 4 roots in $(0, \pi)$ is/are	(R)	3
(D)	The possible value(s) of 'a' for which the smallest value of $x^4 - ax^2 + 2a - 1$ for $x \in [-1, 2]$ is -7 is/are	(S)	5
		(T)	11

SECTION-5

SUBJECTIVE TYPE QUESTIONS

- The number of real values of x satisfying the equation $\sqrt{1 - \sqrt{x^4 - x^2}} = x - 1$ is
- Let ABCD be a rectangle and let E and F be points on CD and BC respectively such that $\text{area}(\triangle ADE) = 16$, $\text{area}(\triangle CEF) = 9$ and $\text{area}(\triangle ABF) = 25$. Then the area $(\triangle AEF)$ is k , find $\frac{k}{5}$.
- The number of integral values of 'a' such that the equation $x^3 - 3x + a = 0$ has three integer roots is
- Let the equation $x^2 - (2a + b)x + (2a^2 + b^2 - b + \frac{1}{2}) = 0$ where $a, b \in \mathbb{R}$ has two real roots. Then the value of $6a + 2b$ is equal to
- Given that the quadratic equation $ax^2 + bx + c = 0$ has no real roots, but Mr. A got two roots 2 and 4 since he wrote down a wrong value of 'a'. Mr. B also got two roots -1 and 4 because he wrote the sign of a term wrongly. Then the value of $2b + 3c$ is equal to

6. Given that m is a real number not less than -1 , such that the equation $x^2 + 2(m - 2)x + m^2 - 3m + 3 = 0$ has two distinct real roots x_1 and x_2 . Find the maximum value of $\frac{1}{2} \left(\frac{mx_1^2}{1 - x_1} + \frac{mx_2^2}{1 - x_2} \right)$.
7. Given that the quadratic equation $x^2 - px + q = 0$, where $p, q \in \mathbb{R}$ has two real roots α and β . If the equation having roots α^3, β^3 is also $x^2 - px + q = 0$, then find the number of possible pairs (p, q) .

8. Let α, β are the roots of the equation $x^2 + x - 3 = 0$. Then the value of $\alpha^3 - 4\beta^2 + 19$ is equal to
9. If a, b are two real numbers satisfying the relations $19a^2 + 99a + 1 = 0$ and $b^2 + 99b + 19 = 0$ and $ab \neq 1$. Then the value of $\left| \frac{ab + 4a + 1}{b} \right|$ is equal to
10. Given that a, b, c are the lengths of three sides of $\triangle ABC$, $a > b > c$, $2b = a + c$ and b is a positive integer. If $a^2 + b^2 + c^2 = 84$, find the value of b .
11. The number of ordered pairs (p, q) , $p, q \in \mathbb{N}$ such that the equation $x^2 - pqx + p + q = 0$ has two integer roots.
12. Let p be an integer such that both roots of the equation $5x^2 - 5px + (66p - 1) = 0$ are positive integers. Then the value of $\left[\frac{p}{10} \right]$ is equal to ($[.]$ denotes greatest integer function)

13. Given that the integers a, b satisfy the equation

$$\left(\frac{\frac{1}{a}}{\frac{1}{a} - \frac{1}{b}} - \frac{\frac{1}{b}}{\frac{1}{a} + \frac{1}{b}} \right) \left(\frac{1}{a} - \frac{1}{b} \right) \times \frac{1}{\left(\frac{1}{a^2} + \frac{1}{b^2} \right)} = \frac{2}{3}. \text{ Then the value of } a + b \text{ is equal to}$$

14. Find the number of integral values of parameter 'a' so that the inequality $\frac{1}{8}(2a - a^2) \leq x^2 - 3x + 2 \leq 3 - a^2$ holds for any real x in the interval $[0, 2]$.

15. Given that a, b are integers and the two real roots α, β of the equation $3x^2 + 3(a + b)x + 4ab = 0$ satisfy the relation $\alpha(\alpha + 1) + \beta(\beta + 1) = (\alpha + 1)(\beta + 1)$. Then the number of ordered pairs (a, b) is equal to

16. Let a, b, c are real numbers and satisfy $a = 8 - b$ and $c^2 = ab - 16$, then $\frac{a}{b}$ is equal to

17. The number of triplets (a, b, c) , where a, b, c are rational numbers such that the equation $x^3 + ax^2 + bx + c = 0$ has roots a, b, c .

18. If α is a real root of equation $x^5 - x^3 + x - 2 = 0$, then $[\alpha^6]$ is equal to ($[.]$ denotes greatest integer function)

19. The number of ordered pairs of integers (x, y) satisfying $x + y = x^2 - xy + y^2$ is

20. How many integer pairs (x, y) satisfy $x^2 + 4y^2 - 2xy - 2x - 4y - 8 = 0$?

21. Given that α, β, γ are all real roots of the equation $x^3 - 2007x + 2002 = 0$, then the value of $\frac{\alpha-1}{\alpha+1} + \frac{\beta-1}{\beta+1} + \frac{\gamma-1}{\gamma+1}$ is equal to
22. The number of integral values of 'a' for which the equation $x^3(x+1) = (x+a)(2x+a)$ has four distinct real solutions is
23. Given that real numbers a, b satisfy $a^3 - 6a^2 + 15a - 21 = 0$ and $b^3 - 6b^2 + 15b - 7 = 0$ respectively, then find the value of $a + b$.
24. Natural numbers k, ℓ , p and q are such that if a and b are roots of $x^2 - kx + \ell = 0$ then $a + \frac{1}{b}$ and $b + \frac{1}{a}$ are the roots of $x^2 - px + q = 0$. What is the sum of all possible values of q?

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25. Given that $a + b + c = 3$, $a^2 + b^2 + c^2 = 5$ and $a^3 + b^3 + c^3 = 7$, then the value of $a^4 + b^4 + c^4$ is equal to
26. The equation $x^4 - (r+1)x^2 + r = 0$ has 4 distinct real solutions which form an arithmetic progression. The number of such real values of r is
27. The smallest value that $x^2 - 7x + 6$ takes in the set $\{x \in \mathbb{R} | x^2 - x - 30 \geq 0\}$ is
28. Given that $x^2 + y^2 = 8x + 6y + 11$, where x and y are integers. What is the smallest

possible value of $|4x-2y|$.

29. The number of different integer triplets (x, y, z) satisfying the equations $x + y^2 - z = 124$ and $x^2 + y - z = 100$ is
30. The polynomial equation $x^4 - 2x^2 + ax + b = 0$ has four distinct real roots. The maximum possible integral value of its root is
31. Find the least possible integral value of $\sqrt{8a + 8b + c}$ for which both the roots of equation $ax^2 - bx + c = 0$ are distinct and lie in interval $(0, 1)$, is, $(a, b, c \in \mathbb{N})$.
32. For a natural number b , let $N(b)$ denotes the number of natural numbers a for which the equation $x^2 + ax + b = 0$ has integer roots. Let x is the sum of digits of the smallest value of b for which $N(b) = 20$. Then $\frac{n+1}{2}$ is equal to

33. Let α, β be the roots of equation $x^2 - 2px + p^2 - 2p - 1 = 0$. If p_1, p_2 be two distinct real values of 'p' for which $\frac{1}{2} \left(\frac{(\alpha - \beta)^2 - 2}{(\alpha + \beta)^2 + 2} \right)$ is an integer. Find $\frac{1}{10} \left(\frac{1}{p_1^2} + \frac{1}{p_2^2} \right)$.
34. The number of real number pairs (x, y) which will satisfy the equation $x^2 - xy + y^2 = 4(x + y - 4)$ is

35. The equation $x^4 - 8x^3 + 24x^2 - 32x - 14 = 0$ has two real roots x_1, x_2 and two imaginary roots x_3, x_4 , then the value of $x_1x_2 + x_3x_4$ is equal to
36. If α and β , $\alpha \neq \beta$ are the roots of the equation $x^2 - p(x + 1) - c = 0$, then find the value of $\frac{\alpha^2 + 2\alpha + 1}{\alpha^2 + 2\alpha + c} + \frac{\beta^2 + 2\beta + 1}{\beta^2 + 2\beta + c}$.
37. The least integral value of 'a' for which the inequality $\frac{x^2 + a^2}{a(4 + x)} \geq 1$, holds true for all real values of x in the interval $(-1, 1)$ is
38. The number of integral values of 'a' for which the inequality $\log_x(x^2 - 4x + a) > 0$ holds for all real values of x in the interval $(0, 1)$.
39. The smallest positive value of 'a' for which every solution of inequality $\log_{\frac{1}{2}} x^2 \geq \log_{\frac{1}{2}} (x + 2)$ is a solution of inequality $49x^2 - 4a^4 \leq 0$ is
40. Let S be the sum of all distinct real values of 'm' for which the equation $mx^3 - 9x^2 + 12x - 5 = 0$ has two equal real roots. Then $[S] = ([.]$ denotes greatest integer function).

41. The number of integral values of 'a' for which any real value of x that satisfies the inequality $ax^2 + (1 - a^2)x - a > 0$ does not exceed two in absolute value.
42. Let $P(x)$ be a polynomial with real coefficients and leading coefficient unity satisfy the identity $(x - 8)P(2x) = 8(x - 1)P(x)$. Then $[\sqrt{P(10)}]$ is equal to ($[.]$ denotes

greatest integer function).

43. Let $P(x) = 0$ be a fifth degree polynomial equation with integer coefficients that has a integral root ' α '. If $P(2) = 13$ and $P(10) = 5$ then $\left[\frac{\alpha}{2}\right]$ is equal to ([.] denotes greatest integer function).
44. The number of integral values of ' a ' for which the equation $\cos 2x + a \sin x = 2a - 7$ possesses solution is
45. The number of ordered pairs (x, y) , $x, y \in \mathbb{R}$ satisfying the equation $4^{|x^2-8x+12|-\log_4 7} = 7^{2y-1}$ and the inequality $|y-3| - 3|y| - 2(y+1)^2 \geq 1$ is
46. The number of integral values of n for which the equation $nx^2 + (n+1)x + (n+2) = 0$ has rational roots only is equal to
47. If p_1, p_2 are the roots of the quadratic equation $ax^2 + bx + c = 0$ and q_1, q_2 are the roots of the quadratic equation $cx^2 + bx + a = 0$ such that p_1, q_1, p_2, q_2 is an A.P. of distinct terms, then $a + c$ is equal to ($a, b, c \in \mathbb{R}$).
48. The number of integral values of ' a ' for which the equation $(x^2 + x + 2)^2 - (a-3)(x^2 + x + 2)(x^2 + x + 1) + (a-4)(x^2 + x + 1)^2 = 0$ has atleast one real root is

49. The smallest integral value of $|a|$ for which the range of function $f(x) = \frac{x^2 + a}{x + 1}$ is $\mathbb{R}$ is

50. The number of distinct values of 'c' for which the equation $x^4 - (c^2 - 7c + 11)x^2 + (18 - 21c + 8c^2 - c^3) = 0$ doesn't have 4 distinct roots is
51. For each real number m, the parabolas $y = (m^2 + 4)x^2 + (m - 2)^2x - 4m + 2$ passes through the same point (a, b), then $a^2 + b^2$ is equal to
52. Suppose 1, 2, 3 are the roots of equation $x^4 + ax^2 + bx = c$, then the value of c is equal to
53. Let a and b be real numbers such that $a \neq 0$. Then the maximum number of possible real roots of the equation $ax^4 + bx^3 + x^2 + x + 1 = 0$ is equal to
54. Let x, y, z are all real numbers such that $x + y + z = 0$ and $xy + yz + zx = -3$, then the value of $|x^3y + y^3z + z^3x|$ is equal to
55. Find the number of positive integers 'n' such that $3^{2n} + 3n^2 + 7$ is a perfect square.

SINGLE CHOICE QUESTIONS

1. D	2. A	3. B	4. C	5. D	6. B
7. D	8. D	9. A	10. B	11. C	12. B
13. D	14. A	15. C	16. C	17. B	18. B
19. D	20. D	21. C	22. B	23. C	24. C
25. A	26. C	27. D	28. B	29. D	30. B
31. B	32. A	33. C	34. B	35. A	36. D
37. C	38. B	39. A	40. B	41. C	

ONE OR MORE THAN ONE CORRECT QUESTIONS

1. B,D	2. A,C	3. A,B,C,D	4. A,B,D	5. B,C,D	6. A,B,C,D
7. B,C	8. C,D	9. A,B	10. A,B,C	11. A,B,D	12. B,D
13. B,D	14. A,C	15. A,B,C	16. B,D	17. A,C,D	18. B,C,D
19. A,B,C,D	20. B,C	21. A,D	22. A,C	23. A,C,D	24. A,B,D
25. A,B,C,D	26. A,B,C,D	27. C,D	28. B,C	29. A,C	30. B,C
31. B,D	32. A,B,C	33. B,D	34. C,D	35. B,C	36. C,D
37. A,B,C	38. C,D	39. B,C	40. A,B,C	41. A,B,D	

COMPREHENSION BASED QUESTIONS

1. C	2. D	3. A	4. A	5. D	6. A
7. A	8. A	9. B	10. A	11. D	12. D
13. C	14. D	15. D	16. C	17. A	18. A
19. D	20. B	21. C	22. D	23. B	24. B
25. D	26. C	27. B	28. B	29. C	30. B
31. A	32. C	33. D	34. D	35. A	

MATCH THE COLUMN

1. $A - T; B - R, S; C - P, S; D - T$
2. $A - Q; B - R; C - P; D - T$
3. $A - S, T; B - T; C - S, T; D - P, R$
4. $A - P, Q; B - R, S; C - R; D - P, T$

SUBJECTIVE TYPE QUESTIONS

- | | | | | | |
|-------|-------|-------|--------|-------|-------|
| 1. 1 | 2. 6 | 3. 2 | 4. 5 | 5. 6 | 6. 5 |
| 7. 6 | 8. 0 | 9. 5 | 10. 5 | 11. 5 | 12. 7 |
| 13. 3 | 14. 1 | 15. 4 | 16. 1 | 17. 3 | 18. 3 |
| 19. 6 | 20. 6 | 21. 2 | 22. 0 | 23. 4 | 24. 4 |
| 25. 9 | 26. 2 | 27. 0 | 28. 2 | 29. 8 | 30. 1 |
| 31. 9 | 32. 8 | 33. 2 | 34. 1 | 35. 8 | 36. 1 |
| 37. 5 | 38. 0 | 39. 3 | 40. 4 | 41. 2 | 42. 9 |
| 43. 7 | 44. 5 | 45. 2 | 46. 3 | 47. 0 | 48. 1 |
| 49. 2 | 50. 4 | 51. 5 | 52. 36 | 53. 2 | 54. 9 |
| 55. 1 | | | | | |

Previous Year Questions

SECTION-1

1. If $\alpha \neq \beta$ but $\alpha^2 = 5\alpha - 3$ and $\beta^2 = 5\beta - 3$ then the equation having α/β and β/α as its roots is **[AIEEE 2002]**
(A) $3x^2 - 19x + 3 = 0$ (B) $3x^2 + 19x - 3 = 0$
(C) $3x^2 - 19x - 3 = 0$ (D) $x^2 - 5x + 3 = 0$
2. Difference between the corresponding roots of $x^2 + ax + b = 0$ and $x^2 + bx + a = 0$ is same and $a \neq b$, then **[AIEEE 2002]**
(A) $a + b + 4 = 0$ (B) $a + b - 4 = 0$
(C) $a - b - 4 = 0$ (D) $a - b + 4 = 0$
3. If p and q are the roots of the equation $x^2 + px + q = 0$, then **[AIEEE 2002]**
(A) $p = 1, q = -2$ (B) $p = 0, q = 1$
(C) $p = -2, q = 0$ (D) $p = -2, q = 1$
4. If a, b, c are distinct +ve real numbers and $a^2 + b^2 + c^2 = 1$ then $ab + bc + ca$ is **[AIEEE 2002]**
(A) less than 1 (B) equal to 1
(C) greater than 1 (D) any real no.

5. The value of 'a' for which one root of the quadratic equation $(a^2 - 5a + 3)x^2 + (3a - 1)x + 2 = 0$ is twice as large as the other is

[AIEEE 2003]

(A) $-\frac{1}{3}$

(B) $\frac{2}{3}$

(C) $-\frac{2}{3}$

(D) $\frac{1}{3}$

6. The real positive number x when added to its inverse gives the minimum value of the sum at x equal to

[AIEEE 2003]

(A) $-\frac{2}{1}$

(B) $\frac{2}{-1}$

(C) 1

(D) -1

7. If $(1 - p)$ is a root of quadratic equation $x^2 + px + (1 - p) = 0$ then its root are

[AIEEE 2004]

(A) $-1, 2$

(B) $-1, 1$

(C) $0, -1$

(D) $0, 1$

8. If one root of the equation $x^2 + px + 12 = 0$ is 4, while the equation $x^2 + px + q = 0$ has equal roots, then the value of 'q' is

[AIEEE 2004]

(A) 4

(B) 12

(C) 3

(D) $\frac{49}{4}$

9. In a triangle PQR, $\angle R = \frac{\pi}{2}$. If $\tan\left(\frac{P}{2}\right)$ and $\tan\left(\frac{Q}{2}\right)$ are the roots of $ax^2 + bx + c = 0$, $a \neq 0$ then

[AIEEE 2005]

(A) $a = b + c$

(B) $c = a + b$

(C) $b = c$

(D) $b = a + c$

10. If both the roots of the quadratic equation $x^2 - 2kx + k^2 + k - 5 = 0$ are less than 5

10. If both the roots of the quadratic equation $x^2 - 2kx + k + k - 5 = 0$ are less than 5, then k lies in the interval [AIEEE 2005]

- (A) $(5, 6]$ (B) $(6, \infty)$
(C) $(-\infty, 4)$ (D) $[4, 5]$

11. If the roots of the quadratic equation $x^2 + px + q = 0$ are $\tan 30^\circ$ and $\tan 15^\circ$, respectively, then the value of $2 + q - p$ is [AIEEE 2006]

- (A) 2 (B) 3
(C) 0 (D) 1

12. All the values of m for which both roots of the equation $x^2 - 2mx + m^2 - 1 = 0$ are greater than -2 but less than 4 , lie in the interval [AIEEE 2006]

- (A) $-2 < m < 0$ (B) $m > 3$
(C) $-1 < m < 3$ (D) $1 < m < 4$

13. If x is real, the maximum value of $\frac{3x^2 + 9x + 17}{3x^2 + 9x + 7}$ is [AIEEE 2006]

- (A) $\frac{1}{4}$ (B) 41
(C) 1 (D) $\frac{17}{7}$

14. If the difference between the roots of the equation $x^2 + ax + 1 = 0$ is less than $\sqrt{5}$, then the set of possible values of a is [AIEEE 2007]

- (A) $(3, \infty)$ (B) $(-\infty, -3)$
(C) $(-3, 3)$ (D) $(-3, \infty)$

15. The quadratic equations $x^2 - 6x + a = 0$ and $x^2 - cx + 6 = 0$ have one root in common. The other roots of the first and second equations are integers in the ratio $4 : 3$. Then the common root is [AIEEE 2009]

(A) 1

(B) 4

(C) 3

(D) 2

16. If the roots of the equation $bx^2 + cx + a = 0$ be imaginary, then for all real values of x , the expression $3b^2x^2 + 6bcx + 2c^2$ is [AIEEE 2009]

(A) less than $4ab$

(B) greater than $-4ab$

(C) less than $-4ab$

(D) greater than $4ab$

17. The equation $e^{\sin x} - e^{-\sin x} - 4 = 0$ has [AIEEE 2012]

(A) infinite number of real roots

(B) no real roots

(C) exactly one real root

(D) exactly four real roots

18. If the equations $x^2 + 2x + 3 = 0$ and $ax^2 + bx + c = 0$, $a, b, c \in \mathbb{R}$, have a common root, then $a : b : c$ is : [IIT Mains 2013]

(A) $3 : 1 : 2$

(B) $1 : 2 : 3$

(C) $3 : 2 : 1$

(D) $1 : 3 : 2$

19. The real number k for which the equation, $2x^3 + 3x + k = 0$ has two distinct real roots in $[0, 1]$ [IIT Mains 2013]

(A) does not exist

(B) lies between 1 and 2

(C) lies between 2 and 3

(D) lies between -1 and 0

20. Let α and β be the roots of equation $x^2 - 6x - 2 = 0$. If $a_n = \alpha^n - \beta^n$, for $n \geq 1$, then the value of $\frac{a_{10} - 2a_8}{2a_9}$ is equal to : [IIT Mains 2015]

(A) -3

(B) 6

(C) -6

(D) 3

21. The sum of all real values of x satisfying the equation

$$(x^2 - 5x + 5)^{x^2 + 4x - 60} = 1 \text{ is :}$$

[IIT Mains 2016]

(A) 3

(B) -4

(C) 6

(D) 5

22. If, for a positive integer, n the quadratic equation,

$$x(x+1) + (x+1)(x+2) + \dots + (x+n-1)(x+n) = 10n$$

has two consecutive integral solutions, then n is equal to :

[IIT Mains 2017]

(A) 10

(B) 11

(C) 12

(D) 9

23. If $\alpha, \beta \in \mathbb{C}$ are the distinct roots, of the equation $x^2 - x + 1 = 0$, then $\alpha^{101} + \beta^{107}$ is equal to :

[IIT Mains 2018]

(A) 2

(B) -1

(C) 0

(D) 1

SECTION-2

1. Find the values of α & β , $0 < \alpha, \beta < \pi/2$, satisfying the following equation,

$$\cos \alpha \cos \beta \cos (\alpha + \beta) = -1/8.$$

[REE '99, 6]

2. If the roots of the equation $x^2 - 2ax + a^2 + a - 3 = 0$ are real & less than 3 then

[JEE '99, 2 + 2]

(A) $a < 2$

(B) $2 \leq a \leq 3$

(C) $3 < a \leq 4$

(D) $a > 4$

3. If α, β are the roots of the equation, $(x-a)(x-b) + c = 0$, find the roots of the equation, $(x-\alpha)(x-\beta) = c$.

[REE 2000 (Mains), 3]

4.(a) For the equation, $3x^2 + px + 3 = 0$, $p > 0$ if one of the roots is square of the other, then p is equal to:

(A) $1/3$

(B) 1

(C) 3

(D) $2/3$

(b) If α & β ($\alpha < \beta$), are the roots of the equation, $x^2 + bx + c = 0$, where $c < 0 < b$, then

(A) $0 < \alpha < \beta$

(B) $\alpha < 0 < \beta < |\alpha|$

(C) $\alpha < \beta < 0$

(D) $\alpha < 0 < |\alpha| < \beta$

(c) If $b > a$, then the equation, $(x - a)(x - b) - 1 = 0$, has :

[JEE 2000 Screening, 1 + 1 + 1 out of 35]

(A) both roots in $[a, b]$

(B) both roots in $(-\infty, a)$

(C) both roots in $[b, \infty)$

(D) one root in $(-\infty, a)$ & other in $(b, +\infty)$

(d) If α, β are the roots of $ax^2 + bx + c = 0$, ($a \neq 0$) and $\alpha + \delta, \beta + \delta$, are the roots of, $Ax^2 + Bx + C = 0$, ($A \neq 0$) for some constant δ , then prove that,

[JEE 2000, Mains, 4 out of 100]

$$\frac{b^2 - 4ac}{a^2} = \frac{B^2 - 4AC}{A^2}.$$

5. Let a, b, c be real numbers with $a \neq 0$ and let α, β be the roots of the equation $ax^2 + bx + c = 0$. Express the roots of $a^3x^2 + abcx + c^3 = 0$ in terms of α, β .

[JEE 2001, Mains, 5 out of 100]

6. The set of all real numbers x for which $x^2 - |x + 2| + x > 0$, is

[JEE 2002 (screening), 3]

(A) $(-\infty, -2) \cup (2, \infty)$

(B) $(-\infty, -\sqrt{2}) \cup (\sqrt{2}, \infty)$

(C) $(-\infty, -1) \cup (1, \infty)$

(D) $(\sqrt{2}, \infty)$

7. If $x^2 + (a - b)x + (1 - a - b) = 0$ where $a, b \in \mathbb{R}$ then find the values of 'a' for

which equation has unequal real roots for all values of 'b'.

[JEE 2003, Mains-4 out of 60]

8. (a) If one root of the equation $x^2 + px + q = 0$ is the square of the other, then

(A) $p^3 + q^2 - q(3p + 1) = 0$
 (C) $p^3 + q^2 + q(3p - 1) = 0$

(B) $p^3 + q^2 + q(1 + 3p) = 0$
 (D) $p^3 + q^2 + q(1 - 3p) = 0$

(b) If $x^2 + 2ax + 10 - 3a > 0$ for all $x \in \mathbb{R}$, then

[JEE 2004 (Screening)]

(A) $-5 < a < 2$

(B) $a < -5$

(C) $a > 5$

(D) $2 < a < 5$

9. Find the range of values of t for which $2 \sin t = \frac{1 - 2x + 5x^2}{3x^2 - 2x - 1}$, $t \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$.

[JEE 2005(Mains), 2]

10. (a) Let a, b, c be the sides of a triangle. No two of them are equal and $\lambda \in \mathbb{R}$. If the roots of the equation $x^2 + 2(a + b + c)x + 3\lambda(ab + bc + ca) = 0$ are real, then

[JEE 2006, 3]

(A) $\lambda < \frac{4}{3}$

(B) $\lambda > \frac{5}{3}$

(C) $\lambda \in \left(\frac{1}{3}, \frac{5}{3}\right)$

(D) $\lambda \in \left(\frac{4}{3}, \frac{5}{3}\right)$

(b) If roots of the equation $x^2 - 10cx - 11d = 0$ are a, b and those of $x^2 - 10ax - 11b = 0$ are c, d , then find the value of $a + b + c + d$.

(a, b, c and d are distinct numbers)

[JEE 2006, 6]

11. (a) Let α, β be the roots of the equation $x^2 - px + r = 0$ and $\alpha/2, 2\beta$ be the roots of the equation $x^2 - qx + r = 0$. Then the value of 'r' is

(A) $\frac{2}{9}(p-q)(2q-p)$

(B) $\frac{2}{9}(q-p)(2p-q)$

(C) $\frac{2}{9}(q-2p)(2q-p)$

(D) $\frac{2}{9}(2p-q)(2q-p)$

Match the column :

(b) Let $f(x) = \frac{x^2 - 6x + 5}{x^2 - 5x + 6}$

Match the expressions / statements in **Column I** with expressions / statements in **Column II**.

	Column-I		Column-II
(A)	If $-1 < x < 1$, then $f(x)$ satisfies	(P)	$0 < f(x) < 1$
(B)	If $1 < x < 2$, the $f(x)$ satisfies	(Q)	$f(x) < 0$
(C)	If $3 < x < 5$, then $f(x)$ satisfies	(R)	$f(x) > 0$
(D)	If $x > 5$, then $f(x)$ satisfies	(S)	$f(x) < 1$

[JEE2007,3+6]

12. The smallest value of k , for which both the roots of the equation

$x^2 - 8kx + 16(k^2 - k + 1) = 0$ are real, distinct and have values at least 4, is

[JEE 2009]

13. Let p and q be real numbers such that $p \neq 0$, $p^3 \neq q$ and $p^3 \neq -$ If α and β are nonzero

complex numbers satisfying $\alpha + \beta = -p$ and $\alpha^3 + \beta^3 = q$, then a quadratic equation

having $\frac{\alpha}{\beta}$ and $\frac{\beta}{\alpha}$ as its roots is

[JEE 2010]

- (A) $(p^3 + q)x^2 - (p^3 + 2q)x + (p^3 + q) = 0$
- (B) $(p^3 + q)x^2 - (p^3 - 2q)x + (p^3 + q) = 0$
- (C) $(p^3 - q)x^2 - (5p^3 - 2q)x + (p^3 - q) = 0$
- (D) $(p^3 - q)x^2 - (5p^3 + 2q)x + (p^3 - q) = 0$

14.. A value of b for which the equations

$$x^2 + bx - 1 = 0$$

$$x^2 + x + b = 0,$$

have one root in common is

[JEE 2011]

- (A) $-\sqrt{2}$
- (B) $-i\sqrt{3}$
- (C) $i\sqrt{5}$
- (D) $\sqrt{2}$

15. Let S be the set of all non-zero real numbers α such that the quadratic equation $\alpha x^2 - x + \alpha = 0$ has two distinct real roots x_1 and x_2 satisfying the inequality

$|x_1 - x_2| < 1$. Which of the following intervals is(are) a subset(s) of S ?

[IIT Advance - 2015]

- (A) $\left(-\frac{1}{2}, -\frac{1}{\sqrt{5}}\right)$
- (B) $\left(-\frac{1}{\sqrt{5}}, 0\right)$
- (C) $\left(0, \frac{1}{\sqrt{5}}\right)$
- (D) $\left(\frac{1}{\sqrt{5}}, \frac{1}{2}\right)$

16. Let $-\frac{\pi}{6} < \theta < -\frac{\pi}{12}$. Suppose α_1 and β_1 are the roots of the equation $x^2 - 2x \sec\theta + 1$

$= 0$ and α_2 and β_2 are the roots of the equation $x^2 + 2x \tan \theta - 1 = 0$. If $\alpha_1 > \beta_1$ and $\alpha_2 > \beta_2$, then $\alpha_1 + \beta_2$ equals [IIT Advance 2016]

- (A) $2(\sec \theta - \tan \theta)$ (B) $2 \sec \theta$
(C) $-2 \tan \theta$ (D) 0

COMPREHENSION : (17 TO 18)

[IIT ADVANCE 2017]

Let p, q be integers and let α, β be the roots of the equation, $x^2 - x - 1 = 0$, where $\alpha \neq \beta$. For $n = 0, 1, 2, \dots$, let $a_n = p\alpha^n + q\beta^n$.

Fact : If a and b are rational numbers and $a + b\sqrt{5} = 0$, then $a = 0 = b$.

17. If $a_4 = 28$, then $p + 2q =$

- (A) 14 (B) 7
(C) 12 (D) 21

18. $a_{12} =$

- (A) $2a_{11} + a_{10}$ (B) $a_{11} - a_{10}$
(C) $a_{11} + a_{10}$ (D) $a_{11} + 2a_{10}$

19. Let a, b, c be three non-zero real numbers such that the equation $\sqrt{3} a \cos x + 2 b$

$\sin x = c$, $x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$, has two distinct real roots α and β with $\alpha + \beta = \frac{\pi}{3}$. Then,

the value of $\frac{b}{a}$ is _____.

[IIT Advance 2018]

Answer Key

SECTION-1

- | | | | | | |
|-------|-------|-------|-------|-------|-------|
| 1. A | 2. A | 3. A | 4. A | 5. B | 6. C |
| 7. C | 8. D | 9. B | 10. C | 11. B | 12. C |
| 13. B | 14. C | 15. D | 16. B | 17. B | 18. B |
| 19. A | 20. D | 21. A | 22. B | 23. D | |

SECTION-2

- | | | | |
|---|--|---------------------|------------------------|
| 1. $\alpha = \beta = \pi/3,$ | 2. A | 3. (a, b) | 4. (a) C, (b) B, (c) D |
| 5. $\gamma = \alpha^2\beta$ and $\delta = \alpha\beta^2$ or $\gamma = \alpha\beta^2$ and $\delta = \alpha^2\beta$ | 6. B | 7. $a > 1$ | |
| 8. (a) D; (b) A | 9. $\left[-\frac{\pi}{2}, -\frac{\pi}{10}\right] \cup \left[\frac{3\pi}{10}, \frac{\pi}{2}\right]$ | 10. (a) A, (b) 1210 | |
| 11. (a) D, (b) (A) P, R, S; (B) Q, S; (C) Q, S; (D) P, R, S | | | |
| 12. $k = 2$ | 13. B | 14. B | 15. A, D |
| | | 16. C | 17. C |
| 18. C | 19. 0.5 | | |



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CHAPTER

2

SECTION-I

SEQUENCE AND PROGRESSION

● SINGLE CHOICE QUESTIONS

1. The first and last two terms of an A.P. are a, b, c respectively in order. Then the sum of all the terms is :

$$(A) \frac{(2c - a - b)(a + c)}{2(c - b)}$$

$$(B) \frac{(2c - a - b)(a + c)}{(c - b)}$$

$$(C) \frac{(4c - 2a - 2b)(a + c)}{(c - b)}$$

$$(D) \frac{(2c + a + b)(c - a)}{(c - b)}$$

$$(A) \log_a b = \log_c b$$

$$(B) \log_b a = \log_c b$$

2. If $a^x = b^y = c^z$ and x, y, z are in G.P., where a, b, c, x, y, z are positive real numbers not equal to unity, then :

$$(C) \log_{10} b = \log_{10} a \log_{10} c$$

$$(D) \log_b a + \log_c b = 0$$

3. The sum of three numbers in H.P. is 11 and the sum of their reciprocals is 1, then their product is equal to

$$(A) 30$$

$$(B) 48$$

$$(C) 24$$

$$(D) 36$$

4. If $p^{\text{th}}, q^{\text{th}}$ and r^{th} terms of an H.P. be respectively x, y, z , then $(p - q)xy + (q - r)yz + (r - p)xz =$

$$(A) xyz + pqr$$

$$(B) pqr$$

(C) xyz

(D) 0

5. If A_1, A_2 be A.M.'s, G_1, G_2 be G.M.'s and H_1, H_2 be H.M.'s between two numbers, then

(A) $\frac{G_1 G_2}{H_1 H_2} = \frac{A_1 + A_2}{H_1 + H_2}$

(B) $\frac{G_1 G_2}{H_1 H_2} = \frac{H_1 + H_2}{A_1 + A_2}$

(C) $\frac{G_1 G_2}{A_1 A_2} = \frac{H_1 + H_2}{A_1 + A_2}$

(D) $\frac{G_1 G_2}{A_1 A_2} = \frac{A_1 + A_2}{H_1 + H_2}$

6. If a, b, c are in H.P., then $\left(\frac{1}{b} + \frac{1}{c} - \frac{1}{a}\right)\left(\frac{1}{c} + \frac{1}{a} - \frac{1}{b}\right)$ is equal to

(A) $\frac{1}{b^2} - \frac{1}{ab}$

(B) $\frac{3}{b^2} - \frac{2}{ab}$

(C) $\frac{4}{b^2} - \frac{2}{ab}$

(D) $\frac{3}{b^2} - \frac{1}{ab}$

7. If A, G, H be respectively the A.M., G.M. and H.M. between two positive numbers and if $xA = yG = zH$ where x, y, z are non-zero positive quantities, then x, y, z are in

(A) A.P.

(B) G.P.

(C) H.P.

(D) Nothing can be said

8. If the $(m+1)^{\text{th}}$, $(n+1)^{\text{th}}$ and $(r+1)^{\text{th}}$ terms of an A.P. are in G.P. and m, n, r are in H.P., then the ratio of the common difference to the first term in the A.P. is equal to

(A) $\frac{2}{n}$

(B) $\frac{1}{n}$

(C) $-\frac{1}{n}$

(D) $-\frac{2}{n}$

9. If ℓ, m, n are three numbers in G.P., then the ratio of first term to the common difference of an A.P. whose ℓ^{th} , m^{th} and n^{th} terms are in H.P. is equal to

(A) $\frac{m+1}{2}$

(B) m

(C) $m+1$

(D) $\frac{1}{m+1}$

10. If between any two quantities there be inserted two arithmetic means A_1, A_2 ; two geometric means G_1, G_2 ; and two harmonic means H_1, H_2 , then $G_1 G_2 : A_1 + A_2$ is equal to

(A) $H_1 H_2 : H_1 + H_2$

(B) $\frac{1}{H_1} + \frac{1}{H_2}$

(C) $H_1 + H_2$

(D) $H_1 H_2$

11. Let S_n be sum of the first 'n' terms of an A.P. $\{a_n\}$. If $S_6 = S_9$, then the ratio $a_3 : a_5$ is equal to

(A) $9 : 5$

(B) $5 : 9$

(C) $3 : 5$

(D) $5 : 3$

12. If in the arithmetic progression $\{a_n\}$, $a_1 > 0$, $3a_8 = 5a_{13}$, then among the following listed partial sums the largest one is

(A) S_{10}

(B) S_{11}

(C) S_{20}

(D) S_{21}

13. Let α, β be the roots of the equation $ax^2 + 2bx + c = 0$ and γ, δ be the roots of the equation $px^2 + 2qx + r = 0$. If $\alpha, \beta, \gamma, \delta$ are in G.P., then

(A) $\frac{ac}{b^2} = \frac{pr}{q^2}$

(B) $\frac{ab}{c^2} = \frac{pq}{r^2}$

(C) $\frac{ac}{q^2} = \frac{pr}{b^2}$

(D) $\frac{pq}{c^2} = \frac{ab}{r^2}$

14. $\sum_{r=1}^{99} r!(r^2 + r + 1)$ is equal to

(A) $102! - 100!$

(B) $100(100!) - 1$

(C) $99(100!) - 1$

(D) $100(99!) - 1$

15. If a, b, c form an A.P. with common difference $d (\neq 0)$ and x, y, z form a G.P. with common ratio $r (\neq 1)$, then the area of triangle with vertices $(a, x), (b, y)$ and (c, z) is independent of

(A) a

(B) d

(C) x

(D) r

16. A G.P. consist of an even number of terms. If the sum of all the terms is 5 times the sum of the terms occupying the odd places. Then the common ratio will be equal to

(A) 3

(B) 4

(C) 5

(D) 6

17. A convex n -sided polygon has a circumcircle and an inscribed circle, its area is B and the area of its circumcircle and inscribed circle are A and C respectively, then

(A) $B = \frac{A+C}{2}$

(B) $B > \frac{A+C}{2}$

(C) $B < \frac{A+C}{2}$

(D) Nothing can be said

18. The sum $\sum_{k=1}^n \frac{k^2 - \frac{1}{2}}{k^4 + \frac{1}{4}}$ is equal to

(A) $\frac{2n^2 - 2n + 1}{2n^2 + 2n + 1}$

(B) $\frac{2n^2 - n}{2n^2 + 2n + 1}$

(C) $\frac{n}{2n^2 + 2n + 1}$

(D) $\frac{2n}{2n^2 + 2n + 1}$

19. The value of $\frac{\left(1^4 + \frac{1}{4}\right)\left(3^4 + \frac{1}{4}\right) \dots \left((2n-1)^4 + \frac{1}{4}\right)}{\left(2^4 + \frac{1}{4}\right)\left(4^4 + \frac{1}{4}\right) \dots \left((2n)^4 + \frac{1}{4}\right)}$ is equal to

(A) $\frac{1}{4n^2 + 2n + 1}$

(B) $\frac{1}{8n^2 + 4n + 1}$

(C) $\frac{1}{4(2n^2 + n + 1)}$

(D) $\frac{n}{8n^2 - 4n + 1}$

20. The sum $\sum_{k=1}^n \frac{1}{(k+1)\sqrt{k} + k\sqrt{k+1}}$ is equal to

(A) $\frac{\sqrt{n+1} - 2}{\sqrt{n+1}}$

(B) $\frac{\sqrt{n+1} - 1}{\sqrt{n+1}}$

(C) $\frac{\sqrt{n+1} + 1}{\sqrt{n+1}}$

(D) $\frac{n+1}{n\sqrt{n+1}}$

21. The sum $\sum_{k=1}^n k!(k^2 + k + 1)$ is equal to

(A) $(n+1)(n+1)! - 1$

(B) $nn! - 1$

(C) $(n+1)! - 1$

(D) $n(n+1)! - 1$

22. Let $a_1, a_2, a_3, \dots, a_n$ be an arithmetic progression with common difference 'd'.

Then $\sum_{k=1}^{n-1} \frac{1}{a_k a_{k+1}}$ is equal to

(A) $\frac{n}{a_1(a_1 + nd)}$

(B) $\frac{(n+1)}{a_1(a_1 + nd)}$

(C) $\frac{(n-1)}{a_1(a_1 + (n-1)d)}$

(D) $\frac{n}{a_1(a_1 + (n-1)d)}$

23. Let S_n, S_{2n}, S_{3n} are respectively the sums of first $n, 2n, 3n$ terms of an arithmetic progression, then $S_{3n} =$
- (A) $2(S_{2n} - S_n)$ (B) $\frac{3}{2}(S_{2n} - S_n)$
 (C) $3(S_{2n} - S_n)$ (D) $6(S_{2n} - S_n)$
24. The sum $\frac{7}{2 \times 3} \left(\frac{1}{3}\right) + \frac{9}{3 \times 4} \left(\frac{1}{3}\right)^2 + \frac{11}{4 \times 5} \left(\frac{1}{3}\right)^3 + \frac{13}{5 \times 6} \left(\frac{1}{3}\right)^4 + \dots$ upto 10 terms is equal to
- (A) $1 - \frac{1}{12 \times 3^{10}}$ (B) $\frac{1}{2} - \frac{1}{12 \times 3^{10}}$
 (C) $\frac{1}{3} - \frac{1}{12 \times 3^{10}}$ (D) $\frac{1}{2} - \frac{1}{11 \times 3^{10}}$
25. $\sum_{0 \leq i < j \leq 10} (i + j)$ is equal to
- (A) 545 (B) 548
 (C) 550 (D) 552

26. Let there be a G.P. whose first term is 'a' and common ratio is 'r'. If A and H are the arithmetic mean and the harmonic mean respectively for the first 'n' terms of the G.P. Then AH is equal to
- (A) ar^{n-1} (B) a^2r^{n-1}
 (C) a^2r^n (D) $ar^{\frac{n-1}{2}}$
27. Let $0 < x < y < 2019$, then the number of ordered pair of integers (x, y) such that A.M. of x and y exceeds their G.M. by 2 is
- (A) 42 (B) 43

(C) 44

(D) 45

28. The sum $\frac{19}{1 \cdot 2 \cdot 3} \frac{1}{4} + \frac{28}{2 \cdot 3 \cdot 4} \frac{1}{8} + \frac{39}{3 \cdot 4 \cdot 5} \frac{1}{16} + \frac{52}{4 \cdot 5 \cdot 6} \frac{1}{32} + \dots +$ upto infinite terms is equal to

(A) $\frac{1}{2}$

(B) 1

(C) 2

(D) 4

29. The sum of infinite series

$1^2 - \frac{2^2}{5} + \frac{3^2}{5^2} - \frac{4^2}{5^3} + \frac{5^2}{5^4} - \frac{6^2}{5^5} + \dots \infty$ is equal to

(A) $\frac{5}{27}$

(B) $\frac{25}{27}$

(C) $\frac{25}{108}$

(D) $\frac{25}{54}$

30. Let x, y are real numbers such that $x, x + 2y, 2x + y$ form an A.P. while the numbers $(y + 1)^2, xy + 5, (x + 1)^2$ form a G.P., then $|x - y|$ is equal to

(A) 0

(B) 1

(C) 2

(D) 4

31. The sum of all values of θ in interval $[0, 4\pi]$ for which $\frac{\sin \theta}{6}, \cos \theta, \tan \theta$ taken in that order constitute a geometric progression is

(A) $\frac{23\pi}{3}$

(B) 24π

(C) 16π

(D) 8π

32. $\frac{n}{1 \cdot 2 \cdot 3} + \frac{n-1}{2 \cdot 3 \cdot 4} + \frac{n-2}{3 \cdot 4 \cdot 5} + \dots$ upto n terms is equal to

$$(A) \frac{1}{2(n+2)} + \frac{n+1}{4} - \frac{1}{2}$$

$$(B) \frac{1}{2(n+2)} + \frac{n+1}{4} + \frac{1}{2}$$

$$(C) \frac{1}{n+2} + \frac{n+1}{4} - \frac{1}{2}$$

$$(D) \frac{1}{2(n+2)} + \frac{n+1}{2} + \frac{1}{2}$$

33. $\frac{1}{1+x} + \frac{2}{1+x^2} + \frac{4}{1+x^4} + \frac{8}{1+x^8} + \dots$ upto 'n' terms is equal to

$$(A) \frac{1}{1-x} - \frac{2^{n-1}}{1-x^{2^{n-1}}}$$

$$(B) \frac{1}{1-x} + \frac{2^n}{1-x^{2^n}}$$

$$(C) \frac{1}{x+1} - \frac{2^n}{x^{2^n}-1}$$

$$(D) \frac{1}{x-1} - \frac{2^n}{x^{2^n}-1}$$

34. If x, y, z are positive and $x + y + z = 1$, then the minimum value of

$$\left(\frac{1}{x}-1\right)\left(\frac{1}{y}-1\right)\left(\frac{1}{z}-1\right) \text{ is}$$

$$(A) 12$$

$$(B) 4$$

$$(C) 8$$

$$(D) 9$$

35. $\frac{a_1}{1+a_1} + \frac{a_2}{(1+a_1)(1+a_2)} + \frac{a_3}{(1+a_1)(1+a_2)(1+a_3)} + \dots + \frac{a_n}{(1+a_1)(1+a_2)\dots(1+a_n)}$ is equal to

$$(A) \frac{1}{1+a_1} - \frac{1}{(1+a_1)(1+a_2)\dots(1+a_n)}$$

$$(B) 1 - \frac{1}{(1+a_1)(1+a_2)\dots(1+a_n)}$$

$$(C) 1 - \frac{a_n}{(1+a_1)(1+a_2)\dots(1+a_{n-1})}$$

$$(D) \frac{1}{2} - \frac{1}{2(1+a_1)(1+a_2)\dots(1+a_n)}$$

36. $\frac{3}{1^2 2^2} + \frac{5}{2^2 3^2} + \frac{7}{3^2 4^2} + \dots$ upto n terms is equal to

$$(A) 1 - \frac{1}{(n+1)^2}$$

$$(B) 1 - \frac{1}{n^2}$$

(C) $1 - \frac{1}{n(n+1)}$

(D) $\frac{n}{(n+1)^2}$

37. The sum $1 + (1+2)x + (1+2+3)x^2 + (1+2+3+4)x^3 + \dots + \dots + (1+2+3+4 + \dots + n)x^{n-1}$ is equal to

(A) $\frac{1-x^n}{(1-x)^3} - \frac{nx^n}{(1-x)^2} - \frac{n(n+1)x^n}{2(1-x)}$

(B) $\frac{(1-x^n)}{(1-x)^3} - \frac{nx^n}{(1-x)^2} + \frac{n(n+1)x^n}{2(1-x)}$

(C) $\frac{1-x^n}{(1-x)^3} + \frac{nx^n}{(1-x)^2} + \frac{n(n+1)x^n}{2(1-x)}$

(D) $\frac{1-x^n}{(1-x)^3} + \frac{nx^n}{(1-x)^2} - \frac{n(n+1)x^n}{2(1-x)}$

38. The sum $2 \cdot 3 + 3 \cdot 6 + 4 \cdot 11 + 5 \cdot 18 + \dots +$ upto n terms is equal to

(A) $\frac{n}{12} (3n^3 + 7n^2 + 21n + 36)$

(B) $\frac{n}{12} (3n^3 - 10n^2 + 22n - 36)$

(C) $\frac{n}{12} (3n^3 + 10n^2 + 21n + 38)$

(D) $\frac{n}{12} (3n^3 + 8n^2 + 20n + 37)$

39. For $x > 1$, $\lim_{n \rightarrow \infty} \prod_{k=0}^n \left(1 + \frac{2}{x^{2^k} + x^{-2^k}} \right)$ is equal to

(A) $\frac{x^2 - 1}{x^2 + 1}$

(B) $\frac{x^2 + 1}{x^2 - 1}$

(C) $\frac{x - 1}{x + 1}$

(D) $\frac{x + 1}{x - 1}$

40. Let G_1, G_2 are two geometric means and A is the arithmetic mean between two positive numbers, then $\frac{G_1^2}{G_2} + \frac{G_2^2}{G_1} =$

(A) A

(B) $2A$

(C) $3A$

(D) $4A$

ONE OR MORE THAN ONE CORRECT QUESTIONS

- For all permissible values of x , consider $y = \frac{\sin 3x(\cos 6x + \cos 4x)}{\sin x(\cos 8x + \cos 2x)}$ and range of y is $(-\infty, a) \cup (b, \infty)$. If $2b$ is the first term of G.P. and 'a' is its common ratio, then (s_∞ denotes the sum of infinite terms of G.P.)
 - $b - a = \frac{10}{3}$
 - $3a + b = 4$
 - $s_\infty = 9$
 - $s_\infty = \frac{27}{10}(a + b)$
- It is given that the sequence $\{a_n\}$ satisfies $a_1 = 0$, $a_{n+1} = a_n + 1 + 2\sqrt{1 + a_n}$ for $n \in \mathbb{N}$. Then
 - $a_{100} = 9999$
 - $a_{2001} = 4004000$
 - $a_{2001} = 4002000$
 - $a_{19} = 360$
- Let $\{a_n\}$ consists of positive numbers and for any positive integer n , $\frac{a_n + 2}{2} = \sqrt{2s_n}$, where $s_n = \sum_{i=1}^n a_i$. Then
 - $a_{21} = 82$
 - $a_{12} = 48$
 - $a_{13} = 50$
 - $a_{14} = 54$
- If x, y, z are three distinct positive real numbers and are in H.P., then $\frac{3x + 2y}{2x - y} + \frac{3z + 2y}{2z - y}$ is greater than
 - 9
 - 10
 - 12
 - 15
- The sequence $\{a_n\}$, $n \in \mathbb{N}$ satisfies $a_1 = 1$ and $5^n = 1 + \frac{1}{n + \frac{2}{3}}$. Then
 - $[a_{501}] = 3$
 - $[a_{207}] = 3$
 - $[a_{223}] = 4$
 - $[a_{625}] = 4$

(where $[\cdot]$ denotes greatest integer function)

6. If a, b, c are three positive real numbers, then $\frac{a^2+1}{b+c} + \frac{b^2+1}{c+a} + \frac{c^2+1}{a+b}$ can never be equal to
- (A) 1 (B) 2
- (C) $\frac{8}{3}$ (D) 3
7. The sum of squares of 3 distinct real numbers which are in G.P. is s^2 . If their sum is αs , then α^2 lie in interval is
- (A) $\left(0, \frac{1}{3}\right)$ (B) $\left(\frac{1}{3}, 1\right)$
- (C) $(1, 3)$ (D) $(3, 9)$
8. Let 'p' be the first of 'n' arithmetic means between two positive numbers and 'q' be first of 'n' harmonic means between same two numbers. Then $\frac{q}{p}$ can lie in interval(s)
- (A) $(-\infty, 1]$ (B) $\left(1, \left(\frac{n+1}{n-1}\right)^2\right)$
- (C) $\left(\left(\frac{n-1}{n+1}\right)^2, \left(\frac{n+1}{n-1}\right)^2\right)$ (D) $\left[\left(\frac{n+1}{n-1}\right)^2, \infty\right)$
9. A sequence of real numbers $a_1, a_2, a_3, \dots, a_{n+1}$ is such that $a_1 = 0, |a_2| = |a_1 + 1|, |a_3| = |a_2 + 1|, \dots, |a_{n+1}| = |a_n + 1|$. Then $\frac{a_1 + a_2 + \dots + a_{n+1}}{n}$ can not be equal to
- (A) -2 (B) -1
- (C) $-\frac{3}{4}$ (D) $-\frac{4}{7}$
10. Let x, y, z are distinct positive integers and
- $$m = \left(\frac{x^2 + y^2 + z^2}{x + y + z}\right)^{(x+y+z)}, n = x^x y^y z^z, P = \left(\frac{x + y + z}{3}\right)^{(x+y+z)}, \text{ then}$$
- (A) $m > n$ (B) $n > p$

(C) $m < n$

(D) $n < p$

11. Let $f(x) = \lim_{n \rightarrow \infty} \left(\frac{1}{x+1} + \frac{2}{x^2+1} + \frac{4}{x^4+1} + \dots + \frac{2^n}{x^{2^n}+1} \right)$ for $x > 1$, then

(A) $\sum_{r=2}^6 \frac{1}{f(r)} = 20$

(B) $f(5) = \frac{1}{6}$

(C) $f(5) = \frac{1}{4}$

(D) $\sum_{r=2}^6 \frac{1}{f(r)} = 15$

12. For a positive integer 'n', let $S(n) = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \dots + \frac{1}{2^n - 1}$. Then

(A) $S(200) > 100$

(B) $S(200) < 100$

(C) $S(100) < 100$

(D) $S(100) > 100$

13. Let $a_1, a_2, a_3, \dots$ be a sequence of integers satisfying $a_n + a_{n-1} = 2n \forall n \geq 2, n \in \mathbb{N}$.
If $a_1 = 100$, then

(A) $a_7 = 106$

(B) $a_{11} = 110$

(C) $a_8 = -90$

(D) $a_{12} = -88$

14. Consider a sequence $\{a_n\}$ with $a_1 = 2$ and $a_n = \frac{a_{n-1}^2}{a_{n-2}} \forall n \geq 3, n \in \mathbb{N}$, terms of the sequence being distinct. Given that a_2 and a_5 are positive integers and $a_5 \leq 200$, then possible value(s) of a_5 can be

(A) 8

(B) 32

(C) 162

(D) 200

15. Let $a_1, a_2, a_3, \dots, a_n$ be first 'n' terms of a G.P. with first term 'a' and common ratio 'r'. Then

(A) $\frac{1}{a_1^2 - a_2^2} + \frac{1}{a_2^2 - a_3^2} + \dots + \frac{1}{a_{n-1}^2 - a_n^2} = \frac{(1-r^{2n})}{a^2 r^{2n-4} (1-r^2)^2}$

(B) $\frac{1}{a_1^2 - a_2^2} + \frac{1}{a_2^2 - a_3^2} + \dots + \frac{1}{a_{n-1}^2 - a_n^2} = \frac{(1-r^{2n-2})}{a^2 r^{2n-4} (1-r^2)^2}$

$$(C) \frac{1}{a_1^m + a_2^m} + \frac{1}{a_2^m + a_3^m} + \dots + \frac{1}{a_{n-1}^m + a_n^m} = \frac{(r^{mn-m} - 1)}{a^m(1+r^m)(r^{mn-m} - r^{mn-2m})}$$

$$(D) \frac{1}{a_1^m + a_2^m} + \frac{1}{a_2^m + a_3^m} + \dots + \frac{1}{a_{n-1}^m + a_n^m} = \frac{(r^{mn-m} - 1)}{a^m(1-r^m)(r^{mn-m} - r^{mn-2m})}$$

16. Let the three non-zero real numbers $x(y-z)$, $y(z-x)$, $z(y-x)$ form a geometric progression in that order, with common ratio 'r', then the possible value(s) of $[r]$ is equal to ($[.]$ denotes greatest integer function)

- (A) -2 (B) -1
(C) 0 (D) 1

17. The sequence $\{a_n\}$ and $\{b_n\}$ satisfy $a_k b_k = 1$, $k = 1, 2, 3, \dots$. If the sum of first 'n' terms of $\{a_n\}$ is equal to $\frac{n(n+1)(n+2)}{3}$. Then

- (A) $\sum_{r=1}^n b_r = \frac{n}{n+1}$ (B) $\lim_{n \rightarrow \infty} \sum_{r=1}^n b_r = 2$
(C) $\sum_{r=1}^n b_r = \frac{2n+1}{n+2}$ (D) $\lim_{n \rightarrow \infty} \sum_{r=1}^n b_r = 1$

18. Let the equation $x^3 + px^2 + qx - q = 0$, where $p, q \in \mathbb{R}$, $q \neq 0$ has 3 real roots α, β, γ in H.P., then

- (A) $\frac{1}{\alpha^2} + \frac{1}{\beta^2} + \frac{1}{\gamma^2} \geq \frac{1}{3}$ (B) $9p + 2q + 27 = 0$
(C) Maximum value of $\frac{1}{\alpha\beta} + \frac{1}{\beta\gamma} + \frac{1}{\gamma\alpha}$ is $\frac{1}{3}$ (D) $\frac{p}{q} \geq -\frac{1}{3}$

19. The sum $\sum_{n=1}^{\infty} \sum_{k=1}^{n-1} \frac{k}{2^{n+k}}$ is equal to

- (A) $\sum_{k=1}^{\infty} \frac{k}{2^k}$ (B) $\sum_{k=1}^{\infty} \frac{k}{4^k}$
(C) $\sum_{m=1}^{\infty} \left(\frac{m}{2} \sum_{n=1}^{\infty} \frac{1}{2^n} \right)$ (D) $\frac{4}{3}$

20. For any natural number $n > 1$, consider the sum $S =$

$$\frac{1}{n^2+1} + \frac{2}{n^2+2} + \frac{3}{n^2+3} + \dots + \frac{n}{n^2+n}, \text{ then}$$

(A) $S < \frac{1}{2} + \frac{1}{2n}$

(B) $S > \frac{1}{2} + \frac{1}{2n}$

(C) $S > \frac{1}{2}$

(D) $S < 1$

Sequence and Progression

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21. If $n \in \mathbb{N}$, $n > 5$ then which of the following holds true ?

(A) $n^n > 1 \cdot 3 \cdot 5 \dots (2n-1)$

(B) $2^n > 1 + n\sqrt{2^{n-1}}$

(C) $\frac{1}{n+1} + \frac{1}{n+2} + \frac{1}{n+3} + \dots + \frac{1}{2n} > \frac{1}{2}$

(D) $2^n < 1 + n\sqrt{2^{n-1}}$

22. Let $\{a_n\}$ be a sequence of real numbers such that $a_1 = 2$, $a_{n+1} = a_n^2 - a_n + 1$ for $n = 1, 2, 3, \dots$. Let $S = \frac{1}{a_1} + \frac{1}{a_2} + \dots + \frac{1}{a_{2018}}$, then

(A) $S < 1 - \frac{1}{(2018)^{2018}}$

(B) $S > 1 - \frac{1}{(2018)^{2018}}$

(C) $S < 1$

(D) $S > 1 - \frac{1}{(2017)^{2017}}$

23. Let $a_k = \frac{k}{(k-1)^{4/3} + k^{4/3} + (k+1)^{4/3}}$ and $S_n = \sum_{k=1}^n a_k$, then

(A) $S_{26} > \frac{17}{4}$

(B) $S_{26} < \frac{17}{4}$

(C) $S^{999} < 50$

(D) $S^{999} > 50$

24. Let $S = \frac{1}{\sqrt{1}+\sqrt{3}} + \frac{1}{\sqrt{5}+\sqrt{7}} + \frac{1}{\sqrt{9}+\sqrt{11}} + \dots + \frac{1}{\sqrt{9997}+\sqrt{9999}}$, then

(A) $S < 24$

(B) $S > 24$

(A) $S < 24$

(B) $S > 24$

(C) $S > 18$

(D) $S < 18$

25. When we divide the ninth term of an arithmetic progression by its second term, we get 5 as quotient and when we divide the thirteenth term of that progression by the sixth term, we get 2 as a quotient and 5 as a remainder, then

(A) First term is 3

(B) Seventh term is 27

(C) First term is 4

(D) Third term is 9

26. Define $f_n(x) = (1+x)(1+2x)(1+4x) \dots (1+2^n x) = a_{n,0} + a_{n,1}x + a_{n,2}x^2 + \dots + a_{n,n}x^n$,

where n is a positive integer, then

(A) $a_{100,2} = \frac{(2^{100} - 1)(2^{102} - 4)}{3}$

(B) $a_{100,2} = \frac{(2^{101} - 1)(2^{101} - 2)}{3}$

(C) $a_{100,2} - a_{99,2} = 2^{201} - 2^{101}$

(D) $a_{100,2} - a_{99,2} = 2^{200} - 2^{100}$

27. Let $\{a_n\}$ be a sequence of real number such that $a_1 = 2$, $a_{n+1} = a_n^2 - a_n + 1 \forall n \in \mathbb{N}$

and $S = \frac{1}{a_1} + \frac{1}{a_2} + \frac{1}{a_3} + \dots + \frac{1}{a_{2019}}$, then

(A) $S < 1$

(B) $S < 1 - \frac{1}{(2018)^{2018}}$

(C) $S > 1 - \frac{1}{(2018)^{2018}}$

(D) $S > 1 - \frac{1}{(2019)^{2019}}$

28. Let a, b, c be positive integers such that $a + b + c = n$, then

(A) $(a^a b^b c^c)^{1/n} \leq \frac{a^2 + b^2 + c^2}{n}$

(B) $(a^b b^c c^a)^{1/n} \leq \frac{ab + bc + ca}{n}$

$$(C) (a^b b^c c^a)^{1/n} \leq \frac{a^2 + b^2 + c^2}{n}$$

$$(D) (a^a b^b c^c)^{1/n} + (a^b b^c c^a)^{1/n} + (a^c b^a c^b)^{1/n} \leq n$$

29. If a, b, c, d are four unequal positive numbers which are in A.P., then

$$(A) \frac{1}{a} + \frac{1}{d} > \frac{1}{b} + \frac{1}{c}$$

$$(B) \frac{1}{a} + \frac{1}{d} < \frac{1}{b} + \frac{1}{c}$$

$$(C) ad < bc$$

$$(D) \frac{1}{b} + \frac{1}{c} > \frac{4}{a+d}$$

30. Let $a_1, a_2, a_3, \dots, a_n$ are in A.P., then

$$(A) a_1 - 2a_2 + a_3 = 0$$

$$(B) a_1 - 2a_2 + 2a_3 - 2a_4 + a_5 = 0$$

$$(C) a_1 + 3a_2 - 3a_3 - a_4 = 0$$

$$(D) a_1 - 4a_2 + 6a_3 - 4a_4 + a_5 = 0$$

31. Let $S = 2016^2 + 2015^2 + 2014^2 - 2013^2 - 2012^2 - 2011^2 + 2010^2 + 2009^2 + 2008^2 - 2007^2 - 2006^2 - 2005^2 + \dots + 6^2 + 5^2 + 4^2 - 3^2 - 2^2 - 1^2$, then S is divisible by

$$(A) 8$$

$$(B) 27$$

$$(C) 112$$

$$(D) 2017$$

32. It is given that the numbers $3, 3 \log_y x, 3 \log_z y, 7 \log_x z$ form an arithmetic progression. Then

$$(A) x^6 = y^8$$

$$(B) y^3 = z^4$$

$$(C) x^6 = y^7$$

$$(D) x^9 = z^{14}$$

33. Consider a geometric progression of real numbers such that sum of the first four terms is equal to 15 and the sum of their squares is equal to 85, then the possible value of sixth terms of the G.P. is/are

$$(A) 32$$

$$(B) 16$$

(C) $\frac{1}{4}$

(D) $\frac{1}{8}$

34. Let $S_n = \frac{1}{3!} + \frac{5}{4!} + \frac{11}{5!} + \frac{19}{6!} + \dots + \text{upto } n \text{ terms}$ and $S = \lim_{n \rightarrow \infty} S_n$, then

(A) $S = \frac{1}{6}$

(B) $S_n = \frac{1}{3!} - \frac{1}{(n+1)!}$

(C) $S_n = \frac{1}{2!} - \frac{(n+1)}{(n+2)!}$

(D) $S = \frac{1}{2}$

35. Let $f_n(x) = \frac{x}{1-x^2} + \frac{x^2}{1-x^4} + \frac{x^4}{1-x^8} + \dots + \frac{x^{2^{n-1}}}{1-x^{2^n}}$ where $x > 0$, $x \neq 1$ and

$f(x) = \lim_{n \rightarrow \infty} f_n(x)$, then

(A) $f\left(\frac{1}{4}\right) = \frac{1}{3}$

(B) $f\left(\frac{1}{2}\right) = 1$

(C) $f(3) = -\frac{1}{2}$

(D) $f(5) = -\frac{1}{6}$

36. Let

$$f(x) = \frac{1}{x+1} + \frac{2x}{(x+1)(x+2)} + \frac{3x^2}{(x+1)(x+2)(x+3)} + \dots + \frac{nx^{n-1}}{(x+1)(x+2)(x+3)\dots(x+n)}$$

then

(A) $f(x) = \frac{x}{1+x} - \frac{x^n}{(x+1)(x+2)\dots(x+n)}$

(B) $1 - \frac{x^n}{(x+1)(x+2)\dots(x+n)}$

(C) $f'(x) = \left(-\frac{x^n}{(x+1)(x+2)\dots(x+n)} \right) \left(\sum_{r=1}^n \frac{r}{x+r} \right)$

(D) $f'(x) = \left(-\frac{x^{n-1}}{(x+1)(x+2)\dots(x+n)} \right) \left(\sum_{r=1}^n \frac{r}{x+r} \right)$

37. The sum to first 'n' terms of the series ($n \geq 3$) $1^2 + 2 \cdot 2^2 + 3^2 + 2 \cdot 4^2 + 5^2 + 2 \cdot 6^2 + \dots +$

37. The sum to first n terms of the series $(1 + 2 + 2^2 + 3 + 2^4 + 5 + 2^6 + \dots)$ upto ' n ' terms is equal to
- (A) $\frac{n^2(n+1)}{2}$ if n is odd (B) $\frac{n(n+1)^2}{2}$ if n is even
- (C) $n^2(n-1)$ if n is odd (D) $2(n+1)^2$ if n is even
38. Let $a_1, a_2, a_3, \dots, a_n$ be the first ' n ' terms of an A.P. having common difference ' d ' ($d \neq 0$), then the greatest value of product of two terms equidistant from the extreme terms is
- (A) $a_1 a_n + \frac{d^2(n-1)^2}{4}$ if n is odd (B) $a_1 a_n + \frac{d^2(n+1)^2}{4}$ if n is odd
- (C) $a_1 a_n + \frac{d^2 n(n+2)}{4}$ if n is even (D) $a_1 a_n + \frac{d^2}{4} n(n-2)$ if n is even
39. If three successive terms of a G.P. having common ratio ' r ' form the sides of a triangle. Then the possible value(s) of $[r]$ is/are (where $[\cdot]$ denotes greatest integer function)
- (A) 2 (B) 0
- (C) 1 (D) 3
40. If a, b, c be in A.P. and a^2, b^2, c^2 be in H.P., then which of the following(s) may be true?
- (A) $\frac{a}{2}, b, c$ are in G.P. (B) $a, -\frac{b}{2}, c$ are in G.P.
- (C) $-\frac{a}{2}, b, c$ are in G.P. (D) $a = b = c$

SECTION-3

COMPREHENSION BASED QUESTIONS

COMPREHENSION: (Q.1 TO Q.3)

Let $\{a_n\}$ and $\{b_n\}$ be two sequence of numbers in A.P. each with common difference 2 such that $a_1 < b_1$ and $c_n = \sum_{k=1}^n a_k, d_n = \sum_{k=1}^n b_k$. Suppose that the points (a_n, c_n) and $(b_n, d_n) \forall n \in \mathbb{N}$ both lie on the curve $y = \alpha x^2 + \beta x + \gamma$ where α, β, γ are real constants, then :

1. α is equal to

- (A) $\frac{1}{8}$ (B) $\frac{1}{4}$ (C) $\frac{1}{2}$ (D) 1

2. If $\gamma = 0$, then a_1 is equal to

- (A) 0 (B) 1 (C) 2 (D) $\frac{1}{2}$

3. If $\gamma = 0$, then b_1 is equal to

- (A) $\frac{1}{4}$ (B) $\frac{1}{2}$ (C) 1 (D) 2

COMPREHENSION (Q.4 TO Q.6) :

$\{a_n\}$, $n \in \mathbb{N}$ is a sequence of real numbers and are arranged in the form

a_1

$a_2 a_3$

$a_4 a_5 a_6$

$a_7 a_8 a_9 a_{10}$

Let $\{b_n\}$, $n \in \mathbb{N}$ is a sequence formed by $a_1, a_2, a_4, a_7, a_{11}, \dots$ where $b_1 = a_1 = 1$.

Consider the sum, $S_n = \sum_{k=1}^n b_k$ satisfies $\frac{2b_n}{b_n S_n - S_n^2} = 1$ for $n \geq 2$. Also in the above table starting from the third row, each row (from left to right) is a G.P. and their

common ratio are all equal to the same positive numbers.

4. b_{12} is equal to

- (A) $-\frac{1}{78}$ (B) $-\frac{1}{66}$ (C) $-\frac{1}{21}$ (D) $-\frac{1}{105}$

5. $\frac{1}{S} + \frac{1}{S} + \frac{1}{S} + \dots + \frac{1}{S}$ is equal to

$$(A) \frac{n(7n-3)}{4}$$

$$(B) \frac{n(n+1)}{2}$$

$$(C) \frac{n}{4}(5n-1)$$

$$(D) \frac{n}{5}(6n-1)$$

6. The sum of all numbers in 10th row, if $a_{81} = -\frac{4}{91}$ is equal to

$$(A) -\frac{1358}{47}$$

$$(B) -\frac{729}{47}$$

$$(C) -\frac{1021}{55}$$

$$(D) -\frac{1023}{55}$$

COMPREHENSION (Q.7 TO Q.9) :

Given that a_1, a_2, a_3 is an A.P. in that order satisfying $a_1 + a_2 + a_3 = 15$; b_1, b_2, b_3 is a G.P. in that order and $b_1 b_2 b_3 = 27$. If $a_1 + b_1, a_2 + b_2, a_3 + b_3$ are positive integers and form a G.P. in that order.

7. The greatest possible value of common difference of A.P. a_1, a_1, a_3 , is equal to

$$(A) \frac{55+7\sqrt{61}}{2}$$

$$(B) \frac{63+7\sqrt{61}}{2}$$

$$(C) \frac{65+7\sqrt{61}}{2}$$

$$(D) \frac{61+7\sqrt{61}}{2}$$

8. The greatest possible value of common ratio of G.P. b_1, b_2, b_3 , is equal to

$$(A) \frac{57+7\sqrt{61}}{6}$$

$$(B) \frac{53+7\sqrt{61}}{6}$$

$$(C) \frac{61+7\sqrt{61}}{6}$$

$$(D) \frac{55+7\sqrt{61}}{6}$$

9. The greatest value of a_3 is equal to

$$(A) \frac{73+7\sqrt{61}}{2}$$

$$(B) \frac{55+7\sqrt{61}}{2}$$

$$(C) \frac{71+7\sqrt{61}}{2}$$

$$(D) \frac{75+7\sqrt{61}}{2}$$

COMPREHENSION (Q.10 TO Q.12) :

Let $\{a_n\}$, $a_n \geq 0$, $n \in \mathbb{N}$, $a_1 = 1$ be a sequence of real numbers and $S_n = a_1 + a_2 + a_3 +$

.... + a_n . Also it is given that $S_n = \frac{1}{2} \left(a_n + \frac{1}{a_n} \right)$.

10. $S_{100} + S_{400} + S_{900}$ is equal to

- (A) 10 (B) 20 (C) 30 (D) 60

11. $\lim_{n \rightarrow \infty} (\sqrt{n} a_n)$ is equal to

- (A) 1 (B) 1/2 (C) $\frac{1}{\sqrt{2}}$ (D) 0

12. $\left[\sum_{k=1}^{100} \frac{1}{S_k} \right]$ is equal to ([.] = denotes greatest integer function)

- (A) 18 (B) 19 (C) 20 (D) 17

COMPREHENSION (Q.13 to Q.15) :

If $\phi(r) = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \dots + \frac{1}{r}$ and $\sum_{r=1}^n (2r+1)\phi(r) = P(n)\phi(n+1) - Q(n)$, where $P(n)$ and $Q(n)$ are polynomial functions of 'n', then

13. $\sum_{r=0}^{10} P(r)$ is equal to

- (A) 235 (B) 506 (C) 285 (D) 385

14. $\sum_{r=0}^{\infty} \frac{1}{Q(r)}$ is equal to

- (A) 1 (B) 2 (C) 4 (D) 8

15. $P(13) - Q(13)$ is equal to

- (A) 81 (B) 78 (C) 91 (D) 65

COMPREHENSION (Q.16 to Q.17):

An arithmetic progression has the following property : for any even number of terms, the ratio of the sum of first half of the terms to the sum of second half is always equal to a constant k . Let the first term of A.P is 1.

16. The sum of all possible values of k is

(A) 1

(B) $\frac{4}{3}$

(C) $\frac{5}{3}$

(D) 3

17. Let the number of terms of A.P. is 20, then the sum of sum of all terms of all possible A.P.'s is

(A) 400

(B) 420

(C) 440

(D) 480

COMPREHENSION (Q.18 to Q.19):

Let $\{a_n\}$ be a sequence of integers such that $a_1 = 1$, $a_2 = 1$ and $a_{n+1} = a_n + a_{n-1}$ for all $n \in \mathbb{N}$, $n > 1$.

18. $\sum_{n=2}^{\infty} \frac{a_n}{a_{n-1} a_{n+1}}$ is equal to

(A) 1

(B) 2

(C) 4

(D) $\frac{1}{2}$

19. $\sum_{n=2}^{\infty} \frac{1}{a_{n-1} a_{n+1}}$ is equal to

(A) 1

(B) 2

(C) 4

(D) $\frac{1}{2}$

COMPREHENSION (Q.20 TO Q.22) :

Two A.P. s have the same number of terms. The ratio of the last term of the first A.P. to the first term of second A.P. is equal to the ratio of the last term of the second A.P to the first term of the first A.P. and is equal to 4. The ratio of the sum of the first 'n' terms of first A.P. to the sum of first 'n' terms of the second A.P. is equal to 2.

20. The ratio of first term of second A.P. to the first term of first A.P. is equal to
 (A) 4 (B) $\frac{2}{7}$ (C) $\frac{5}{2}$ (D) $\frac{7}{2}$
21. The ratio of the common difference of first A.P. to the common difference of second A.P. is equal to
 (A) 13 (B) 26 (C) 12 (D) 8
22. If the first term of first A.P. is equal to 1, then the sum of its first 14 terms is equal to
 (A) 84 (B) 98 (C) 105 (D) 120

COMPREHENSION (Q.23 TO Q.25)

In the following arrangement of n^2 positive numbers,

$a_{11}, a_{12}, \dots, a_{1n}$

$a_{21}, a_{22}, \dots, a_{2n}$

$\vdots$

$a_{n1}, a_{n2}, \dots, a_{nn}$

each horizontal row is an arithmetic progression and each vertical column is a geometric progression. It is known that geometric progressions have the same

positive common ratio. Given that $a_{24} = 1$, $a_{42} = \frac{1}{8}$ and $a_{43} = \frac{3}{16}$.

23. Let d_i ($i = 1, 2, 3, \dots, n$) be the common difference of i^{th} row, then $\lim_{n \rightarrow \infty} \sum_{i=1}^n d_i =$

- (A) 1 (B) 2 (C) 4 (D) 6

24. The value of $\lim_{n \rightarrow \infty} (1.a^{11} + 2a^{22} + 3a^{33} + \dots + na^{nn}) =$

- (A) 2 (B) 4 (C) 6 (D) 8

25. The value of the sum of all n^2 elements is equal to

- (A) $\frac{n(n-1)}{2} \left(1 - \frac{1}{2^n}\right)$ (B) $\frac{n(n+1)}{4} \left(1 - \frac{1}{2^{n-1}}\right)$
 (C) $\frac{n(n+1)}{2} \left(1 - \frac{1}{2^n}\right)$ (D) $\frac{n(n+1)}{2} \left(1 - \frac{1}{2^{n+1}}\right)$

COMPREHENSION (Q.26 TO Q.28)

Let a_m ($m = 1, 2, 3 \dots, p$) be the possible integral values of a for which the graphs of $f(x) = ax^2 + 2bx + b$ and $g(x) = 5x^2 - 3bx - a$ meets at some point for all real values

of b . Let $t_r = \prod_{m=1}^p (r - a_m)$ and $S_n = \sum_{r=1}^n t_r$, $n \in \mathbb{N}$.

26. The minimum possible value of a is

- (A) $\frac{1}{5}$ (B) $\frac{5}{26}$ (C) $\frac{3}{38}$ (D) $\frac{2}{43}$

27. The sum of values of n for which S_n vanishes is

- (A) 8 (B) 9 (C) 10 (D) 11

28. The value of $\sum_{r=5}^{\infty} \frac{1}{t_r}$ is equal to

- (A) $\frac{1}{3}$ (B) $\frac{1}{6}$ (C) $\frac{1}{15}$ (D) $\frac{1}{18}$

COMPREHENSION (Q.29 TO Q.30)

Let $x, x^{\log_{10} x}, y^{\log_{10} y}$ and $(xy)^{\log_{10}(xy)}$ are four consecutive terms of a geometric progression, $x, y > 0$, then

29. The number of ordered pair (x, y) is equal to

- (A) 1 (B) 2 (C) 3 (D) 4

30. If $(x_i, y_i), 1 \leq i \leq n$ be the ordered pair satisfying the given condition, $x_i, y_i \notin \mathbb{N}$,

then $\prod_{i=1}^n \frac{x_i}{y_i}$ is equal to

- (A) $\sqrt{10}$ (B) $100\sqrt{10}$ (C) 10 (D) $\sqrt{1000}$

SECTION-4

MATCH THE COLUMN

1.	Column-I		Column-II
(A)	If the roots of equation $x^3 - 9x^2 + 26x - k = 0$ are positive and in A.P., then k is equal to	(P)	7
(B)	If the roots of equation $x^3 - 14x^2 + kx - 64 = 0$ are positive and in G.P., then k is equal to	(Q)	11
(C)	If the roots of equation $6x^3 - kx^2 + 6x - 1 = 0$ are positive and in H.P., then k is equal to	(R)	24

(D)	In the equation $x^3 - kx + 6 = 0$, the sum of two roots is 3, then the value of k is equal to	(S)	26
		(T)	56

:

2.

	Column-I		Column-II
(A)	If $A = \sum_{r=1}^n r^2$, $B = \sum_{m=1}^n \sum_{r=1}^m r - \frac{1}{2} \sum_{r=1}^n r$, then $\frac{A}{B}$ is equal to	(P)	1
(B)	For positive numbers a, b, c the minimum value of $\frac{a(b^2 + c^2) + b(c^2 + a^2) + c(a^2 + b^2)}{abc}$ is equal to	(Q)	2
(C)	If $x + y + z = 1$, $x, y, z > 0$, then the minimum value of $\frac{2x^2}{y+z} + \frac{2y^2}{z+x} + \frac{2z^2}{x+y}$ is equal to	(R)	3
(D)	If $\frac{1}{xy} + \frac{1}{yz} + \frac{1}{zx} = 3$ where $x, y, z \in \mathbb{N}$, then $x + y + z$ is equal to	(S)	4
		(T)	6

3.

	Column-I		Column-II
(A)	If $\log_2 x + 4\log_4 y = 4 - 6\log_8 z$, then $x + y + z$ can be equal to	(P)	2
(B)	Let $x^2 - 3x + p = 0$ has two positive roots 'a'	(Q)	3

	and 'b', then $\frac{4}{a} + \frac{1}{b}$ can be equal to		
(C)	For triangle ABC, if cosecA, cosecB, cosecC are in H.P., then possible value(s) of $\frac{2b}{c}$ will be (where a, b, c denote lengths of sides of ΔABC as in usual notation)	(R)	5
(D)	If $3^{ \sin x } 2^{- \sec y } = a - 5\cos z$, where $x, y, z \in \mathbb{R}$ and $y \neq (2n + 1)\frac{\pi}{2}$, $n \in \mathbb{I}$, then possible value(s) of a can be	(S)	6
		(T)	7

4.	Column-I		Column-II
(A)	Let a, b, c are positive real numbers such that $a^3b^2c = 12$, then the minimum value of $49a + 3b + c$ is equal to	(P)	1
(B)	The minimum value of $\left 2x^3 - \frac{3}{x^2} \right $ for $x < 0$ is equal to	(Q)	5
(C)	The maximum value of $\frac{x^5(8-x^3)}{\sqrt[3]{25}}$ for $0 < x < 2$ is equal to	(R)	7
(D)	If $x^7y^5 = a$ and $7x + 5y \geq 12 \forall x, y > 0$, then the minimum value of 'a' is equal to	(S)	15
		(T)	42

5. Consider a sequence $\{b_n\}$ of integers such that b_1, b_2, b_3 are in G.P. b_2, b_3, b_4 are in A.P., b_3, b_4, b_5 are in G.P., b_4, b_5, b_6 are in A.P., b_5, b_6, b_7 are in G.P. and so on. Also

given that $b_1 = 1$ and $b_5 + b_6 = 198$. Then

	Column-I		Column-II
(A)	$\sqrt{b_7}$ is equal to	(P)	5
(B)	Sum of digits of b^8 is equal to	(Q)	15
(C)	$\sqrt{b_9}$ is equal to	(R)	9
(D)	Sum of digits of b_{10} is equal to	(S)	17
		(T)	13

SECTION-5

SUBJECTIVE TYPE QUESTIONS

1. If $\sum_{r=1}^n a_r = \sum_{k=1}^n \sum_{j=1}^k \sum_{i=1}^j 2$ and $\lambda = \lim_{n \rightarrow \infty} \left(\sum_{r=1}^n \frac{1}{a_r} \right)^n$ then $\left[\frac{1}{\lambda} \right]$ is equal to (where $[.]$ denotes greatest integer function).

2. If $a = 1^2 + \frac{2^2}{3} + \frac{3^2}{5} + \frac{4^2}{7} + \dots + \frac{(1001)^2}{2001}$, $b = \frac{1^2}{3} + \frac{2^2}{5} + \frac{3^2}{7} + \dots + \frac{(1001)^2}{2003}$ then $[a - b]$ is equal to ($[.]$ denotes greatest integer function)

3. The sum of the first 2018 terms of the sequence 1, 2, 1, 2, 2, 1, 2, 2, 2, 1, 2, 2, 2, 2, 1, 2, 2, 2, 2, 2, 1, 2, 2, is equal to

4. If $a, b, c, d, e \in \mathbb{R}$ satisfy $a + b + c + d + e = 8$ and $a^2 + b^2 + c^2 + d^2 + e^2 = 16$, then the complete range of e is $[a, b]$. Then $5(a + b)$ is equal to
5. Let $a_i + b_i = 1 \forall i = 1, 2, \dots, 6$ and $a = \frac{1}{6}(a_1 + a_2 + \dots + a_6)$, $b = \frac{1}{6}(b_1 + b_2 + \dots + b_6)$. Then
 $a_1b_1 + a_2b_2 + \dots + a_6b_6 = nab - (a_1 - a)^2 - (a_2 - a)^2 - \dots - (a_6 - a)^2$ where n is equal to
6. It is given that for the sequence $\{a_n\}$, its sum of first 'n' terms $S_n = n^2 + 3n + 4$, $n \in \mathbb{N}$. Find the value of $a_1 + a_3 + a_5 + a_7 + \dots + a_{21}$.
7. The common difference 'd' of an A.P. $\{a_n\}$ is not zero, the common ratio of a G.P. $\{b_n\}$ is a positive rational less than 1. If $a_1 = d$, $b_1 = d^2$ and $\frac{a_1^2 + a_2^2 + a_3^2}{b_1 + b_2 + b_3} = m$, where m is a positive integer, then m is equal to :

8. The value of expression $[\sqrt{1}] + [\sqrt{2}] + [\sqrt{3}] + \dots + [\sqrt{100^2}]$, (where $[.]$ denotes greatest integer function) is equal to $+$
9. If the value of $\sum_{i=0}^{\infty} \sum_{j=0}^{\infty} \sum_{k=0}^{\infty} \frac{1}{3^i 3^j 3^k}$ ($i \neq j \neq k$) is equal to $\frac{m}{n}$, where m, n are coprime natural numbers, then $m + n$ is equal to
10. If the sum $\sum_{m=1}^{\infty} \sum_{n=1}^{\infty} \frac{m^2 n}{3^m (n3^m + m3^n)} = \frac{p}{q}$ where p, q are relatively prime natural numbers, then $m + n$ is equal to :

11. Suppose x is a positive real number such that $\{x\}$, $[x]$ and x are in a geometric progression. Find the least positive integer n such that $x^n > 100$. (Here $[x]$ denotes the integral part of x and $\{x\} = x - [x]$).
12. Integers $1, 2, 3, \dots, n$, where $n > 2$ are written on a board. Two numbers m, k such that $1 < m < n$, $1 < k < n$ are removed and the average of the remaining numbers is found to be 17. What is the maximum sum of the two removed numbers?
13. Five distinct 2-digit numbers are in a geometric progression. Find the middle term.
14. Let the equation $x^4 - 16x^3 + px^2 - 256x + q = 0$ has four positive real roots in G.P., then $p + q$ is equal to
15. Let $x_1, x_2, x_3, \dots, x_{2018}$ be real numbers different from 1, such that $x_1 + x_2 + \dots + x_{2018} = 1$ and $\frac{x_1}{1-x_1} + \frac{x_2}{1-x_2} + \dots + \frac{x_{2018}}{1-x_{2018}} = 1$. Then the value of $\frac{x_1^2}{1-x_1} + \frac{x_2^2}{1-x_2} + \dots + \frac{x_{2018}^2}{1-x_{2018}}$ is equal to

16. The first term of a sequence is 2014. Each succeeding term is the sum of the cubes of the digits of the previous term. What is the 2018th term of the sequence.
17. Let $a_1, a_2, a_3, \dots, a_{2018}$ be an arithmetic progression of positive real numbers with common difference d . Let $a_1^2 + a_3^2 + \dots + a_{2017}^2 = x$, $a_2^2 + a_4^2 + \dots + a_{2018}^2 = y$ and $a_{1009} + a_{1010} = z$, then $\frac{y-x}{dz}$ is equal to

18. Let $x_1, x_2, \dots, x_{2018}$ be positive real numbers such that $x_1 + x_2 + \dots + x_{2018} = 1$.

Determine the smallest constant k such that $k \sum_{i=1}^{2018} \frac{x_i^2}{1-x_i} \geq 1$

19. Let $S = \frac{1}{2[\sqrt{1}] + 1} + \frac{1}{2[\sqrt{2}] + 1} + \frac{1}{2[\sqrt{3}] + 1} + \dots + \frac{1}{2[\sqrt{1000}] + 1}$ (where $[.]$ denotes greatest integer function), then $[S]$ is equal to

20. Let x, y, z are positive real numbers satisfy $2x - 2y + \frac{1}{z} = \frac{1}{2018}$, $2y - 2z + \frac{1}{x} = \frac{1}{2018}$, $2z - 2x + \frac{1}{y} = \frac{1}{2018}$ then $x + y - z$ is equal to

21. Let a, b, c be positive real numbers such that $\frac{a}{1+b} + \frac{b}{1+c} + \frac{c}{1+a} = 1$. Then the maximum value of abc is $\frac{p}{q}$ where p, q are relatively prime natural numbers, where $p + q$ is equal to

22. Let a, b, c be positive real numbers such that $a^3 + b^3 + c^3 \geq 4$, then find the minimum value of $\frac{a^3}{(a-b)(a-c)} + \frac{b^3}{(b-c)(b-a)} + \frac{c^3}{(c-a)(c-b)}$.

23. Let k be a positive real number such that $\frac{1}{k+a} + \frac{1}{k+b} + \frac{1}{k+c} < 1$ for any positive real numbers a, b and c with $abc = 1$. Then the least integral value of k is equal to

24. Let $x_1, x_2, x_3, \dots, x_{2018}$ be positive real numbers satisfying the condition

$\frac{1}{1+x_1} + \frac{1}{1+x_2} + \dots + \frac{1}{1+x_{2018}} = 1$. Then the minimum value of $\prod_{r=1}^{2018} x_r$ is equals to $(k)^{k+1}$, where k is equal to

25. The value of $\left[\sum_{n=1}^{\infty} \frac{1}{(n+1)\sqrt{n}} \right]$ (where $[.]$ denotes greatest integer function) is equal to

26. The product $\prod_{n=1}^{\infty} \left(1 + \frac{1}{2^{2^n}} \right)$ is equal to

27. The sum $\sum_{k=1}^{\infty} \frac{6^k}{(3^k - 2^k)(3^{k+1} - 2^{k+1})}$ is equal to

28. The sequence $\{x^n\}$ is defined by $x^1 = \frac{1}{2}$ and $x^{k+1} = x_k^2 + x^k \forall k \in \mathbb{N}$. Then $\left[\frac{1}{x_1+1} + \frac{1}{x_2+1} + \frac{1}{x_3+1} + \dots + \frac{1}{x_{100}+1} \right]$, where $[.]$ denotes greatest integer function, is equal to

29. Let $n > 4$ be a natural number and let P be a polygon with 'n' sides. Let $a_1, a_2, a_3, \dots, a^n$ be the lengths of the sides of P and let 'p' be its perimeter. Then find the value of $\left[\sum_{i=1}^n \frac{a_i}{p-a_i} \right]$, where $[.]$ denotes greatest integer function.

30. If S_1, S_2, S_3 denote the sums of first twenty terms of three non constant sequence in A.P., whose first terms are unity and common differences are in H.P. Then

$\frac{2S_3S_1 - S_1S_2 - S_2S_3}{S_1 - 2S_2 + S_3}$ is equal to

31. Let 'p' is the coefficient of x^2 in the expansion $(1 + x)(1 - 3x)(1 + 5x)(1 - 7x)....(1 - 23x)(1 + 25x)$, then the sum of the digits of $|p|$ is equal to
32. Find the number of triplets (a, b, c) such that a, b, c are three distinct positive numbers and $a, b, c, b + c - a, c + a - b, a + b - c$ and $a + b + c$ form a seven term arithmetic progression in some order.
33. Given that $a_1, a_2, a_3, ..., a_{15}, a_{16}$ are positive numbers constituting a geometric progression. If $a_1 + a_2 + a_3 + a_4 = 20$, $a_5 + a_6 + a_7 + a_8 = 320$, then $a_{13} + a_{14} + a_{15} + a_{16} = N$, find number of divisors of N .
34. The sum upto infinite terms of the series $1^2 + \frac{3^2}{2} + \frac{5^2}{2^2} + \frac{7^2}{2^3} + \infty$ is equal to
35. Let $\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{k^3 + 6k^2 + 11k + 5}{(k+3)!} = \frac{p}{q}$, where p and q are coprime natural numbers, then $p + q$ is equal to
36. If $\frac{a^n + b^n}{a^{n-1} + b^{n-1}}$ is the G.M. between two distinct positive numbers a and b , then n is equal to

SINGLE CHOICE QUESTIONS

- | | | | | | |
|-------|-------|-------|-------|-------|-------|
| 1. A | 2. B | 3. D | 4. D | 5. A | 6. B |
| 7. B | 8. D | 9. C | 10. A | 11. D | 12. C |
| 13. A | 14. B | 15. A | 16. B | 17. C | 18. D |
| 19. B | 20. B | 21. A | 22. C | 23. C | 24. B |
| 25. C | 26. B | 27. A | 28. B | 29. D | 30. C |
| 31. D | 32. A | 33. D | 34. C | 35. B | 36. A |
| 37. A | 38. C | 39. D | 40. B | | |

ONE OR MORE THAN ONE CORRECT QUESTIONS

- | | | | | | |
|-------------|-----------|------------|-------------|-----------|-------------|
| 1. B,C,D | 2. A,B,D | 3. C,D | 4. A,B | 5. A,B,D | 6. A,B,C |
| 7. B,C | 8. A,D | 9. A,B,C,D | 10. A,B | 11. C,D | 12. A,C |
| 13. A,B,C | 14. B,C | 15. B,C | 16. B,D | 17. A,D | 18. A,B,C,D |
| 19. B,C | 20. A,C,D | 21. A,B,C | 22. B,C,D | 23. B,C | 24. B,C |
| 25. A,B | 26. B,D | 27. A,C,D | 28. A,B,C,D | 29. A,C,D | 30. A,B |
| 31. A,B,C,D | 32. B,C,D | 33. A,C | 34. C,D | 35. A,B,C | 36. B,D |
| 37. A,B | 38. A,D | 39. B,C | 40. C,D | | |

COMPREHENSION BASED QUESTIONS

- | | | | | | |
|-------|-------|-------|-------|-------|-------|
| 1. B | 2. A | 3. D | 4. A | 5. C | 6. D |
| 7. B | 8. D | 9. A | 10. D | 11. B | 12. A |
| 13. D | 14. B | 15. C | 16. B | 17. B | 18. B |
| 19. A | 20. D | 21. B | 22. C | 23. A | 24. C |
| 25. C | 26. B | 27. C | 28. D | 29. C | 30. D |

MATCH THE COLUMN

1. $A \rightarrow R ; B \rightarrow T ; C \rightarrow Q ; D \rightarrow P$
2. $A \rightarrow Q ; B \rightarrow T ; C \rightarrow P ; D \rightarrow R$
3. $A \rightarrow R, S, T ; B \rightarrow Q, R, S, T ; C \rightarrow P, Q ; D \rightarrow P, Q, R, S$
4. $A \rightarrow T ; B \rightarrow Q ; C \rightarrow S ; D \rightarrow P$
5. $A \rightarrow T ; B \rightarrow P ; C \rightarrow S ; D \rightarrow Q$

SUBJECTIVE TYPE QUESTIONS

- | | | | | | |
|--------|-----------|---------|---------|----------|----------|
| 1. 2 | 2. 500 | 3. 3974 | 4. 16 | 5. 6 | 6. 268 |
| 7. 8 | 8. 661750 | 9. 289 | 10. 41 | 11. 10 | 12. 69 |
| 13. 36 | 14. 352 | 15. 0 | 16. 370 | 17. 1009 | 18. 2017 |
| 19. 30 | 20. 2018 | 21. 9 | 22. 4 | 23. 3 | 24. 2017 |
| 25. 1 | 26. 2 | 27. 2 | 28. 1 | 29. 1 | 30. 20 |
| 31. 19 | 32. 0 | 33. 30 | 34. 34 | 35. 8 | 36. 0.5 |

Previous Year Questions

SECTION-1

1. If $1, \log_9(3^{1-x} + 2), \log_3(4 \cdot 3^x - 1)$ are in A.P. then x equals **[AIEEE 2002]**
(A) $\log_3 4$ (B) $1 - \log_3 4$
(C) $1 - \log_4 3$ (D) $\log_4 3$
2. The value of $2^{1/4} \cdot 4^{1/8} \cdot 8^{1/16} \dots \infty$ is **[AIEEE 2002]**
(A) 1 (B) 2
(C) $3/2$ (D) 4
3. Fifth term of a GP is 2, then the product of its 9 terms is **[AIEEE 2002]**
(A) 256 (B) 512
(C) 1024 (D) None of these
4. Sum of infinite number of terms of GP is 20 and sum of their squares is 100. The common ratio of GP is **[AIEEE 2002]**
(A) 5 (B) $3/5$
(C) $8/5$ (D) $1/5$

5. $1^3 - 2^3 + 3^3 - 4^3 + \dots + 9^3 =$

[AIEEE 2002]

(A) 425

(B) -425

(C) 475

(D) -475

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Advance Problems in Algebra for JEE

6. Let T_r be the r^{th} term of an A.P. whose first term is a and common difference is d .

If for some positive integers $m, n, m \neq n, T_m = \frac{1}{n}$ and $T_n = \frac{1}{m}$, then $a - d$ equals-

[AIEEE 2004]

(A) 0

(B) 1

(C) $1/mn$

(D) $\frac{1}{m} + \frac{1}{n}$

7. The sum of the first n terms of the series $1^2 + 2 \cdot 2^2 + 3^2 + 2 \cdot 4^2 + 5^2 + 2 \cdot 6^2 + \dots$ is

$\frac{n(n+1)^2}{2}$ when n is even. When n is odd the sum is-

[AIEEE 2004]

(A) $\frac{3n(n+1)}{2}$

(B) $\frac{n^2(n+1)}{2}$

(C) $\frac{n(n+1)^2}{4}$

(D) $\left[\frac{n(n+1)}{2} \right]^2$

8. If $x = \sum_{n=0}^{\infty} a^n, y = \sum_{n=0}^{\infty} b^n, z = \sum_{n=0}^{\infty} c^n$ where a, b, c are in A.P. and $|a| < 1, |b| < 1,$

$|c| < 1$ then x, y, z are in -

[AIEEE 2005]

(A) GP

(B) AP

(C) Arithmetic - Geometric Progression

(D) HP

9. Let $a_1, a_2, a_3, \dots$ be terms of an A.P. If $\frac{a_1 + a_2 + \dots + a_p}{a_1 + a_2 + \dots + a_q} = \frac{p^2}{q^2}, p \neq q$ then $\frac{a_6}{a_{21}}$ equals -

[AIEEE 2006]

(A) $\frac{7}{2}$
(C) $\frac{11}{41}$

(B) $\frac{2}{7}$
(D) $\frac{41}{11}$

10. If $a_1, a_2, \dots, a_n$ are in H.P., then the expression $a_1 a_2 + a_2 a_3 + \dots + a_{n-1} a_n$ is equal to

[AIEEE 2006]

(A) $(n-1)(a_1 - a_n)$

(B) $na_1 a_n$

(C) $(n-1)a_1 a_n$

(D) $n(a_1 - a_n)$

11. In a geometric progression consisting of positive terms, each term equals the sum of the next two terms. Then the common ratio of this progression equals-

[AIEEE 2007]

(A) $\frac{1}{2}(1 - \sqrt{5})$

(B) $\sqrt{5}$

(C) $\frac{1}{2}\sqrt{5}$

(D) $\frac{1}{2}(\sqrt{5} - 1)$

12. The first two terms of a geometric progression add up to 12. The sum of the third and the fourth terms is 48. If the terms of the geometric progression are alternately positive and negative, then the first term is

[AIEEE 2008]

(A) -4

(B) -12

(C) 12

(D) 4

13. The sum to infinity of the series

$$1 + \frac{2}{3} + \frac{6}{3^2} + \frac{10}{3^3} + \frac{14}{3^4} + \dots$$

[AIEEE 2009]

(A) 2

(B) 3

(C) 4

(D) 6

14. A person is to count 4500 currency notes. Let a_n denote the number of notes he counts in the n^{th} minute. If $a_1 = a_2 = \dots = a_{10} = 150$ and $a_{10}, a_{11}, \dots$ are in an AP with

[AIEEE 2010]

- (A) 34 minutes (B) 125 minutes
(C) 135 minutes (D) 24 minutes

15. A man saves ₹ 200 in each of the first three months of his service. In each of the subsequent months his saving increases by ₹ 40 more than the saving of immediately previous month. His total saving from the start of service will be ₹ 11040 after

[AIEEE 2011]

- (A) 19 months (B) 20 months
(C) 21 months (D) 18 months

16. **Statement-1 :** The sum of the series

$1 + (1 + 2 + 4) + (4 + 6 + 9) + (9 + 12 + 16) + \dots + (361 + 380 + 400)$ is 8000.

Statement-2 : $\sum_{k=1}^n (k^3 - (k-1)^3) = n^3$, for any natural number n . [AIEEE 2012]

- (A) Statement-1 is true, Statement-2 is true, Statement-2 is a correct explanation for Statement-1
(B) Statement-1 is true, Statement-2 is true, Statement-2 is a **NOT** the correct explanation for Statement-1
(C) Statement-1 is true, Statement-2 is false
(D) Statement-1 is false, Statement-2 is true

17. The sum of first 20 terms of the sequence $0.7, 0.77, 0.777, \dots$, is :

[JEE (Mains) 2013]

- (A) $\frac{7}{9} (99 + 10^{-20})$ (B) $\frac{7}{81} (179 - 10^{-20})$
(C) $\frac{7}{9} (99 - 10^{-20})$ (D) $\frac{7}{81} (179 + 10^{-20})$

18. Let α and β be the roots of equation $px^2 + qx + r = 0$, $p \neq 0$. If p, q, r are in A.P. and $\frac{1}{\alpha} + \frac{1}{\beta} = 4$, then the value of $|\alpha - \beta|$ is [IIT Main 2014]
- (A) $\frac{2\sqrt{13}}{9}$ (B) $\frac{\sqrt{61}}{9}$
 (C) $\frac{2\sqrt{17}}{9}$ (D) $\frac{\sqrt{34}}{9}$
19. Three positive numbers form an increasing G.P. If the middle term in this G.P. is doubled, the new numbers are in A.P. Then the common ratio of the G.P. is : [IIT Main 2014]
- (A) $2 + \sqrt{3}$ (B) $\sqrt{2} + \sqrt{3}$
 (C) $3 + \sqrt{2}$ (D) $2 - \sqrt{3}$

20. If $(10)^9 + 2(11)^1(10)^8 + 3(11)^2(10)^7 + \dots + 10(11)^9 = k(10)^9$, then k is equal to [IIT Main 2014]
- (A) 110 (B) $\frac{121}{10}$
 (C) $\frac{441}{100}$ (D) 100
21. The sum of first 9 terms of the series $\frac{1^3}{1} + \frac{1^3 + 2^3}{1+3} + \frac{1^3 + 2^3 + 3^3}{1+3+5} + \dots$ is : [JEE Mains 2015]
- (A) 192 (B) 71
 (C) 96 (D) 142
22. If m is the A.M. of two distinct real numbers ℓ and n ($\ell, n > 1$) and G_1, G_2 and G_3 are three geometric means between ℓ and n , then $G_1^4 + 2G_2^4 + G_3^4$ equals.

(A) $4 \ell^2 m^2 n^2$

(B) $4 \ell^2 mn$

(C) $4 \ell m^2 n$

(D) $4 \ell mn^2$

23. If the 2nd, 5th and 9th terms of a non-constant A.P. are in G.P., then the common ratio of this G.P., is :

[JEE Mains 2016]

(A) $\frac{8}{5}$

(B) $\frac{4}{3}$

(C) 1

(D) $\frac{7}{4}$

24. If the sum of the first ten terms of the series

[JEE Mains 2016]

$$\left(1\frac{3}{5}\right)^2 + \left(2\frac{2}{5}\right)^2 + \left(3\frac{1}{5}\right)^2 + 4^2 + \left(4\frac{4}{5}\right)^2 + \dots, \text{ is } \frac{16}{5} m, \text{ then } m \text{ is equal to :}$$

(A) 102

(B) 101

(C) 100

(D) 99

25. If, for a positive integer, n the quadratic equation,
 $x(x+1) + (x+1)(x+2) + \dots + (x+n-1)(x+n) = 10n$
 has two consecutive integral solutions, then n is equal to :

[JEE Mains 2017]

(A) 10

(B) 11

(C) 12

(D) 9

26. For any three positive real numbers a , b and c

$$9(25a^2 + b^2) + 25(c^2 - 3ac) = 15b(3a + c). \text{ Then}$$

[JEE Mains 2017]

(A) a , b and c are in A.P.

(B) a , b and c are in G.P.

(C) b, c and a are in G.P.

(D) b, c and a are in A.P.

27. Let A be the sum of the first 20 terms and B be the sum of the first 40 terms of the series $1^2 + 2 \cdot 2^2 + 3^2 + 2 \cdot 4^2 + 5^2 + 2 \cdot 6^2 + \dots$

If $B - 2A = 100\lambda$, then λ is equal to

[JEE Mains 2018]

(A) 496

(B) 232

(C) 248

(D) 464

28. Let $a_1, a_2, a_3, \dots, a_{49}$ be in A.P. such that $\sum_{k=0}^{12} a_{4k+1} = 416$ and $a_9 + a_{43} = 66$.

If $a_1^2 + a_2^2 + \dots + a_{17}^2 = 140m$, then m is equal to

[JEE Mains 2018]

(A) 33

(B) 66

(C) 68

(D) 34

SECTION-2

1. (a) The harmonic mean of the roots of the equation

$$(5 + \sqrt{2})x^2 - (4 + \sqrt{5})x + 8 + 2\sqrt{5} = 0 \text{ is}$$

(A) 2

(B) 4

(C) 6

(D) 8

(b) Let $a_1, a_2, \dots, a_{10}$ be in A.P. & $h_1, h_2, \dots, h_{10}$ be in H.P. If $a_1 = h_1 = 2$ and

$a_{10} = h_{10} = 3$ then $a_4 h_7$ is:

[JEE '99, 2 + 2 out of 200]

(A) 2

(B) 3

(C) 5

(D) 6

2. The sum of an infinite geometric series is 162 and the sum of its first n terms is 160. If the inverse of its common ratio is an integer, find all possible values of the common ratio, n and the first terms of the series.

[REE '99, 6]

3. (a) Consider an infinite geometric series with first term 'a' and common ratio r . If the sum is 4 and the second term is $3/4$, then :

(A) $a = \frac{7}{4}, r = \frac{3}{7}$

(B) $a = 2, r = \frac{3}{8}$

(C) $a = \frac{3}{2}, r = \frac{1}{2}$

(D) $a = 3, r = \frac{1}{4}$

- (b) If a, b, c, d are positive real numbers such that $a + b + c + d = 2$, then $M = \frac{a+b}{c+d}$ satisfies the relation :

[JEE 2000, Screening, 1 + 1 out of 35]

(A) $0 \leq M \leq 1$

(B) $1 \leq M \leq 2$

(C) $2 \leq M \leq 3$

(D) $3 \leq M \leq 4$

- (c) The fourth power of the common difference of an arithmetic progression with integer entries added to the product of any four consecutive terms of it. Prove that the resulting sum is the square of an integer.

[JEE 2000, Mains, 4 out of 100]

4. Given that α, γ are roots of the equation, $Ax^2 - 4x + 1 = 0$ and β, δ the roots of the equation, $Bx^2 - 6x + 1 = 0$, find values of A and B , such that α, β, γ & δ are in H.P.

[REE 2000, 5 out of 100]

5. The sum of roots of the equation $ax^2 + bx + c = 0$ is equal to the sum of squares of their reciprocals. Find whether bc^2, ca^2 and ab^2 in A.P., G.P. or H.P.?

6. Solve the following equations for x and y

$$\log_2 x + \log_4 x + \log_{16} x + \dots = y$$

$$\frac{5+9+13+\dots+(4y+1)}{1+3+5+\dots+(2y-1)} = 4\log_4 x$$

[REE 2001, 5 out of 100]

7. (a) Let α, β be the roots of $x^2 - x + p = 0$ and γ, δ be the roots of $x^2 - 4x + q = 0$.

If $\alpha, \beta, \gamma, \delta$ are in G.P., then the integral values of p and q respectively, are

(A) -2, -32

(B) -2, 3

(C) -6, 3

(D) -6, -32

(b) If the sum of the first 2n terms of the A.P. 2, 5, 8, is equal to the sum of the first n terms of the A.P. 57, 59, 61,, then n equals

(A) 10

(B) 12

(C) 11

(D) 13

(c) Let the positive numbers a, b, c, d be in A.P. Then abc, abd, acd, bcd are

[JEE 2001, Screening, 1 + 1 + 1 out of 35]

(A) NOT in A.P./G.P./H.P.

(B) in A.P.

(C) in G.P.

(D) H.P.

(d) Let $a_1, a_2, \dots$ be positive real numbers in G.P. For each n, let A_n, G_n, H_n be respectively, the arithmetic mean, geometric mean and harmonic mean of $a_1, a_2, a_3, \dots, a_n$. Find an expression for the G.M. of $G_1, G_2, \dots, G_n$ in terms of $A_1, A_2, \dots, A_n, H_1, H_2, \dots, H_n$.

[JEE 2001 (Mains); 5]

8. (a) Suppose a, b, c are in A.P. and a^2, b^2, c^2 are in G.P. If $a < b < c$ and

$a + b + c = \frac{3}{2}$, then the value of a is

[JEE 2002 (Screening), 3]

(A) $\frac{1}{2\sqrt{2}}$

(B) $\frac{1}{2\sqrt{3}}$

(C) $\frac{1}{2} - \frac{1}{\sqrt{3}}$

(D) $\frac{1}{2} - \frac{1}{\sqrt{2}}$

(b) Let a, b be positive real numbers. If a, A_1, A_2, b are in A.P. ; a, G_1, G_2, b are in G.P. and a, H_1, H_2, b are in H.P. , show that

$$\frac{G_1 G_2}{H_1 H_2} = \frac{A_1 + A_2}{H_1 + H_2} = \frac{(2a + b)(a + 2b)}{9ab}$$

[JEE 2002 , Mains , 5 out of 60]

9. If a, b, c are in A.P., a^2, b^2, c^2 are in H.P. , then prove that either $a = b = c$ or $a, b, -\frac{c}{2}$ form a G.P. [JEE-03, Mains-4 out of 60]

10. The first term of an infinite geometric progression is x and its sum is 5. Then

[JEE 2004 (Screening)]

(A) $0 \leq x \leq 10$

(B) $0 < x < 10$

(C) $-10 < x < 0$

(D) $x > 10$

11. If a, b, c are positive real numbers, then prove that $[(1 + a)(1 + b)(1 + c)]^7 > 7^7 a^4 b^4 c^4$. [JEE 2004, 4 out of 60]

12. (a) In the quadratic equation $ax^2 + bx + c = 0$, if $\Delta = b^2 - 4ac$ and $\alpha + \beta, \alpha^2 + \beta^2, \alpha^3 + \beta^3$ are in G.P. where α, β are the roots of $ax^2 + bx + c = 0$, then

[JEE 2005 (Screening)]

(A) $\Delta \neq 0$

(B) $b\Delta = 0$

(C) $c\Delta = 0$

(D) $\Delta = 0$

(b) If total number of runs scored in n matches is $\left(\frac{n+1}{4}\right)(2^{n+1} - n - 2)$ where $n > 1$, and the runs scored in the k^{th} match are given by $k \cdot 2^{n+1-k}$, where $1 \leq k \leq n$. Find n . [JEE 2005 (Mains), 2]

13. If $A_n = \left(\frac{3}{4}\right) - \left(\frac{3}{4}\right)^2 + \left(\frac{3}{4}\right)^3 + \dots + (-1)^{n-1} \left(\frac{3}{4}\right)^n$ and $B_n = 1 - A_n$, then find the minimum natural number n_0 such that $B_n > A_n, \forall n > n_0$. [JEE 2006, 6]

COMPREHENSION (Q.(A) TO Q.(c)) :

14. Let V_r denote the sum of the first 'r' terms of an arithmetic progression (A.P.) whose first term is 'r' and the common difference is $(2r - 1)$.

Let $T_r = V_{r+1} - V_r - 2$ and $Q_r = T_{r+1} - T_r$ for $r = 1, 2, \dots$

- (a) The sum $V_1 + V_2 + \dots + V_n$ is

(A) $\frac{1}{12}n(n+1)(3n^2 - n + 1)$

(B) $\frac{1}{12}n(n+1)(3n^2 + n + 2)$

(C) $\frac{1}{2}n(2n^2 - n + 1)$

(D) $\frac{1}{3}(2n^3 - 2n + 3)$

- (b) T_r is always

(A) an odd number

(B) an even number

(C) a prime number

(D) a composite number

- (c) Which one of the following is a correct statement?

[JEE 2007, 4 + 4 + 4]

(A) $Q_1, Q_2, Q_3, \dots$ are in A.P. with common difference 5.

(B) $Q_1, Q_2, Q_3, \dots$ are in A.P. with common difference 6.

(C) $Q_1, Q_2, Q_3, \dots$ are in A.P. with common difference 11.

(D) $Q_1 = Q_2 = Q_3 = \dots$

COMPREHENSION (Q.(A) TO Q.(c)) :

15. Let A_1, G_1, H_1 denote the arithmetic, geometric and harmonic means, respectively, of two distinct positive numbers. For $n \geq 2$, let A_{n-1} and H_{n-1} have arithmetic, geometric and harmonic means as A^n, G^n, H^n respectively.

(a) Which one of the following statements is correct?

- (A) $G_1 > G_2 > G_3 > \dots$
- (B) $G_1 < G_2 < G_3 < \dots$
- (C) $G_1 = G_2 = G_3 = \dots$
- (D) $G_1 < G_3 < G_5 < \dots$ and $G_2 > G_4 > G_6 > \dots$

(b) Which one of the following statements is correct?

- (A) $A_1 > A_2 > A_3 > \dots$
- (B) $A_1 < A_2 < A_3 < \dots$
- (C) $A_1 > A_3 > A_5 > \dots$ and $A_2 < A_4 < A_6 < \dots$
- (D) $A_1 < A_3 < A_5 < \dots$ and $A_2 > A_4 > A_6 > \dots$

(c) Which one of the following statements is correct?

[JEE 2007, 4 + 4 + 4]

- (A) $H_1 > H_2 > H_3 > \dots$
- (B) $H_1 < H_2 < H_3 < \dots$
- (C) $H_1 > H_3 > H_5 > \dots$ and $H_2 < H_4 < H_6 < \dots$
- (D) $H_1 < H_3 < H_5 < \dots$ and $H_2 > H_4 > H_6 > \dots$

16. (a) A straight line through the vertex P of a triangle PQR intersects the side QR at the point S and the circumcircle of the triangle PQR at the point T. If S is not the centre of the circumcircle, then [JEE 2008, 4]

(A) $\frac{1}{PS} + \frac{1}{ST} < \frac{2}{\sqrt{QS \times SR}}$ (B) $\frac{1}{PS} + \frac{1}{ST} > \frac{2}{\sqrt{QS \times SR}}$
 (C) $\frac{1}{PS} + \frac{1}{ST} < \frac{4}{QR}$ (D) $\frac{1}{PS} + \frac{1}{ST} > \frac{4}{QR}$

■ ASSERTION & REASON

- (b) Suppose four distinct positive numbers a_1, a_2, a_3, a_4 are in G.P. Let $b_1 = a_1, b_2 = a_2, b_3 = a_3, b_4 = a_4$ and $b_1 = a_1, b_2 = a_2, b_3 = a_3, b_4 = a_4$. [JEE 2008, 3 (-1)]

STATEMENT-1 : The numbers b_1, b_2, b_3, b_4 are neither in A.P. nor in G.P.

and

STATEMENT-2 : The numbers b_1, b_2, b_3, b_4 are in H.P.

- (A) Statement-1 is True, Statement-2 is True; statement-2 is a correct explanation for statement-1
 (B) Statement-1 is True, Statement-2 is True; statement-2 is **NOT** a correct explanation for statement-1
 (C) Statement-1 is True, Statement-2 is False
 (D) Statement-1 is False, Statement-2 is True

17. Let $S_k, k = 1, 2, \dots, 100$, denote the sum of the infinite geometric series whose first term is $\frac{k-1}{k!}$ and the common ratio is $\frac{1}{k}$. Then the value of $\frac{100^2}{100!} +$

$$\sum_{k=1}^{100} (k^2 - 3k + 1) S_k$$

18. Let $a_1, a_2, a_3, \dots, a_{11}$ be real numbers satisfying $a_1 = 15, 27 - 2a_2 > 0$ and $a_k = 2a_{k-1} - a_{k-2}$ for $k = 3, 4, \dots, 11$. If $\frac{a_1^2 + a_2^2 + \dots + a_{11}^2}{11} = 90$, then the value of $\frac{a_1 + a_2 + \dots + a_{11}}{11}$ is equal to

[JEE 2010]

Sequence and Progression

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19. Let $a_1, a_2, a_3, \dots, a_{100}$ be an arithmetic progression with $a_1 = 3$ and $S_p = \sum_{i=1}^p a_i, 1 \leq p \leq 100$. For any integer n with $1 \leq n \leq 20$, let $m = 5n$. If $\frac{S_m}{S_n}$ does not depend on n , then a_2 is
- [JEE 2011]
20. The minimum value of the sum of real numbers $a^{-5}, a^{-4}, 3a^{-3}, 1, a^8$ and a^{10} with $a > 0$ is
- [JEE 2011]
21. Let $a_1, a_2, a_3, \dots$ be in harmonic progression with $a_1 = 5$ and $a_{20} = 25$. The least positive integer n for which $a_n < 0$ is :
- [JEE 2012]
- (A) 22 (B) 23
(C) 24 (D) 25
22. Let $S_n = \sum_{k=1}^{4n} (-1)^{\frac{k(k+1)}{2}} k^2$. Then S_n can take value(s)
- [JEE Advance 2013]
- (A) 1056 (B) 1088
(C) 1120 (D) 1332
23. Let a, b, c be positive integers such that $\frac{b}{a}$ is an integer. If a, b, c are in geometric progression and the arithmetic mean of a, b, c is $b + 2$, then the value of $\frac{a^2 + a - 14}{a + 1}$ is
- [JEE Advance 2014]
24. Suppose that all the terms of an arithmetic progression (A.P.) are natural numbers. If the ratio of the sum of the first seven terms to the sum of the first eleven terms

is $6 : 11$ and the seventh term lies in between 130 and 140, then the common difference of this A.P. is

[JEE Advance 2015]

25. Let $b_i > 1$ for $i = 1, 2, \dots, 101$. Suppose $\log_e b_1, \log_e b_2, \dots, \log_e b_{101}$ are in Arithmetic Progression (A.P.) with the common difference $\log_e 2$. Suppose $a_1, a_2, \dots, a_{101}$ are in A.P. such that $a_1 = b_1$ and $a_{51} = b_{51}$.

If $t = b_1 + b_2 + \dots + b_{51}$ and $s = a_1 + a_2 + \dots + a_{51}$, then [JEE Advance 2016]

- (A) $s > t$ and $a_{101} > b_{101}$ (B) $s > t$ and $a_{101} < b_{101}$
(C) $s < t$ and $a_{101} > b_{101}$ (D) $s < t$ and $a_{101} < b_{101}$

26. The sides of a right angled triangle are in arithmetic progression. If the triangle has area 24, then what is the length of its smallest side? [JEE Advance 2017]

Answer Key

SECTION-1

- | | | | | | |
|-------|-------|-------|-------|-------|-------|
| 1. B | 2. B | 3. B | 4. B | 5. A | 6. A |
| 7. B | 8. D | 9. C | 10. C | 11. D | 12. B |
| 13. B | 14. A | 15. C | 16. A | 17. D | 18. A |
| 19. A | 20. D | 21. C | 22. C | 23. B | 24. B |
| 25. B | 26. D | 27. C | 28. D | | |

SECTION-2

1. (a) B (b) D
2. $(r, n, a) \in \left\{ \left(\frac{1}{3}, 4, 108 \right), \left(\frac{-1}{3}, 4, 216 \right), \left(\frac{1}{3}, 2, 144 \right), \left(\frac{-1}{3}, 2, 180 \right), \left(\frac{1}{3}, 1, 160 \right) \right\}$
3. (a) D (b) A
4. $A = 3$; $B = 8$
5. A.P.
6. $x = 2\sqrt{2}$ and $y = 3$

7. (a) A, (b) C, (c) D, (d) $\left[(A_1, A_2, \dots, A_n) (H_1, H_2, \dots, H_n) \right]^{\frac{1}{2n}}$
8. (a) D 10. B 12. (a) C, (b) $n = 7$
13. $n_0 = 5$ 14. (a) B; (b) D; (c) B 15. (a) C; (b) A; (c) B
16. (a) B, D; (b) C 17. 3 18. 0
19. 9 or 3 20. 8 21. D 22. A, D
23. 4 24. 9 25. B 26. 6



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3

CHAPTER

PERMUTATION AND
COMBINATION

SECTION-1

● SINGLE CHOICE QUESTIONS

1. From a group of 10 professors $A_1, A_2, \dots, A_{10}$ how many ways can a committee of 5 members be formed so that atleast one of professor A_1 and professor A_2 will be included.

(A) 196

(B) 252

(C) 140

(D) 212

2. Let $S = \{1, 2, 3, \dots, n\}$. Find number of unordered pairs (A, B) of subsets of S such that A and B are disjoint, where A or B both may be empty.

(A) 3^n

(B) $\frac{3^n - 1}{2}$

(C) $\frac{3^n + 1}{2}$

(D) $2^n + 1$

3. In how many ways can we choose a black square and a white square from a chessboard so that they are neither in the same row nor the same column.

(A) 658

(B) 768

(C) 1024

(D) 972

4. Consider a rational number $\frac{a}{b}$ in its lowest form a, b are integers, with $0 < \frac{a}{b} < 1$,

$b > 1$. How many of these have $ab = 15!$

(A) 64

(B) 32

(C) 256

(D) 16

5. How many permutations of the digits of the number 123456789 are there in which none of the blocks 12, 34 or 567 appear.

(A) $9! - 661 \times 5!$

(B) $9! - 110 \times 6!$

(C) $9! - 41 \times 7!$

(D) $9! - 55 \times 7!$

6. Five boys and 6 girls must sit around a round table. How many arrangements are possible if Ram (a boy) must always be adjacent to seeta and geeta (two girls).

(A) $10! - 2 \times 6!$

(B) $6 \times 8!$

(C) $9! \times 6$

(D) $2 \times 8!$

7. In how many ways may one seat 100 people into 20 distinct round tables in such a way that there are 5 people per table.

(A) $\frac{100! \times (4!)^5}{(20!)^5}$

(B) $\frac{100!}{(20!)^5}$

(C) $\frac{100! (4!)^5}{(20!)^5 5!}$

(D) $\frac{100! (5!)^4}{(20!)^5}$

8. The number of ordered triplets (a, b, c) from the set $\{1, 2, 3, 4, \dots, 100\}$ such that $a \leq b \leq c$ is equal to

(A) $^{102}C^3$

(B) $^{100}C^3$

(C) $^{101}C_2 + ^{100}C_3$

(D) $^{100}C_1 + 3(^{100}C_2) + 6(^{100}C_3)$

9. Given 'n' different objects arranged in a row. Then the number of ways of choosing k of them so that no two of them are consecutive is equal to

(A) nC_k

(B) $^{n-k}C_k$

(C) $^nC_k - (n - k + 1)$

(D) $^{n-k+1}C_k$

10. In how many ways can we distribute k identical balls into n different boxes so that each box contains atmost one ball and no two consecutive boxes are empty.

(A) $^{k+1}C_{n-k}$

(B) $^{k+(n-1)}C_{n-1}$

(C) nC_k

(D) $^{n-k+1}C_k$

11. How many different sets of 'k' numbers $\{a_1, a_2, \dots, a_k\}$ with integral

$a_i \in \{1, 2, 3, \dots, n\}$ are there satisfying $1 \leq a_1 \leq a_2 \leq a_3 \leq \dots \leq a_k \leq n$.

(A) ${}^{k+n-1}C_{n-1}$

(B) ${}^{n+k-1}C_{k-1}$

(C) ${}^{n-k+1}C_n$

(D) ${}^{n-k+1}C_k$

12. Find number of points (x, y) in xy plane with both x and y integers which satisfy

$|x| + |y| < 100$

(A) 19801

(B) 20000

(C) 19404

(D) 19602

13. The number of three element subsets from the set $\{1, 2, 3, \dots, 298, 299, 300\}$, such that the sum of the elements is a multiple of 3 is equal to

(A) 4851

(B) 1004851

(C) 1014881

(D) 998624

14. How many integers are there between 0 and 10^5 having the digit sum equal to 8.

(A) 560

(B) 495

(C) 35

(D) 640

15. In how many ways $2n + 1$ identical oranges can be distributed among 3 persons so that the sum of number of oranges received by any two persons should exceed the number of oranges received by the other.

(A) $n(2n - 1)$

(B) $\frac{n(n-1)}{2}$

(C) $\frac{n(n-5)}{2}$

(D) $\frac{n(n+1)}{2}$

16. The number of triples (A, B, C) where A, B, C are subsets of $\{1, 2, 3, \dots, n\}$ such that $A \cap B \cap C = \phi$, $A \cap B \neq \phi$, $B \cap C \neq \phi$ is equal to

(A) $7^n - 2 \cdot 6^n + 5^n$

(B) $7^n - 6^n + 5^n$

(C) $7^n - 5^n$

(D) $7^n - 3 \cdot 6^n - 5^n$

17. A postman has to deliver five letters to five different houses. Mischievously, he posts one letter through each door without looking to see if it is the correct address. In how many different ways could he do this so that exactly two of the five houses receive the correct letters.

(A) 20

(B) 40

(C) 60

(D) 15

18. There are 10 bags $B_1, B_2, \dots, B_{10}$ which contain 31, 32, ..., 40 distinct articles respectively. The total number of ways to draw 10 articles all from a single bag is

(A) ${}^{41}C_{30}$

(B) ${}^{41}C_{10}$

(C) ${}^{41}C_{30} - {}^{31}C_{20}$

(D) ${}^{40}C_{29}$

19. Let $\vec{a} = \hat{i} + \hat{j} + \hat{k}$ and $\vec{r}$ be a variable vector such that $\vec{r} \cdot \hat{i}, \vec{r} \cdot \hat{j}$ and $\vec{r} \cdot \hat{k}$ are positive integers. If $\vec{r} \cdot \vec{a} \leq 12$, then the number of values of $\vec{r}$ is

(A) ${}^{11}C_2$

(B) ${}^{14}C_2$

(C) ${}^{15}C_3$

(D) ${}^{12}C_3$

20. How many line segments have both their endpoints located at the vertices of a given cube.

(A) 36

(B) 28

(C) 45

(D) 21

21. At a couple dance party, each man danced with exactly four women and each

woman danced with exactly three men. Nine men attended the party. How many women attended the party.

- (A) 12
- (B) 11
- (C) 10
- (D) 14

22. The number of all 6-digit natural numbers having exactly three odd digits and three even digits is equal to

- (A) 321150
- (B) 271250
- (C) 182510
- (D) 281250

23. There are eight rooms on the first floor of a hotel, with four rooms on each side of the corridor, symmetrically situated (that is each room is exactly opposite one other room). Four guests have to be accommodated in four of the eight rooms (that is, one in each) such that no two guests are in adjacent rooms or in opposite rooms. In how many ways can the guests be accommodated.

- (A) 24
- (B) 48
- (C) 96
- (D) 72

24. There are five cities A, B, C, D, E on a certain island. Each city is connected to every other city by road. In how many ways can a person starting from city A come back to A after visiting some cities without visiting a city more than once and without taking the same road more than once. (The order in which he visits the cities also matters)

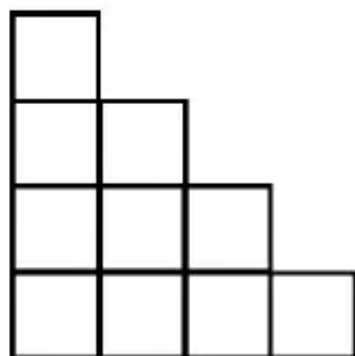
- (A) 60
- (B) 61
- (C) 63
- (D) 64

25. Find number of ways to distribute 8 distinct balls among 3 children so that every child receives atleast one ball is

- (A) 6308
- (B) 6561
- (C) 5796
- (D) 5793

26. The greatest of d so that 12^d divides in $100!$ is equal to ($d \in \mathbb{N}$)
- (A) 47 (B) 48
(C) 49 (D) 97
27. How many different 9 digit numbers can be formed from the number 223355888 by rearranging its digits so that the odd digits occupy odd positions.
- (A) 300 (B) 60
(C) 240 (D) 360
28. Let $A = \{x_1, x_2, x_3, x_4, x_5, x_6, x_7\}$ and $B = \{y_1, y_2, y_3\}$. The total number of functions $f: A \rightarrow B$ that are onto and there are exactly three elements x in A such that $f(x) = y_1$ is equal to
- (A) $12 \times {}^7C_3$ (B) $15 \times {}^7C_3$
(C) $16 \times {}^7C_3$ (D) $14 \times {}^7C_3$
29. The number of interior points where diagonals of a convex polygon of n sides intersect if no three diagonal pass through the same interior point is equal to
- (A) $\frac{n(n-1)(n-2)(n-3)}{8}$ (B) $\frac{n(n-1)(n-2)(2n-1)}{24}$
(C) nC_4 (D) $\frac{n(n-1)(n-2)(n-3)}{12}$
30. The number of ways to select n objects from $3n$ objects of which n are identical and rest are different is equal to
- (A) $2^{2n-1} - \frac{(2n)!}{2(n!)^2}$ (B) $2^{2n-1} + \frac{(2n)!}{2(n!)^2}$
(C) $2^{2n-1} + \frac{(2n)!}{(n!)^2}$ (D) $2^{2n-1} - \frac{(2n)!}{(n!)^2 2}$

31. The number of common divisors of 10800 and 9000 is equal to
(A) 36 (B) 48
(C) 60 (D) 45
32. A, B are two students in a group of n students. If the number of ways of assigning the n students to a line of n single rooms such that A and B are not in adjacent rooms is 3600, then n is equal to
(A) 9 (B) 8
(C) 7 (D) 6
33. The number of three digit numbers of the form xyz such that $x < y$ and $z \leq y$ is
(A) 221 (B) 240
(C) 256 (D) 276
34. The number of all possible four digit numbers having exactly two fives and no two consecutive digits identical is equal to
(A) 216 (B) 225
(C) 228 (D) 235
35. The number of ways in which all the letters of the word 'SAKSHAM' can be placed in the squares of the figure shown, so that no row remains empty is equal to



- (A) 95760 (B) 94640
(C) 97840 (D) 82120
36. If three dice are rolled and we make a set of numbers shown on the three dice. How

many different sets are possible.

(A) 56

(B) 15

(C) 216

(D) 224

SECTION-2

ONE OR MORE THAN ONE CORRECT QUESTIONS

1. Let $A = \{1, 2, 3, \dots, n\}$. The number of functions $f: A \rightarrow A$ which is strictly increasing such that $f(i) \leq i \forall i \in A$ is equal to

(A) ${}^{2n-2}C_{n-1} - {}^{2n-2}C_n$

(B) ${}^{2n}C_n - {}^{2n}C_{n+1}$

(C) $\frac{{}^{2n-2}C_{n-1}}{n}$

(D) $\frac{{}^{2n}C_n}{n+1}$

2. Let $S = \{1, 2, 3, 4, \dots, (n+1)\}$, where $n \geq 6$ and let $T = \{(x, y, z) \mid x, y, z \in S, x < z, y < z\}$, then number of members of T is equal to

(A) ${}^{n+1}C_2 + {}^{n+1}C_3$

(B) ${}^{n+1}C_2 + 2 {}^{n+1}C_3$

(C) $\sum_{k=1}^n k^2$

(D) $\sum_{k=1}^{n-1} k^2$

3. A decimal code is declared legal if it has an even number of zeros. For example 1900200 is a legal code, but 10002 is not. Let a_n be the number of legal decimal codes of length 'n'. Then

(A) $a_n = 8a_{n-1} + 10^{n-1}$

(B) $a_n = \frac{8^n + 10^n}{2}$

(C) $a_n = 9a_{n-1} + 10^n$

(D) $a_n = 9^n + 10^n$

4. If Mr. A can climb either one step or two steps at a time. Let a_n is the number of ways, he climbs a n-step staircase then

(A) $a_8 = 35$

(B) $a_n = a_{n-1} + a_{n-2}$

(C) $a_9 = 55$

(D) $a_{11} = 144$

5. The number of ways to choose a subset of two elements $\{a, b\}$ from the set $\{1, 2, 3, 4, \dots, 49, 50\}$ such that (The pair $\{a, b\}$ is indistinguishable from the pair $\{b, a\}$)

(A) $|a - b| \leq 5$ is 245

(B) $|a - b| \leq 5$ is 235

(C) $|a - b| = 5$ is 45

(D) $|a - b| \geq 5$ is 1035

6. The sequence of all positive palindromes are written in ascending order 1, 2, 3, 4, 5, 6, 7, 8, 9, 11, 22,.....

Then

(A) the 2018th positive palindrome is 1019101

(B) the 1998th positive palindrome is 1001001

(C) the rank of 2140412 is 3139

(D) the rank of 1999999 is 2998

7. Given '2n' different objects arranged around a circle, the number of ways of choosing k of them so that no two of them are consecutive is equal to ($0 \leq k \leq n$)

(A) ${}^{2n-k+1}C_k - {}^{2n-k-1}C_{k-2}$

(B) $\frac{2n}{k} {}^{2n-k-1}C_{k-1}$

(C) $\frac{2n}{(2n-k)} {}^{2n-k}C_k$

(D) $\frac{n(4n-4k+1)}{(2n-2k+1)} {}^{2n-k}C_k$

8. The number of ordered triplets (x, y, z) of non negative integers satisfying the conditions.

$$x + y + z \leq 100 \quad \text{and} \quad x \leq y \leq z$$

(A) if x is odd is 14724

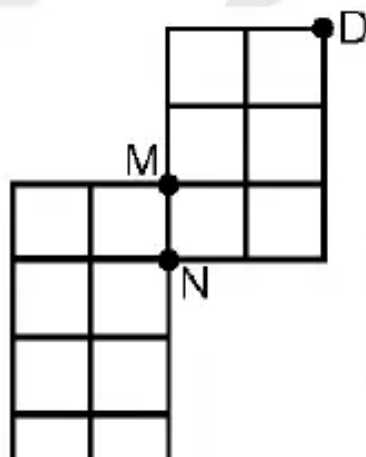
(B) if x is odd is 14722

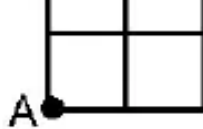
(C) if x is even is 16065

(D) if x is even is 16164

9. Let x be the number of 6 digit numbers, the sum of whose digits is even and y be the number of 6 digit numbers, the sum of whose digits is odd, then
- (A) $x + y = 9 \times 10^5$ (B) $x < y$
 (C) $x = y$ (D) $y = 450000$
10. There are n lines in a plane, no two of which are parallel and no three of them are concurrent. Let the plane be divided by n lines in a_n parts, then
- (A) $a_n = a_{n-1} + (n - 1)$ (B) $a_n = a_{n-1} + n$
 (C) $a_6 = 22$ (D) $a_{10} = 56$

11. There are n married couples at a party. Each person shakes hand with every person other than her or his spouse. The total number of handshakes must be
- (A) ${}^{2n}C_2 - 2n$ (B) ${}^{2n}C_2 - n$
 (C) $2n(2n - 1)$ (D) $2n(n - 1)$
12. Let 5 letter words are formed using the letters of the word 'CALCULUS'. Then
- (A) The number of all such possible words is 1110.
 (B) The number of words with exactly one alike pair of letters is 720.
 (C) The number of words with exactly two alike pair of letters is 270.
 (D) The number of words with exactly one alike pair of letters is 480.
13. Let a person has to go from A to D moving along horizontal and vertical grids using shortest path (as shown in figure). Then





- (A) The number of paths is equal to 276.
 (B) The number of paths is equal to 186.
 (C) The number of paths in which he doesn't cross point N is equal to 36.
 (D) The number of paths in which he doesn't cross point M is equal to 60.
14. Consider all possible permutations of all the letters of the word 'CONTINUITY', then the number of permutations.
- (A) Which have 'COUNT' in all of them is $2(5!)$.
 (B) Which have 'COUNT' in all of them is $3(5!)$.
 (C) Which have all vowels separated is $15(7!)$.
 (D) Which have all vowels separated is $7(7!)$.
15. Let a_n be the number of non empty subsets of $S = \{1, 2, 3, 4, \dots, n-1, n\}$ such that there are no two consecutive numbers in one and the same set, then
- (A) $a_n = a_{n-1} + a_{n-2}$ (B) $a_n = a_{n-1} + a_{n-2} + 1$
 (C) $a_7 = 33$ (D) $a_7 = 21$
16. 'A' has cans of paint in eight different colours. He wants to paint the four unit squares of a 2×2 board in such a way that neighbouring unit squares are painted in different colours.
- (A) The number of distinct colouring schemes 'A' can make is equal to 2072.
 (B) The number of distinct colouring schemes 'A' can make is equal to 2036.
 (C) The number of distinct colouring schemes 'A' can make in which two colouring schemes are considered the same if one can be obtained from the other by rotation is equal to 532.
 (D) The number of distinct colouring schemes 'A' can make in which two colouring schemes are considered the same if one can be obtained from the other by rotation is equal to 616.
17. The integers from 1 to 1000 are written in order around a circle. Starting at 1, every

fifteenth number is marked (i.e. 1, 16, 31,). This process is continued until a number is reached which has already been marked, then marked numbers are

- (A) 11 (B) 17
(C) 986 (D) 996

18. The number of selections of 4 letters taken from the word COLLEGE is equal to

- (A) 24
(B) 18

(C) Coefficient of x^4 is $(1+x)^4 (1+x)^2 (1+x)^2$
(D) Coefficient of x^4 in $(1+x)^3 (1+x+x^2)^2$

19. All the five digit numbers in which each successive digit exceeds its predecessor are arranged in the increasing order. The 105th number does not contain the digit

- (A) 1 (B) 2
(C) 3 (D) 4

SECTION-3

COMPREHENSION BASED QUESTIONS

COMPREHENSION: (Q.1 TO Q.3)

A library has 5 indistinguishable physics books, 4 indistinguishable mathematics books and 3 indistinguishable chemistry books. In how many distinguishable ways can a student take home.

1. 6 books

- (A) 16 (B) 18 (C) 19 (D) 20

2. 6 books taking atleast one of each subject.

(A) 11

(B) 8

(C) 10

(D) 9

3. 6 books taking not more than 2 of each subject.

(A) 1

(B) 2

(C) 4

(D) 6

COMPREHENSION (Q.4 TO Q.5) :

Each of 5 woman who attend a banquet checks her coat and hat with the receptionist on arrival. Upon leaving, each woman is given a coat and a hat at random.

4. The number of ways these coats and hats may be distributed such that nobody gets back both her coat and her hat is

(A) 2048

(B) 1936

(C) 1956

(D) 1978

5. The number of ways these coats and hats may be distributed such that nobody gets back either her coat or her hat is

(A) 14400

(B) 118000

(C) 11842

(D) 11844

COMPREHENSION (Q.6 TO Q.8) :

A point in the x-y plane whose coordinates are integers is called a lattice point. Consider a path from the origin to the lattice point $A(n, n)$, where n is non negative, that

(i) starts from origin.

(ii) is always parallel to the x-axis or y-axis.

(iii) makes turns only at a lattice point, either along positive x-axis or along the positive y-axis.

(iv) terminates at A.

6. Either $x > y$ at all interior lattice points or $y > x$ at all interior lattice points.

(A) ${}^{2n-2}C_{n-1} - {}^{2n-2}C_n$

(B) $2 \frac{{}^{2n-1}C_n}{(n+1)}$

(C) $\frac{2}{n} {}^{2n-2}C_{n-1}$

(D) $2 \frac{{}^{2n}C_n}{n+1}$

7. $y \leq x$ at every lattice point on the path

(A) $\frac{{}^{2n}C_n}{n+1}$

(B) $\frac{{}^{2n-2}C_{n-1}}{n}$

(C) ${}^{2n-2}C_{n-1} - {}^{2n-2}C_n$

(D) $\frac{{}^{2n-1}C_{n-1}}{n}$

8. The path never crosses the line $y = x$.

(A) $2 \frac{{}^{2n-1}C_n}{n+1}$

(B) $2 \left({}^{2n-2}C_{n-1} - {}^{2n-2}C_n \right)$

(C) $\frac{2}{n} {}^{2n-2}C_{n-1}$

(D) $\frac{{}^{2n}C_n}{n+1}$

COMPREHENSION (Q.9 TO Q.11) :

Consider a binary string which consist only of digits 0 and 1. Let a_n be the number of binary strings of length 'n' that do not contain the sequence 11 and b_n be the number of binary strings of length 'n' that do not contain the string 111. Then

(For example 0001001010 is a binary string of length 10)

9. a_n is equal to (where $n \geq 5$)

(A) $a_{n-1} + 2a_{n-2}$

(C) $2a_{n-1} + a_{n-2}$

(B) $a_{n-1} + a_{n-3}$

(D) $a_{n-1} + a_{n-2}$

10. b_n is equal to (where $n \geq 5$)

(A) $b_{n-1} + 2b_{n-2} + 3b_{n-3}$

(C) $4b_{n-1} + 2b_{n-2} + b_{n-3}$

(B) $b_{n-3} + b_{n-2} + b_{n-1}$

(D) $b_{n-1} + b_{n-2}$

11. b_6 is equal to

(A) 64

(C) 26

(B) 58

(D) 44

COMPREHENSION (Q.12 TO Q.13) :

A bag has 5 white marbles, 3 red marbles, and 5 blue marbles, marbles are drawn one by one and without replacement till all marbles are drawn. Marbles of each colour are indistinguishable. Then

12. In how many ways may one draw the marbles out of the bag if all blue are drawn in odd numbered draws, or all blue are drawn in even numbered draw.

(A) 1202

(C) 1512

(B) 1448

(D) 1668

13. In how many ways will all red marbles come before any of the blue marbles.

(A) 1496

(C) 1024

(B) 1287

(D) 967

COMPREHENSION (Q.14 TO Q.16) :

Consider the word 'ASSASSINATION'. How many different arrangements are there of all the letters of the word given so that

14. The first N precedes O.

(A) $\frac{13!}{3!3!4!}$

(B) $\frac{13!}{3!3!4!2!}$

(C) $\frac{10! \times {}^{13}C_3}{3!}$

(D) $\frac{13!}{3!4!}$

15. The first A precedes the first S.

(A) $2700 \times {}^{13}C_7$

(B) $150 \times {}^{13}C_7$

(C) ${}^{13}C_7 {}^6C_3$

(D) ${}^{13}C_7 \frac{6!}{4}$

16. First A precedes the first S and the first N precedes O

(A) $\frac{3 \times 13!}{4 \times 5!}$

(B) $\frac{13!}{7 \times 4!}$

(C) $\frac{13! \times 11!}{7}$

(D) $\frac{5 \times 13!}{2 \times 7!}$

COMPREHENSION (Q.17 TO Q.18) :

Eleven criminals want to keep the location of their master criminal in a safe. They want to be able to open the safe only when any 6 of them are present. The safe is thus equipped with a number of different locks, and each criminal is given the keys to some of these locks.

17. What is the minimum number of locks required

(A) 256

(B) 504

(C) 462

(D) 420

18. What is the minimum number of keys each criminal must carry

(A) 216

(B) 252

(C) 126

(D) 504

Let there is a large pile of red, white, green and blue balls, where balls are all alike except for the colour. Then find the number of ways to select 20 balls from them so that

19. The selection has atmost 3 green balls.

- | | |
|---------|---------|
| (A) 802 | (B) 969 |
| (C) 973 | (D) 879 |

20. The selection has an even number of white balls

- | | |
|---------|----------|
| (A) 844 | (B) 929 |
| (C) 948 | (D) 1017 |

COMPREHENSION (Q.21 TO Q.22) :

A committee of 10 person is to be selected from 9 women and 8 men consisting of atleast 4 women and 4 men. Find number of ways committee if can be formed if

21. Miss X refuses to work with Mr. Y

- | | |
|-----------|-----------|
| (A) 12182 | (B) 5586 |
| (C) 10876 | (D) 10878 |

22. Miss X and Mr. Y insist to work together

- | | |
|-----------|----------|
| (A) 9128 | (B) 6486 |
| (C) 10876 | (D) 8232 |

COMPREHENSION (Q.23 TO Q.25) :

There are 'm' seats in the first row of a theatre, of which 'n' are to be occupied, where m is odd and n is even and $n < \frac{m}{2}$. Then find the number of ways of arranging 'n' persons so that.

23. N two persons sit side by side.

(A) ${}^{m-n+1}C_n (n-1)!$

(B) ${}^{m-n+1}P_n$

(C) ${}^{m-n}P_n$

(D) mP_n

24. Each person has exactly one neighbour

(A) ${}^{m-n+1}C_{\frac{n}{2}} n!$

(B) ${}^{m-n+1}C_{\frac{n}{2}} \left(\frac{n}{2}\right)!$

(C) ${}^{m-n}C_{\frac{n}{2}} \left(\frac{n}{2}\right)!$

(D) ${}^{m-n+1}C_{\frac{n}{2}} \left(\frac{n}{2}\right)! 2^{\frac{n}{2}}$

25. Out of any two seats located symmetrically about the middle of the row, atleast one is empty

(A) ${}^{m+1}C_n 2^n n! + {}^{m+1}C_{n-1} n! 2^{n-1}$

(B) ${}^{m+1}C_n 2^n n!$

(C) $2^{\frac{m-1}{2}} C_n 2^n n!$

(D) $2^{n-1} n! \left(2^{\frac{m-1}{2}} C_n + 2^{\frac{m-1}{2}} C_{n-1} \right)$

COMPREHENSION (Q.26 TO Q.28) :

Consider the word 'MATHEMATICS'. Find the number of all possible words that can be formed using all letters of the given word so that.

26. The odd numbered places do not contain all distinct letters.

(A) $\frac{11!}{2! 2! 2!} - \frac{9!}{2! 2!}$

(B) $\frac{(111) 5! 6!}{2! 2! 2!}$

(C) $\frac{11!}{2! 2! 2!}$

(D) $\frac{(113) 5! 6!}{2! 2! 2!}$

27. Two M's and two T's are together but 2A's are not together.

(A) $(28) 7!$

(B) $\frac{6! 4!}{2! 2!}$

(C) $(24) 7!$

(D) $\frac{8!}{2!}$

28. Two vowels are never together.

(A) $115 \times 8!$

(B) $210 \times 7!$

(C) $220 \times 7!$

(D) $240 \times 7!$

SECTION-4

MATCH THE COLUMN

1. Five balls are to be placed in three boxes. Each can hold all the five balls. The number of different ways can we place the balls so that no box remain empty if

	Column-I		Column-II
(A)	balls and boxes are all different is equal to	(P)	2
(B)	balls are identical but boxes are different is equal to	(Q)	6
(C)	balls are different but boxes are identical is equal to	(R)	25
(D)	balls as well as boxes are identical is equal to	(S)	50
		(T)	150

2. The number of 10 letter permutations comprising 4a's, 3b's and 3c's such that

	Column-I		Column-II
(A)	a's are separated and all b's are together is equal to	(P)	18
(B)	a's are separated and exactly two b's are together is equal to	(Q)	20
(C)	no two adjacent letters are identical is equal to	(R)	150
(D)	no two b's are together, no two c's are	(S)	180

	together and all 'a's are		
		(T)	248

3. A person moves in the x - y plane moving along points with integer coordinates x and y only. When she is at point (x, y) , she takes a step based on the following rules:
- (a) if $x + y$ is even, she moves to either $(x + 1, y)$ or $(x + 1, y + 1)$.
 - (b) if $x + y$ is odd, she moves to either $(x, y + 1)$ or $(x + 1, y + 1)$.

The number of distinct paths can she take to go from $(0, 0)$ to $(8, 8)$ given that

	Column-I		Column-II
(A)	she took exactly three steps to the right $((x, y) \text{ to } (x + 1, y))$	(P)	210
(B)	she took exactly two steps up $((x, y) \text{ to } (x, y + 1))$ is equal to	(Q)	320
(C)	she took exactly four steps to the right $((x, y) \text{ to } (x + 1, y))$ is equal to	(R)	364
(D)	she took exactly three steps to the right $((x, y) \text{ to } (x + 1, y))$ and exactly three steps up $((x, y) \text{ to } (x, y + 1))$ is equal to	(S)	462
		(T)	495

4. A is set containing 'n' elements. A subset A_1 of A is chosen. The set A is reconstructed by replacing the elements of A_1 . Then a subset A_2 of A is chosen and again set A is reconstructed by replacing elements of A_2 . In this way we choose 'm' subsets $A_1, A_2, A_3, \dots, A_m$, $m > 3$. Then the number of ways of choosing $A_1, A_2, A_3, \dots, A_m$ such that

	Column-I		Column-II
--	----------	--	-----------

(A)	$A_1 \cup A_2 \cup \dots \cup A_m$ contains exactly 'r' elements of A ($r < n$) is	(P)	$(2^m - 1)^n$
(B)	$A_1 \cap A_2 \cap \dots \cap A_m$ contains exactly 'r' elements of A ($r < n$) is	(Q)	${}^nC_r(2^m - 1)^{n-r}$
(C)	$A_i \cap A_j = \phi \quad \forall i \neq j$ is	(R)	${}^nC_r(m + 1)^r$
(D)	$A_1 \cup A_2 \cup A_3 \cup \dots \cup A_m = A$ is	(S)	$(m + 1)^n$
		(T)	${}^nC_r(2^m - 1)^r$

5.	Column-I		Column-II
(A)	The number of ways of selecting two parallel lines from the edges and face diagonals of a cube is equal to	(P)	18
(B)	The number of ways of selecting two skew face diagonals from the edges and face diagonals of a cube is equal to	(Q)	24
(C)	The number of ways of selecting an edge and a face diagonal from the edges and face diagonals of a cube such that they are skew is equal to	(R)	30
(D)	The number of ways of selecting two coplanar and perpendicular lines from the edges and face diagonals of a cube is equal to	(S)	54
		(T)	72

6. Let N be the number of 4 digit numbers, abcd (where a, b, c, d are digits) satisfying the condition in column-I and the last digit of N in column-II.

	Column-I		Column-II
(A)	a, b, c, d are distinct and $a > \max \{b, c, d\}$	(P)	0
(B)	a, b, c, d are distinct and $a < \min \{b, c, d\}$	(Q)	2
(C)	$a \leq b \leq c \leq d$	(R)	4
(D)	$a \geq b \geq c \geq d$	(S)	5
		(T)	6

7. Let $A = \{1, 2, 3, 4\}$, $B = \{1, 2, 3, 4, 5, 6, 7, 8\}$

	Column-I		Column-II
(A)	The number of functions $f : B \rightarrow A$, such that if $x_1 > x_2$, then $f(x_1) \geq f(x_2) \forall x_1 \in B$ is N . Then the sum of digits of N is	(P)	7
(B)	The number of onto functions $f : B \rightarrow A$ such that if $x_1 > x_2$, then $f(x_1) \leq f(x_2) \forall x_1 \in B$ is N . Then the sum of digits of N is	(Q)	8
(C)	The numbers of one-one function $f : A \rightarrow B$ such that $f(x) \neq x \forall x \in A$ is N . Then the sum of digits of N is	(R)	9
(D)	The number of one-one function $f : A$	(S)	10

	→ B such that there are exactly two, x , $x \in A$ for which $f(x) = x$ is N. Then the sum of digits of N is		
		(T)	12

SECTION-5

SUBJECTIVE TYPE QUESTIONS

- Let $0 < a < b < c < d < e < f < g$ be a geometric sequence of integers. Let $*(k)$ denote the number of divisors of k . For example $*(6) = 4$ because 1, 2, 3, 6 are divisors of 6. If $*(a) = 7$, $*(g) = 13$ and $d - c = 432$, then find $\left[\frac{b}{100} \right]$ ([.] denote greatest integer function)
- Let N is the number of times digit 5 is written when listing all natural numbers from 1 to 10^5 . Then the sum of digits of N is
- Let N be number of ways four different integers be chosen from the set $\{1, 2, 3, 4, \dots, 104, 105\}$ so that their sum is divisible by 4, then $\left[\frac{N}{10^5} \right]$ is equal to

([.] denote greatest integer function)

4. Let N be the number of ordered 6-tuples $(x_1, x_2, x_3, x_4, x_5, x_6)$ of positive integers satisfying $x_1 + x_2 + x_3 + 3x_4 + 3x_5 + 5x_6 = 21$, then $\sqrt{N}$ is equal to
5. Let N is the number of 4 element subsets $\{a, b, c, d\}$ of $\{1, 2, 3, 4, \dots, 20\}$ such that $a + b + c + d$ is divisible by 3. Then the sum of the digits of N is equal to
6. Suppose $A_1, A_2, \dots, A_6$ are six sets, each with four elements and $B_1, B_2, \dots, B_n$ are n sets each with two elements. Let $S = A_1 \cup A_2 \cup \dots \cup A_6 = B_1 \cup B_2 \cup \dots \cup B_n$. Given that each element of S belongs to exactly four of the A 's and to exactly three of the B 's, then n is equal to
7. Suppose $A_1, A_2, \dots, A_{20}$ is a 20 sided regular polygon. Let N be the number of non isosceles triangles that can be formed whose vertices are the vertices of the polygon but whose sides are not the sides of the polygon. Then $\sqrt{\frac{N}{10}}$ is equal to

8. Let $A_1 A_2 A_3 \dots A_{200}$ be a regular 200-sided convex polygon. Make the diagonals $A_i A_{i+9}$, $i = 1, 2, 3, \dots, 200$, where $A_{i+200} = A_i$ for $1 \leq i \leq 9$. Let N be the number of distinct points of intersection formed inside the polygon by these 200 diagonals, then the sum of the digits of N is
9. There are 6 given points on a circle and each two points are connected by a segment. Suppose that any three segments are not concurrent, so any three intersecting segments form a triangle inside the circle. Let N be the number of triangles formed, then sum of digits of N is
10. All the 7 digit numbers containing each of the digits 1, 2, 3, 4, 5, 6, 7 exactly once and not divisible by 5, are arranged in the increasing order. Let N be the 2018th

number in this list, then the last digit of N is

11. An insect moves from $(0, 0)$ to $(6, 3)$ by moving through lattice points (lattice points are point (x, y) such that $x, y \in \mathbb{I}$), moving either one unit right or one unit up at each step. Let N be the number of paths in which the line segments joining $(2, 1)$ and $(2, 2)$ and $(3, 2)$ and $(4, 2)$ are avoided, then $\left[\frac{N}{6} \right]$ is equal to ([.] denote greatest integer function).
12. Let N be the number of all 5 digit numbers each of which contains the block 15 and is divisible by 15. Then the last digit of N is
13. Let N be the number of 6 digit numbers such that the digits of each number are all from the set $\{1, 2, 3, 4, 5\}$ and any digit that appears in the number appears atleast twice. Then the last digit of N is
14. Let N be the number of all 6 digit numbers such that the sum of their digits is 10 and each of the digits 0, 1, 2, 3 occurs atleast once in them, then the last digit of N is

15. Let N be the number of all 4 digit numbers having non zero digits and which are divisible by 4 but not by 8. Then the last digit of N is
16. Let $X = \{1, 2, 3, \dots, 12\}$ and N be the number of pairs $\{A, B\}$ such that $A \subseteq X, B \subseteq X, A \neq B$ and $A \cap B = \{2, 3, 5, 7, 8\}$. Then the value of N is
17. Find the number of eight digit numbers the sum of whose digits is 4.

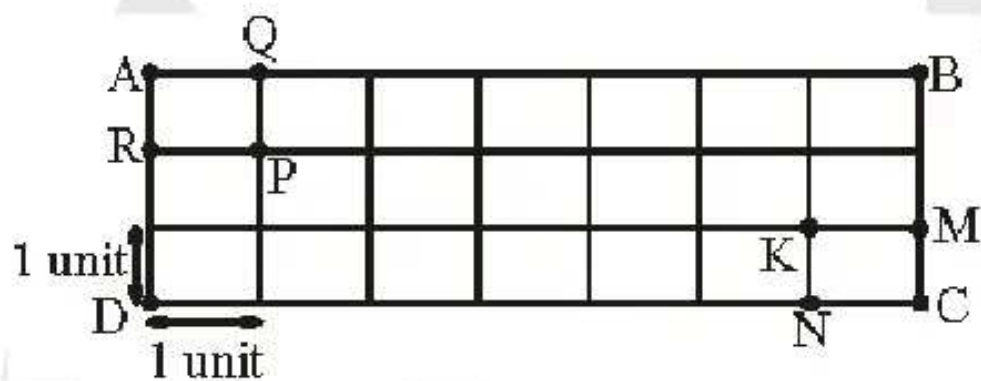
18. Find number of all 4-tuples (a, b, c, d) of natural numbers with $a \leq b \leq c$ and $a! + b! + c! = 3^d$.
19. Let N be the number of non empty subsets S of the set $\{0, 1, 2, 3, \dots, 9\}$ so that the sum of elements of 'S' is divisible by 3. Then the sum of the digits of N is
20. Let N be the number of ordered pairs (A, B) where A and B are subsets of $\{1, 2, 3, 4, 5\}$ such that neither $A \subseteq B$ nor $B \subseteq A$. Then N is equal to
21. Suppose 32 objects are placed along a circle at equal distances. Let N be the number of ways can 3 objects be chosen from among them so that no two of the three chosen objects are adjacent nor diametrically opposite. Then the value of N is
22. Let N be the number of 3-digit numbers having atleast one 5 and atmost one 3. Then N is equal to
23. There are four basket ball players A, B, C, D . Initially, the ball is with A . The ball is always passed from one person to a different person. Let N be the number of ways the ball come back to A after seven passes, (For example $A \rightarrow C \rightarrow B \rightarrow D \rightarrow A \rightarrow B \rightarrow C \rightarrow A$ and so on). Then N is equal to

24. Let N be the number of positive integers less than 1000 have the property that the sum of the digits of each such number is divisible by 7 and the number itself is divisible by 3. Then N is equal to
25. Let N be the number of triplets (a, b, c) , where $a, b, c \in \{1, 2, 3, 4, 5\}$ such that $2^a + 3^b + 5^c$ is divisible by 4. Then the sum of digits of N is

26. Let $A = \{1, 2, 3, 4, 5, \dots, 99, 100\}$ and B be a subset of A such that the sum of no two elements in B is divisible by 7. Let N be maximum number of elements in B , then $\frac{N}{5}$ is equal to
27. Let 5 letter words are formed using the letters of the word 'CALCULUS' and N is the rank of word CALCUL among them as arranged in dictionary. Then the sum of digit of N is
28. Let N be the number of 5 letter words can you spell using the letters S, I and T only, if a 'word' is defined as any sequence of letters that doesn't contain 3 consecutive consonants. Then the sum of digits of N is
29. A number has four divisors and sum of its divisors excluding 1 and itself is 30, then the number of such numbers is equal to
30. The number of order triplets (a, b, c) such that $\text{LCM}(a, b) = 1000$, $\text{LCM}(b, c) = 2000$ and $\text{LCM}(c, a) = 2000$ is
31. Let N be the number of eight digit numbers that can be formed using the digits 1, 2, 3, 4 only such, that sum of the eight digits is 12. Then the sum of digits of N .

32. Consider all 5 digit numbers such that the sum of their digits is 7 and each of the digits 0, 1, 2 occurs atleast once in them, the number of such numbers is equal to

33. A class contains three girls and four boys. Every Saturday, five students go on a picnic, a different group being sent each week. During the picnic, each girl in the group is given a doll by the accompanying teacher. After all possible groups have gone once, the total number of dolls the girls have got is equal to
34. The total number of divisors of $3^5 5^7 7^9$ which are of the form $4k + 1$, $k \geq 0$, $k \in I$ is equal to
35. A is the set of three digit numbers formed using the digits 1, 2, 3, ..., 8, 9 using each digit at the most once. Also every number in A has exactly one even digit. Find the sum of all the numbers in A.
36. An ant has to travel from point C to A along the gridlines. It gets lost on its way but manages to reach A by travelling 12 units in all. Its first two moves are CM and MK and its last two moves are PQ and QA. In how many ways could the ant have travelled from C to A, if it never travels downwards.



37. Find the number of quadruples of positive integers (a, b, c, d) such that $(a + b)(c + d) = 2016$.
38. For any set S , let $|S|$ denote the number of elements in S and let $n(S)$ be the number of subsets of S , including the empty set and S itself. If A , B and C are sets for which $n(A) + n(B) + n(C) = n(A \cup B \cup C)$ and $|A| = |B| = 100$, then what is the minimum possible value of $|A \cap B \cap C|$.

39. Let N be the number of ordered triples of (A, B, C) such that $A \cup B \cup C = \{1, 2, 3, \dots, 2003\}$ and $A \cap B \cap C = \phi$. Then number of divisors of N is equal to
40. For how many pairs of consecutive integers in $\{1000, 1001, 1002, \dots, 2000\}$ is no carrying required when the two integers are added.
41. Find the number of ways 10 can be expressed as a sum of positive integer. For example '3' can be expressed as a sum of positive integers in 4 ways 3, $2 + 1$, $1 + 2$, $1 + 1 + 1$.

Answer Key

SINGLE CHOICE QUESTIONS

- | | | | | | |
|-------|-------|-------|-------|-------|-------|
| 1. A | 2. C | 3. B | 4. B | 5. A | 6. D |
| 7. A | 8. A | 9. D | 10. A | 11. A | 12. A |
| 13. B | 14. B | 15. D | 16. A | 17. A | 18. C |
| 19. D | 20. B | 21. A | 22. D | 23. B | 24. A |
| 25. C | 26. B | 27. A | 28. D | 29. C | 30. B |
| 31. A | 32. C | 33. D | 34. B | 35. A | 36. A |

ONE OR MORE THAN ONE CORRECT QUESTIONS

- | | | | | | |
|-----------|---------|----------|-----------|-----------|-----------|
| 1. B,D | 2. B,C | 3. A,B | 4. B,C,D | 5. B,C,D | 6. A,CD |
| 7. A,B,C | 8. B,C | 9. A,C,D | 10. B,C,D | 11. B,D | 12. A,B,C |
| 13. B,C,D | 14. B,C | 15. B,C | 16. A,D | 17. A,C,D | 18. B,D |
| 19. A,C,D | | | | | |

COMPREHENSION BASED QUESTIONS

- | | | | | | |
|-------|-------|-------|-------|-------|-------|
| 1. B | 2. D | 3. A | 4. B | 5. D | 6. C |
| 7. A | 8. D | 9. D | 10. B | 11. D | 12. C |
| 13. B | 14. A | 15. A | 16. D | 17. C | 18. B |
| 19. A | 20. C | 21. D | 22. D | 23. B | 24. A |
| 25. D | 26. D | 27. A | 28. B | | |

MATCH THE COLUMN

- | | |
|---|---|
| 1. $A \rightarrow T; B \rightarrow Q; C \rightarrow P; D \rightarrow R$ | 2. $A \rightarrow Q; B \rightarrow S; C \rightarrow T; D \rightarrow P$ |
| 3. $A \rightarrow S; B \rightarrow P; C \rightarrow T; D \rightarrow S$ | 4. $A \rightarrow T; B \rightarrow Q; C \rightarrow S; D \rightarrow P$ |
| 5. $A \rightarrow Q; B \rightarrow R; C \rightarrow T; D \rightarrow S$ | 6. $A \rightarrow P; B \rightarrow T; C \rightarrow S; D \rightarrow R$ |
| 7. $A \rightarrow T; B \rightarrow Q; C \rightarrow P; D \rightarrow R$ | |

SUBJECTIVE TYPE QUESTIONS

- | | | | | | |
|-----------|---------|-------------|----------|------------|--------|
| 1. 9 | 2. 5 | 3. 7 | 4. 9 | 5. 9 | 6. 9 |
| 7. 8 | 8. 7 | 9. 3 | 10. 1 | 11. 8 | 12. 9 |
| 13. 5 | 14. 0 | 15. 9 | 16. 2186 | 17. 120 | 18. 3 |
| 19. 9 | 20. 570 | 21. 3616 | 22. 249 | 23. 546 | 24. 28 |
| 25. 7 | 26. 9 | 27. 8 | 28. 6 | 29. 4 | 30. 70 |
| 31. 7 | 32. 100 | 33. 45 | 34. 240 | 35. 133200 | 36. 56 |
| 37. 59508 | 38. 98 | 39. 4016016 | 40. 156 | 41. 512 | |

Previous Year Questions

SECTION-1

1. Let T_n be the number of all possible triangles formed by joining vertices of an n -sided regular polygon.

If $T_{n+1} - T_n = 10$, then the value of n is

[IIT JEE Main 2013]

(A) 8

(B) 7

(C) 5

(D) 10

2. The number of integers greater than 6,000 that can be formed, using the digits 3, 5, 6, 7 and 8, without repetition, is:

[IIT JEE Main 2015]

(A) 72

(B) 216

(C) 192

(D) 120

3. Let A and B be two sets containing four and two elements respectively. Then the number of subsets of the set $A \times B$, each having at least three elements is :

[IIT JEE Main 2015]

(A) 510

(B) 219

(C) 256

(D) 275

4. If all the words (with or without meaning) having five letters, formed using the

letters of the word SMALL and arranged as in a dictionary : then the position of the

letters of the word SMALL and arranged as in a dictionary, then the position of the word SMALL is : [IIT JEE Main 2016]

- (A) 46th (B) 59th
(C) 52nd (D) 58th

120

Advance Problems in Algebra for JEE

5. A man X has 7 friends, 4 of them are ladies and 3 are men. His wife Y also has 7 friends, 3 of them are ladies and 4 are men. Assume X and Y have no common friends. Then the total number of ways in which X and Y together can throw a party inviting 3 ladies and 3 men, so that 3 friends of each of X and Y are in this party, is : [IIT JEE Main 2017]

- (A) 469 (B) 484
(C) 485 (D) 468

6. From 6 different novels and 3 different dictionaries, 4 novels and 1 dictionary are to be selected and arranged in a row on a shelf so that the dictionary is always in the middle. The number of such arrangements is : [JEE Main 2018]

- (A) at least 750 but less than 1000
(B) at least 1000
(C) less than 500
(D) at least 500 but less than 750

SECTION-2

1. How many different nine digit numbers can be formed from the number 223355888 by rearranging its digits so that the odd digits occupy even positions ?

[JEE '2000, (Scr)]

- (A) 16 (B) 36

(C) 60

(D) 180

2. Let T_n denote the number of triangles which can be formed using the vertices of a regular polygon of 'n' sides. If $T_{n+1} - T_n = 21$, then 'n' equals:

[JEE '2001, (Scr)]

(A) 5

(B) 7

(C) 6

(D) 4

3. The number of arrangements of the letters of the word BANANA in which the two N's do not appear adjacently is

[JEE 2002 (Screening), 3]

(A) 40
(C) 80

(B) 60
(D) 100

4. Number of points with integral co-ordinates that lie inside a triangle whose co-ordinates are (0, 0), (0, 21) and (21, 0)

[JEE 2003 (Screening), 3]

(A) 210

(B) 190

(C) 220

(D) None

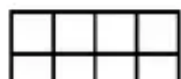
5. Using permutation or otherwise, prove that $\frac{(n^2)!}{(n!)^n}$ is an integer, where n is a positive integer.

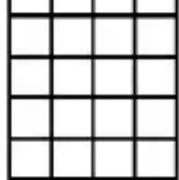
[JEE 2004, 2 out of 60]

6. A rectangle with sides $2m - 1$ and $2n - 1$ is divided into squares of unit length

by drawing parallel lines as shown in the diagram, then the number of rectangles possible with odd side lengths is

[JEE 2005 (Screening), 3]





(A) $(m + n + 1)^2$

(B) 4^{m+n-1}

(C) m^2n^2

(D) $mn(m + 1)(n + 1)$

7. If r, s, t are prime numbers and p, q are the positive integers such that their LCM of p, q is $r^2t^4s^2$, then the numbers of ordered pair of (p, q) is [JEE 2006, 3]

(A) 252

(B) 254

(C) 225

(D) 224

122

Advance Problems in Algebra for JEE

8. The letters of the word **COCHIN** are permuted and all the permutations are arranged in an alphabetical order as in an English dictionary. The number of words that appear before the word **COCHIN** is [JEE 2007, 3]

(A) 360

(B) 192

(C) 96

(D) 48

9. Consider all possible permutations of the letters of the word ENDEANOEL

Match the statements / Expression in **Column-I** with the statements / Expressions in **Column-II**.

	Column-I		Column-II
(A)	The number of permutations containing the word ENDEA is	(P)	$5!$
(B)	The number of permutations in which the letter E occurs in the first and the last position is	(Q)	$2 \times 5!$

(C)	The number of permutations in which none of the letters D, L, N occurs in the last five positions is	(R)	$7 \times 5!$
(D)	The number of permutations in which the letters A, E, O occurs only in odd positions is	(S)	$21 \times 5!$

[JEE 2008, 6]

10. The number of seven digit integers, with sum of the digits equal to 10 and formed by using the digits 1, 2 and 3 only, is [JEE 2009]

- (A) 55 (B) 66
(C) 77 (D) 88

11. Let $S = \{1, 2, 3, 4\}$. The total number of unordered pairs of disjoint subsets of S is equal to [JEE 2010]

- (A) 25 (B) 34
(C) 42 (D) 41

12. The total number of ways in which 5 balls of different colours can be distributed among 3 persons so that each person gets at least one ball is [JEE 2012]

- (A) 75 (B) 150
(C) 210 (D) 243

13. Let $n_1 < n_2 < n_3 < n_4 < n_5$ be positive integers such that $n_1 + n_2 + n_3 + n_4 + n_5 = 20$.

Then the number of such distinct arrangements $(n_1, n_2, n_3, n_4, n_5)$ is

14. Let $n \geq 2$ be an integer. Take n distinct points on a circle and join each pair of points by a line segment. Colour the line segment joining every pair of adjacent points by blue and the rest by red. If the number of red and blue line segments are equal, then the value of n is

[IIT JEE Adv. 2014]

15. Six cards and six envelopes are numbered 1, 2, 3, 4, 5, 6, and cards are to be placed in envelopes so that each envelope contains exactly one card and no card is placed in the envelope bearing the same number and moreover the card numbered 1 is always placed in envelope numbered 2. Then the number of ways it can be done is :

[IIT JEE Advance 2014]

(A) 264

(B) 265

(C) 53

(D) 67

16. Let n be the number of ways in which 5 boys and 5 girls can stand in a queue in such a way that all the girls stand consecutively in the queue. Let m be the number of ways in which 5 boys and 5 girls can stand in a queue in such a way that exactly four girls stand consecutively in the queue. Then the value of $\frac{m}{n}$ is

[IIT JEE Advance 2015]

17. A debate club consists of 6 girls and 4 boys. A team of 4 members is to be selected from this club including the selection of a captain (from among these 4 members) for the team. If the team has to include at most one boy, then the number of ways

of selecting the team is :

[IIT JEE Adv. 2016]

(A) 380

(B) 320

(C) 260

(D) 95

18. Let $S = \{1, 2, 3, \dots, 9\}$. For $k = 1, 2, \dots, 5$, let N_k be the number of subsets of S , each containing five elements out of which exactly k are odd. Then

$$N_1 + N_2 + N_3 + N_4 + N_5 =$$

[JEE Advance 2017]

(A) 125

(B) 252

(C) 210

(D) 126

19. Words of length 10 are formed using the letters A, B, C, D, E, F, G, H, I, J. Let x be the number of such words where no letter is repeated; and let y be the number of such words where exactly one letter is repeated twice and no other letter is

repeated. Then $\frac{y}{9x} =$

[JEE Advance 2017]

20. The number of 5 digit numbers which are divisible by 4, with digits from the set $\{1, 2, 3, 4, 5\}$ and the repetition of digits is allowed, is _____.

[JEE Advanced 2018]

21. Let X be a set with exactly 5 elements and Y be a set with exactly 7 elements. If α is the number of one-one functions from X to Y and β is the number of onto

functions from Y to X , then the value of $\frac{1}{\beta - \alpha}$ is _____.